

Constraining disk-to-corona power transfer fraction, soft x-ray excess origin, and black hole spin population of type-1 AGN across mass scales

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Understanding the nature of the accretion disk, its interplay with the x-ray corona, and assessing black hole spin demographics remain open challenges in astrophysics. In this paper, we examine the predictions of the standard α -disk model, origin of the puzzling soft x-ray excess, and measure the black hole spin parameter by applying an updated high-density disk reflection model to the *XMM-Newton*/*NuSTAR* broadband (0.3–78 keV) x-ray spectra of a sample of 11 type-1 active galactic nuclei (AGN). Our Bayesian analysis confirms that a variable-density relativistic disk reflection model with a broken power-law emissivity profile can simultaneously fit the soft x-ray excess, broad iron K line emission, and Compton hump in three out of 11 AGN. For the remaining sources, a distinct warm Comptonization component is still required, which supports a hybrid origin for the soft x-ray excess. The measured temperature and optical depth of the warm corona span nearly the entire theoretically allowed range, with median values of $0.43^{+0.40}_{-0.18}$ keV and $12.5^{+3.1}_{-3.9}$, respectively. Our first systematic calculation of the disk-to-corona power transfer fraction reveals that the fraction of power released from the accretion disk into the hot corona spans a wide range, with a sample median of $0.68^{+0.25}_{-0.25}$. The sample median values for the hot coronal plasma temperature and optical depth are 54^{+11}_{-12} keV and $0.98^{+0.22}_{-0.28}$, respectively. Finally, through both hard x-ray (3–78 keV) and broadband (0.3–78 keV) relativistic reflection spectroscopy, we systematically constrain the black hole spin parameter across the mass scales of $\log(M_{\text{BH}}/M_{\odot}) \sim 5.5\text{--}9.0$, thereby increasing or refining the available spin measurements in the AGN population by $\sim 20\%$.

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I. INTRODUCTION

Active galactic nuclei (AGN) are the most luminous ($10^{41}\text{--}10^{48}$ erg s^{−1}), compact regions at the center of galaxies, fed by mass accretion from their host galaxies onto super-massive black holes (SMBHs) of mass $M_{\text{BH}} \sim 10^{5\text{--}10} M_{\odot}$ (e.g. [1–4]). The gradual loss of angular momentum causes the

inflowing matter to move toward the center of gravity, forming an accretion disk around the SMBH, which is believed to be the engine that powers AGN (e.g. [5]). The observed correlation between SMBH mass and host galaxy-bulge velocity dispersion (e.g. [6–8]) implies that the growth of the host galaxy is coupled with the energy output from the central SMBH, and feedback from AGN can play a key role in the evolution of galaxies (e.g. [9–12]). Indeed, the AGN accretion disk can influence the evolution of the host galaxy via a sustained release of gravitational energy in the form of

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radiation or outflows (e.g. [13–16]). However, understanding the nature and dynamics of the accretion disk remains an open question in both theoretical and observational astrophysics.

AGN are inherently multiwavelength phenomena [17], and their spectral energy distributions at different energies are the outcome of various physical processes arising from distinct regions, dominating the observed emission for different AGN subclasses. However, the most effective way to probe the immediate vicinity of the central SMBH or innermost regions of AGN is to study the x-ray emission from type-1 AGN, the line of sight of which is not obscured by the molecular torus (e.g. [18–20]). In the unified view of AGN [21,22], the primary x-ray emission is produced by Compton upscattering of the optical/UV seed photons in an optically thin, hot plasma [23,24] called “corona” surrounding the SMBH, as also suggested by the global radiation magnetohydrodynamic simulations [25], where the seed photons are thought to be supplied by the accretion disk [26]. However, it is not well known how the accretion disk and corona are coupled and what fraction of power from the accretion disk gets transported into the corona. When this coronal radiation illuminates the accretion disk, it produces an intrinsic reflection spectrum containing a forest of fluorescent emission lines below ~ 2 keV, narrow iron (Fe) K emission lines (K_α/K_β), and a Compton scattering hump above 10 keV [27–29]. If the disk is close to the black hole, the intrinsic reflection spectrum gets smeared by the strong gravitational field and produces the blurred or relativistic reflection spectrum containing a soft x-ray excess below around 1–2 keV, a broad Fe K emission line (6–7 keV), and the Compton hump (15–30 keV) (e.g. [30–32]). Additionally, the Compton shoulder arising from line broadening in the disk photosphere due to Compton scattering in ionized gas can significantly contribute to the apparent red wing of the broad Fe K emission line [33]. Although the RELXILL family of reflection models incorporates Compton broadening, it still requires a more realistic treatment of the ionization structure of the accretion disk, both vertically and radially. Thus, measuring spin is not a straightforward diagnostic of general relativity; it relies on complex astrophysical modeling. Currently, relativistic reflection spectroscopy of the innermost accretion disk is the most widely adopted technique for measuring the spins of massive black holes in AGN (e.g. [34–37]), which is a crucial step in studying the formation and growth channels of SMBHs (e.g. [38–41]). Therefore, it is of central importance to precisely model disk reflection signatures and systematically assess the black hole spin population in AGN.

The origin of the soft x-ray excess observed below ~ 1 –2 keV in AGN is still unknown despite its discovery nearly 40 years ago by [42] and [43]. At first, it was thought to be the high-energy tail of the standard disk emission. However, the soft excess temperature is much higher than

the maximum disk temperature and remains constant within the range of ~ 0.1 –0.3 keV, irrespective of the black hole mass (see [44] for high-mass quasars and [37] for low-mass dwarf AGN).

Currently, two models are used to describe the soft x-ray excess emission. One model requires low-temperature Comptonization of optical/UV disk photons in an optically thick, warm corona (e.g. [45–47]). On the other hand, the reflection model naturally produces soft x-ray excess as relativistically smeared, broadened fluorescent lines arising from the innermost accretion disk (e.g. [31,48,49]). However, in some AGN, relativistic reflection alone cannot model the entire soft x-ray excess, and a warm Comptonization component is still required, particularly when the density of the accretion disk is fixed at the canonical value of $n_e = 10^{15} \text{ cm}^{-3}$ (e.g. Ark 120: [45,50], Mrk 110: [51]). This issue was tackled by employing a high-density disk reflection model [52], which can potentially fit the entire soft x-ray excess by boosting the strength of the excess emission below ~ 1 keV in the model, where the density parameter can reach as high as $\log[n_e/\text{cm}^{-3}] = 20$, first incorporated by [37]. For a geometrically thin and optically thick standard accretion disk ([53]; hereafter SS73), the electron density of the inner disk is likely to be higher than 10^{15} cm^{-3} . Moreover, the theoretical work of [54] (hereafter SZ94) proposed that the inner disk density would be even higher than that predicted by the SS73 model after including the fraction of power transferred from the accretion disk to the corona, i.e., the disk-to-corona power transfer fraction. While higher-density disk reflection has become more prevalent in explaining the origin of the soft x-ray excess in several AGN [37,55,56], the disk-to-corona power transfer fraction has yet to be explored.

In this paper, we study the broadband (0.3–78 keV) spectra of a sample of type-1 AGN across the central SMBH mass scales of $M_{\text{BH}} \approx 10^{5.5-9} M_\odot$, employing both *XMM-Newton* (0.3–10 keV; [57]) and *NuSTAR* (3–78 keV; [58]) data available in the public archive as of July 2024. Previously, [59] (hereafter JJ19) fit the averaged 0.5–10 keV spectra extracted solely from the *XMM-Newton* observations conducted before 2016 for the sample. They utilized a preliminary version of the high-density disk reflection model (*relxillD* v.1.2.0), where gas density was variable in the range of $\log[n_e/\text{cm}^{-3}] = 15$ –19. The earlier versions of the model did not have coronal temperature as a parameter, as it assumed a simple power law without a cutoff energy for the incident continuum. Our work employs the updated high-density disk reflection model (*relxillCp* v.2.3),¹ where the disk density can freely vary from $\log[n_e/\text{cm}^{-3}] = 15$ –20. This new model also incorporates the coronal temperature as a variable parameter that can only be measured via *NuSTAR* spectroscopy. We will verify whether high-density disk reflection is robust enough to explain both hard and soft x-ray excess self-consistently or

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if an extra warm Comptonization model is still required to fit the observed soft x-ray excess. We will calculate the disk-to-corona power transfer fraction and test the validity of the standard α -disk model for the AGN sample. We will utilize the unique capability of *NuSTAR* to unambiguously probe the hard band reflection associated with the broad Fe K emission together with *XMM-Newton*'s lower-energy coverage to reveal the soft x-ray reflected continuum and measure the SMBH spin population spanning almost the complete SMBH mass range. Modeling the broadband reflection components through joint *XMM-Newton*+*NuSTAR* spectroscopy is currently the most effective technique for constraining the SMBH spin parameter in AGN.

The paper is organized in the following way. In Sec. II, we describe the sample selection criteria with source properties, details of the observations analyzed in this work, and spectral extraction methods from the raw *XMM-Newton* and *NuSTAR* data. Section III describes our comprehensive broadband (0.3–78 keV) x-ray spectral fitting methodology, where we employ a Bayesian framework to assess the significance of both the variable-density disk reflection and warm Comptonization models for the origin of the observed soft x-ray excess. In Sec. IV, we discuss our results and their implications. We summarize our conclusions in Sec. V. The future prospects of this study are outlined in Sec. VI.

II. SAMPLE SELECTION AND DATA PROCESSING

We selected our AGN sample from JJ19, who conducted high-density disk reflection modeling of 17 AGN using only *XMM-Newton* data in the energy range of 0.5–10 keV.

To better understand accretion disk/corona coupling and to unambiguously investigate the origin of the puzzling soft x-ray excess with the updated high-density disk reflection model, we selected 11 AGN from the JJ19 sample that do not show any absorption features in spectra extracted from the European Photon Imaging Camera (EPIC) on *XMM-Newton*. The presence of absorption complicates spectral modeling and can lead to significant degeneracies. Therefore, we plan to study the six remaining AGN that exhibit absorption in the EPIC spectra in a follow-up paper. For the selected AGN sample, we utilize archival data from both the *XMM-Newton* (0.3–10 keV) and *NuSTAR* (3–78 keV) observatories. This enables us to apply the updated variable-density relativistic disk reflection model to the broadband (0.3–78 keV) x-ray spectra of the sample for the first time in a systematic manner. We will investigate the origin of both the hard and soft x-ray excess emission by assessing whether broadband relativistic reflection from a higher-density inner disk alone can adequately explain all the excess emission, or if an additional warm Comptonization component is still required to account for the soft x-ray excess in these AGN. This approach will allow us to constrain the SMBH spin population across mass scales using both hard x-ray (3–78 keV) and broadband (0.3–78 keV) relativistic reflection spectroscopy. Additionally, we will calculate the disk-to-corona power transfer fraction for the first time in any class of accreting objects. The observation details of the sample are provided in Table A1.

The basic characteristics of the selected 11 AGN are presented in Table I. Two key parameters for this study, black hole mass and dimensionless mass accretion rate, are listed

TABLE I. Details of the AGN sample employed. The columns show (1) source name, (2) right ascension (RA), (3) declination (DEC), (4) logarithmic of the central black hole mass, (5) source redshift, (6) Galactic hydrogen column density along the source line of sight in units of 10^{20} cm^{-2} [65], (7) dimensionless mass accretion rate ($\dot{m} = \frac{\dot{M}}{\dot{M}_E}$) taken from JJ19, and (8) source type based on optical classification. The RA, DEC, redshift, and source type are obtained from NED [66]. The black hole masses are taken from the AGN Black Hole Mass Database [60] and measured through optical reverberation mapping using the most updated scaling factor of 4.8 given by [67].

Source (1)	α_{2000} [Degree] (2)	δ_{2000} [Degree] (3)	$\log[\frac{M_{\text{BH}}}{M_{\odot}}]$ (4)	Redshift (z) (5)	$N_{\text{H,Gal}}$ [10^{20} cm^{-2}] (6)	Dimensionless Mass Accretion Rate ($\dot{m}$) (7)	Optical Type (8)
UGC 6728	176.317	79.682	$5.91^{+0.19}_{-0.42}$	0.0065	5.48	$0.58^{+0.76}_{-0.21}$	Sy1.2
Mrk 1310	180.31	−3.678	$6.26^{+0.07}_{-0.09}$	0.0196	2.66	$0.6^{+0.14}_{-0.1}$	BLS1
NGC 4748	193.052	−13.415	$6.46^{+0.11}_{-0.18}$	0.0146	4.07	$2.2^{+1.2}_{-0.5}$	NLS1
Mrk 110	141.304	52.286	$7.34^{+0.1}_{-0.1}$	0.0353	1.39	$0.9^{+0.23}_{-0.19}$	NLS1
Mrk 279	208.264	69.308	$7.48^{+0.1}_{-0.13}$	0.0305	1.72	$0.75^{+0.27}_{-0.16}$	BLS1
Mrk 590	33.64	−0.767	$7.62^{+0.06}_{-0.07}$	0.0264	2.92	$0.31^{+0.06}_{-0.05}$	BLS1
Mrk 79	115.637	49.81	$7.66^{+0.11}_{-0.14}$	0.0222	6.73	$0.13^{+0.05}_{-0.07}$	Sy 1.2
PG 1229 + 204	188.015	20.158	$7.81^{+0.18}_{-0.22}$	0.0636	2.92	$0.5^{+0.4}_{-0.2}$	BLS1
PG 0844 + 349	131.927	34.751	$7.91^{+0.15}_{-0.23}$	0.064	3.13	$1.2^{+1.0}_{-0.4}$	BLS1
PG 0804 + 761	122.744	76.045	$8.78^{+0.05}_{-0.05}$	0.1	3.31	$1.13^{+0.15}_{-0.12}$	BLS1
PG 1426 + 015	217.277	1.285	$9.06^{+0.11}_{-0.16}$	0.0866	2.88	$0.28^{+0.13}_{-0.06}$	BLS1

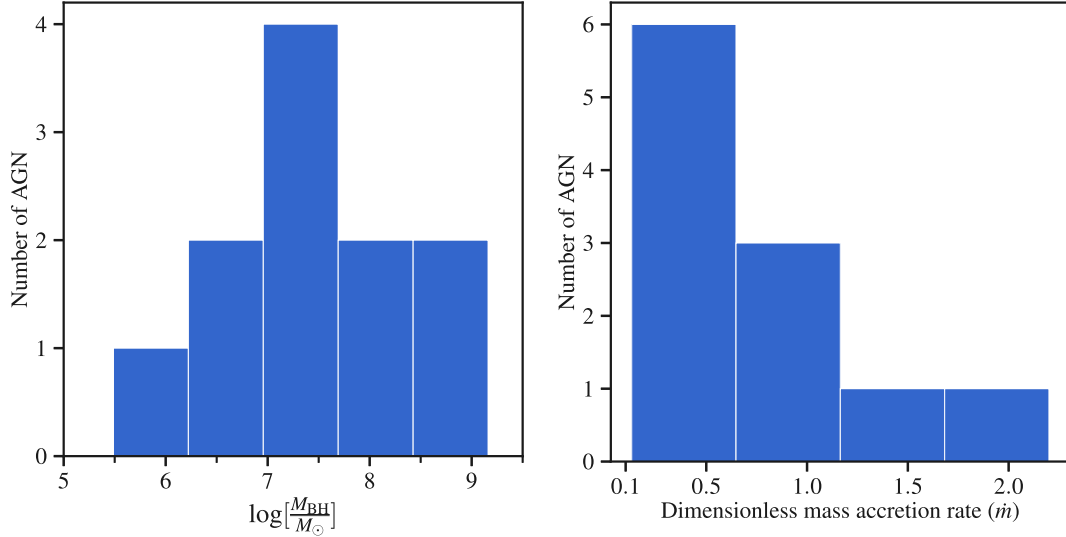


FIG. 1. Distribution of black hole mass (left panel) and dimensionless mass accretion rate (right panel) of the AGN sample employed in this work. We obtain black hole masses from the AGN Black Hole Mass Database [60] and the dimensionless mass accretion rates from JJ19 [59].

in columns (4) and (7), respectively. Black hole masses are determined using optical reverberation mapping and obtained from the AGN Black Hole Mass Database [60]. The dimensionless mass accretion rate is calculated from the observed optical luminosity for each source (see JJ19 for details) and not from the bolometric luminosity due to large uncertainties in the bolometric conversion factor [61] and radiative efficiency in AGN [62]. Figure 1 presents the distributions of black hole mass, $\log[M_{\text{BH}}/M_{\odot}]$, and dimensionless mass accretion rate ($\dot{m}$) of the 11 AGN employed in this work.

A. XMM-Newton data extraction

We start our work by collecting all the raw data of our sample from the *XMM-Newton* observatory available in the HEASARC² archive. We process the raw data from EPIC onboard *XMM-Newton* in the Scientific Analysis System (SAS v.21.0.0) with the most recent calibration files as of July 2024. First, we generated raw event files from EPIC-pn and MOS data with SAS tasks `epproc` and `emproc`, respectively. To exclude the background flares, we created good time intervals (GTI) above 10 keV for the full field using the technique detailed in [63]. We extracted flare-filtered clean event files by applying the flare-corrected GTI and unflagged events with `PATTERN` ≤ 4 for EPIC-pn and `PATTERN` ≤ 12 for EPIC-MOS. The EPIC-pn data have much better sensitivity above 6 keV and suffer less from pileup effects compared to the EPIC-MOS data. Therefore, we concentrate on the EPIC-pn data of the sample. However, we notice that the EPIC-pn events of UGC 6728 are flaring background dominated. Hence, we

consider the EPIC-MOS data only for UGC 6728. The source and background events are extracted from a circular region of radius 35 arcsec centered on the point source and nearby source-free area, respectively. We checked for the presence of pileup effects using the `epatplot` task. Whenever pileup was detected, we removed the central bright pixels by choosing an annular source region. We choose the inner radius of the annulus such that the distributions of the observed data match the model curves produced by `epatplot`. We generate the redistribution matrix file (`rmf`) and ancillary response file (`arf`), source, and background spectra using the SAS task `especget`. We binned the source spectra including background with the FTOOLS task `ftgroupPHA`, where we set `group-type=optsnmin` and `groupscale=5`. The `group-type=optsnmin` uses the optimal binning algorithm of [64] with an additional requirement of a minimum signal-to-noise ratio (SNR) per grouped bin set by `groupscale`. We model the *XMM-Newton*/EPIC spectra across the entire energy range of 0.3–10 keV. The EPIC camera, Obs. ID, start time of the observation, total elapsed time, net exposure, net count rate, and net counts in the 0.3–10 keV band for each source are listed in Table A1.

B. NuSTAR data extraction

NuSTAR observed all the sources in our sample with its two coaligned Focal Plane Modules, A (FPMA) and B (FPMB). We acquired all the available data from the HEASARC archive and reduced the raw (level 1) data in the *NuSTAR* Data Analysis Software (*NuSTARDAS* v.2.1.2). We produced the cleaned and calibrated event files with the `nupipeline` task using the latest (as of 2024 July 24) calibration database (CALDB version: 20240701). We

²<https://heasarc.gsfc.nasa.gov/cgi-bin/W3Browse>

employed conservative criteria, `saamode=optimized` and `tentacle=yes`, to treat the high-background levels induced by the South Atlantic Anomaly region. To maximize the SNR, we determine the optimal radius of a circular extraction region centered on the source using the *NuSTAR* tool `nustar-gen-utils`³ for each observation. The corresponding background extraction region was selected from the same-sized circular off-source region. We produced the response matrices (`rmf` and `arf`), source, and background spectra for both FPMA and FPMB with the `nuproducts` task. Finally, we generated background-subtracted binned spectra using the `ftgrouppha` tool, where we employ the optimal binning algorithm of [64] and a minimum SNR of 5 per grouped bin for both FPMA and FPMB. We fit the calibrated energy range of 3–78 keV for *NuSTAR*/FPMA and FPMB spectra. The Obs. ID, observation start time, total elapsed time, net exposure time, net count rate, and net counts in the 3–78 keV range for both modules are shown in Table A1.

III. BROADBAND X-RAY SPECTROSCOPY

A. Spectral modeling procedure

We jointly fit the *XMM-Newton* and *NuSTAR* spectral data for each source in our sample in the software package `XSPEC` v.12.15.0 [68]. We model neutral photoelectric absorption ($N_{\text{H,Gal}}$) along the line of sight (LOS) to the source using the Galactic absorption model `TBabs`, adopting the cosmic elemental abundances of [69] and photoionization cross sections of [70]. The Galactic hydrogen column density ($N_{\text{H,Gal}}$) incorporates both atomic (N_{H}) and molecular (N_{H_2}) hydrogen. The $N_{\text{H,Gal}}$ value for each source is obtained from [65] (listed in Table I) and kept fixed throughout the spectral fitting. We employ the Cash statistic (*C*-stat, [71]) to find the best-fit model parameters, and the χ^2 statistic to test the goodness of fit. Once the best-fit model parameters are found, we perform Markov Chain Monte Carlo (MCMC) analyses to determine the parameter uncertainties. The quoted uncertainties on best-fit parameters represent the 90% confidence intervals.

We include a constant component (`constant`) to consider cross-calibration uncertainties between FPMA and FPMB throughout our analysis. For simultaneous or quasi-simultaneous *XMM-Newton* and *NuSTAR* observations, we multiply a constant component (`constant`) to account for cross-calibration uncertainties between *XMM-Newton*'s EPIC and *NuSTAR*'s FPM spectra. The constant component is allowed to vary, while all other model parameters are tied between simultaneous or quasisimultaneous EPIC and FPM spectra. For nonsimultaneous *XMM-Newton* and *NuSTAR* observations, we allow the photon index of the primary power-law continuum and the flux or normalization of each model component to vary

independently between observations. The other two parameters likely to vary between nonsimultaneous observations are the disk ionization parameter and the coronal temperature. However, when we allow them to vary independently across nonsimultaneous observations, their values become unconstrained. This usually results in an upper limit for the ionization parameter and a lower limit for the coronal temperature. Moreover, we find that the fit statistics do not improve when these two parameters are allowed to vary independently between observations. Consequently, we have decided not to vary them independently across all observations of a source.

B. Bayesian analysis for model selection

While we use the χ^2 -statistic to assess goodness of fit, comparing only the fit statistics between two models may not always be conclusive, as it is possible for a model to overfit the data due to having more free parameters, resulting in a lower χ^2 -statistic. Therefore, we implement a Bayesian model selection approach, where the posterior distributions of the models are computed from MCMC simulations. We employ the Deviance Information Criterion, DIC [72], which is a Bayesian model selection metric and is defined by

$$\text{DIC} = \overline{D(\theta)} + p_D,$$

where $D(\theta) = -2 \log(p(y|\theta))$ and $p_D = \overline{D(\theta)} - D(\bar{\theta})$. Here $p(y|\theta)$ is the likelihood function, $D(\theta)$ is the deviance of a model parameter θ , y represents the data, p_D is the effective number of parameters in the model, $\overline{D(\theta)}$ represents the average of $D(\theta)$ calculated over the samples of θ , and $D(\bar{\theta})$ is the value of $D(\theta)$ evaluated at the average ($\bar{\theta}$) of the samples of θ .

DIC considers both the goodness of fit evaluated by the likelihood function and the effective number of model parameters. DIC is a hierarchical modeling generalization of the Akaike information criterion, AIC [73]. However, DIC does not penalize a model for parameters unconstrained by the data [74] because it uses an effective number of parameters, unlike AIC. DIC considers a model parameter only if it affects the goodness of fit to the data and is altered by varying that parameter. The degree to which different parameters are constrained is reflected in the noninteger nature of the effective number of model parameters.

In order to confirm whether one model is preferred over another, we compare the DIC values calculated from the two models. Statistically, the model with a lower DIC is preferred by the data, and the difference in DIC values, ΔDIC , between the two models measures the strength of the preference. According to a scale proposed by [75] and updated by [74], the ΔDIC values between 0 and 2 hint only marginal evidence, ΔDIC between 2 and 6 provides positive evidence, ΔDIC between 6 and 10 suggests strong evidence,

³<https://github.com/NuSTAR/nustar-gen-utils>

and ΔDIC greater than 10 shows very strong evidence for one model over another.

C. Basic spectral exploration

To unambiguously detect any spectral features, we first consider the continuum bands, 3–5 keV and 7–10 keV, solely dominated by the primary coronal emission. We fit the 3–5 keV and 7–10 keV bands by the Galactic absorption corrected power-law model (TBabs*zpowerlw). In search of various spectral features, we extrapolate the best-fit primary power-law continuum model over the whole 0.3–78 keV energy range. The *XMM-Newton* EPIC-pn (MOS for UGC 6728) and *NuSTAR* FPMA/B spectral data, the Galactic absorption corrected power-law continuum model, and data-to-model ratio plots for the sample are shown in Fig. A1. The ratio plots reveal a soft x-ray excess below ~ 1 –2 keV and Fe K emission in the ~ 5 –7 keV band for all AGN, as well as a Compton hump in the ~ 15 –30 keV range for most sources in the sample through visual inspection. However, the detection of these features, especially the Compton hump and Fe K emission, through visual inspection of ratio plots is insufficient, as the presence or strength of these residual features depends on the choice of bands presumed to be dominated by the primary power-law continuum.

D. Probing the hard x-ray excess emission: 3–78 keV spectral modeling

As evident from Fig. A1, the x-ray photon count spectra of the sample are complex, with an excess emission in the soft x-ray band. Therefore, we start our spectral modeling first considering the hard x-ray (3–78 keV) photon count spectra and probe the Fe K emission as well as the Compton hump. We describe the hard x-ray primary continuum by the physically motivated *nthComp* model [76], which produces a power-law like continuum due to the thermal Comptonization of disk seed photons in a hot corona of electrons. The free parameters of the *nthComp* model are photon index (Γ), electron temperature (kT_e), and normalization.

In the 6–7 keV band, we can obtain either narrow or broad or both narrow and broad Fe K emission features. However, the narrow Fe K_α/K_β emission line cores are never resolved in low-resolution CCD data. Therefore, while performing progressive spectral fitting to assess the presence of Fe K_α/K_β emission line core(s) from the torus or other distant material, we first add simple Gaussian line(s) [zGauss_N for Fe K_α core or zGauss_N + zGauss2_N for Fe K_α plus K_β cores] with their width fixed at 10 eV (see e.g. [45]), which is smaller than the resolution of EPIC-pn at ~ 6 –7 keV. On the other hand, the broad emission feature in the Fe K band is characteristic of relativistic reflection from the inner accretion disk [77,78]. Therefore, to fit the broad Fe K line emission, we apply the relativistic disk reflection model *relxillCp* [31,52,79].

In the *relxillCp* model, we set `refl_frac` = -1 to fit only the reprocessed emission from the accretion disk. The parameters (photon index Γ and electron temperature kT_e) that describe the properties of the corona are tied between *relxillCp* and *nthComp* models. The *relxillCp* table model is calculated with the seed photon temperature of the accretion disk fixed at 50 eV. Accordingly, we set the input disk seed photon temperature at 50 eV in the *nthComp* model for consistency. The relevant input parameters of the *relxillCp* model are:

- (i) Electron density (n_e) of the accretion disk.
- (ii) Iron abundance (A_{Fe}) in the disk relative to solar.
- (iii) The dimensionless spin (a^*) of the black hole, which is measured by setting the inner radius (r_{in}) of the accretion disk at the innermost stable circular orbit (r_{isco}). To meet the criterion, we need to set `Rin` = -1 in the model.
- (iv) Inclination angle (θ) of the disk relative to the line of sight.
- (v) Disk ionization parameter, $\xi = \frac{4\pi F}{n_e}$, where F is the illuminating continuum flux.
- (vi) The emissivity profile of the accretion disk, which is a measure of the reflected flux as a function of disk radius and is parametrized by a broken power law: $\epsilon(r) \propto r^{-q_{\text{in}}}$ for $r_{\text{in}} \leq r \leq r_{\text{br}}$, and $\epsilon(r) \propto r^{-q_{\text{out}}}$ for $r_{\text{br}} \leq r \leq r_{\text{out}}$. The inner emissivity index (q_{in}) is a free parameter in the model. Over the outer disk, the emissivity profile falls as r^{-3} , as expected in flat spacetime. Therefore, we fix the outer emissivity index at $q_{\text{out}} = 3$. The break radius corresponds to the radial extent of the corona and is fixed at $r_{\text{br}} = 6r_g$, a typical value for the coronal radius in AGN [37,63].

To quantitatively assess the presence of the broad Fe K emission feature, we evaluate the DIC values for models with and without the relativistic disk reflection (*relxillCp*) component in the 3–10 keV band. Additionally, we estimated the significance of the broad Fe K emission feature by employing the maximum likelihood ratio (MLR) test. The MLR test results agree with those obtained from the Bayesian model selection metric (Table A2). Both Bayesian analysis and MLR tests confirm that the relativistic disk reflection responsible for the broad Fe K emission is evident in all 11 AGN with at least 90% confidence.

We then determine the presence of Compton hump above 10 keV by fitting the 7–78 keV spectra with the Compton reflection model *xillverCp* [80]. Within *xillverCp*, we set `refl_frac` = -1 and tied the coronal parameters (photon index Γ and electron temperature kT_e) to those in *nthComp*. The density of the reflector in the distant reflection model (*xillverCp*) is kept fixed at the canonical value of $n_e = 10^{15} \text{ cm}^{-3}$ throughout the spectral fitting. The distant reflector is assumed to be near-neutral ($\log \xi = 0$) and has a high-inclination angle of $\theta = 60$ degree (e.g. [55,81]). For quantitative assessment of the Compton hump, we compute the DIC values for the primary

power-law continuum (nthComp) and the primary continuum plus Compton reflection (nthComp+xillverCp) models fitted to the 7–78 keV spectra. The MLR test evaluates the detection significance of the Compton hump, which is in agreement with the result obtained from the Bayesian model selection metric, presented in Table A3. To check the relativistic nature of the Compton hump, we then replace the distant reflection model xillverCp with the relativistic disk reflection model relxillCp and refit the 7–78 keV spectra with nthComp+relxillCp model. The fit statistics, along with the confidence levels and DIC values, are shown in Table A4.

Our statistical analyses show that the relativistically broad Fe K emission feature is present in all 11 AGN. However, the Compton hump is either very weak or lacks statistical significance in some AGN, which can be entirely attributed to insufficient SNR in their *NuSTAR* observations. As shown in column (9) of Table A1, all cases where the Compton reflection hump is weak or undetected correspond to low net source counts ($\lesssim 10^4$) in the *NuSTAR* data. When the Compton hump is weak or undetected, and the Fe K band contains only a broad Fe K emission feature, the model that best describes the hard x-ray (3–78 keV) spectra is nthComp+relxillCp. In the absence of a Compton hump, when the Fe K band contains both broad and narrow Fe K emission features, the hard x-ray spectra are best described by models nthComp+zGauss_N+relxillCp or nthComp+zGauss_N+zGauss2_N+relxillCp. The second narrow Gaussian line zGauss2_N is included when the Fe K_β emission line is detected at around 7 keV. We employ the distant reflection model xillverCp only when the Compton hump is statistically significant. Therefore, the model nthComp+relxillCp+xillverCp best fits the hard x-ray spectra when both broad and narrow Fe K emission features, as well as the Compton hump, are detected. The iron abundance (A_{Fe}) parameter was tied between relxillCp and xillverCp. Figure 2 (Step 1) illustrates our methods of fitting various spectral features in the hard x-ray (3–78 keV) photon count spectra. The following four sets of models best fit the joint *XMM-Newton*+*NuSTAR* hard x-ray spectra of the sample:

- (i) relxillCp+nthComp for UGC 6728, Mrk 1310, PG 1229 + 204 and PG 0844 + 349.
- (ii) relxillCp+zGauss_N+zGauss2_N+nthComp for Mrk 590.
- (iii) relxillCp+zGauss_N+nthComp for PG 0804 + 761.
- (iv) relxillCp+xillverCp+nthComp for NGC 4748, Mrk 110, Mrk 279, Mrk 79 and PG 1426 + 015.

We note that even when narrow Fe K emission lines are detected, the Compton reflection hump remains undetected or weak in two AGN, Mrk 590 and PG 0804 + 761, indicating that a weak or undetected Compton hump is not exclusive to sources exhibiting only broad Fe K emission

lines. Some AGN lack a statistically significant Compton hump above 10 keV, even in the presence of Fe K emission lines (broad or narrow), which can be fully explained by low net source counts ($\lesssim 10^4$) in their *NuSTAR* observations, reported in column (9) of Table A1.

E. Probing both soft and hard x-ray excess: 0.3–78 keV spectral modeling

Once the hard x-ray (3–78 keV) best-fit spectral models have been found for the sample, we extrapolate that model down to 0.3 keV, to see whether the same model can fit the whole 0.3–78 keV spectra or not. When the hard x-ray best-fit spectral model of an AGN can also explain the soft x-ray excess emission without the need for any extra blackbody or low-temperature Comptonization model, it will justify that the same physical mechanism is responsible for the origin of the soft x-ray excess emission, i.e., the relativistic reflection from an accretion disk with higher density. If the hard x-ray spectral model cannot fully describe the soft x-ray band and a warm Comptonization model is indeed required, we can conclude that the origin of the observed soft x-ray excess is relativistic disk reflection together with the warm coronal emission. Figure 2 (Step 2) illustrates our methodology for fitting the hard-to-soft x-ray spectra.

For all sources in the sample, we quantitatively assess the possible presence of warm coronal emission by adding the warm Comptonization (compTT) component to the hard x-ray (3–78 keV) best-fit model and refit the broadband (0.3–78 keV) spectra. The resulting fit statistics and DIC values, both without and with the warm Comptonization model, are presented in Table A5. We also estimated the confidence levels of the warm coronal emission through the MLR test, which agree with those obtained from the Bayesian model selection metric (Table A5). Both Bayesian analysis and MLR tests confirm that the observed soft x-ray excess requires an additional compTT for eight out of 11 AGN in the sample. For the remaining three AGN, variable-density relxillCp is sufficient to fit the broad Fe K emission feature and soft x-ray excess emission simultaneously. We find the following five sets of models best describe the joint *XMM-Newton*+*NuSTAR* broadband x-ray spectra of the AGN sample:

- (i) relxillCp+nthComp for UGC 6728 and PG 1229 + 204.
- (ii) relxillCp+zGauss_N+zGauss2_N+nthComp for Mrk 590.
- (iii) compTT+relxillCp+nthComp for Mrk 1310 and PG 0844 + 349.
- (iv) compTT+relxillCp+zGauss_N+nthComp for PG 0804 + 761.
- (v) compTT+relxillCp+xillverCp+nthComp for NGC 4748, Mrk 110, Mrk 279, Mrk 79, and PG 1426 + 015.

Figure A2 shows the *XMM-Newton*/EPIC, *NuSTAR*/FPMA, and FPMB photon count spectra, along with the

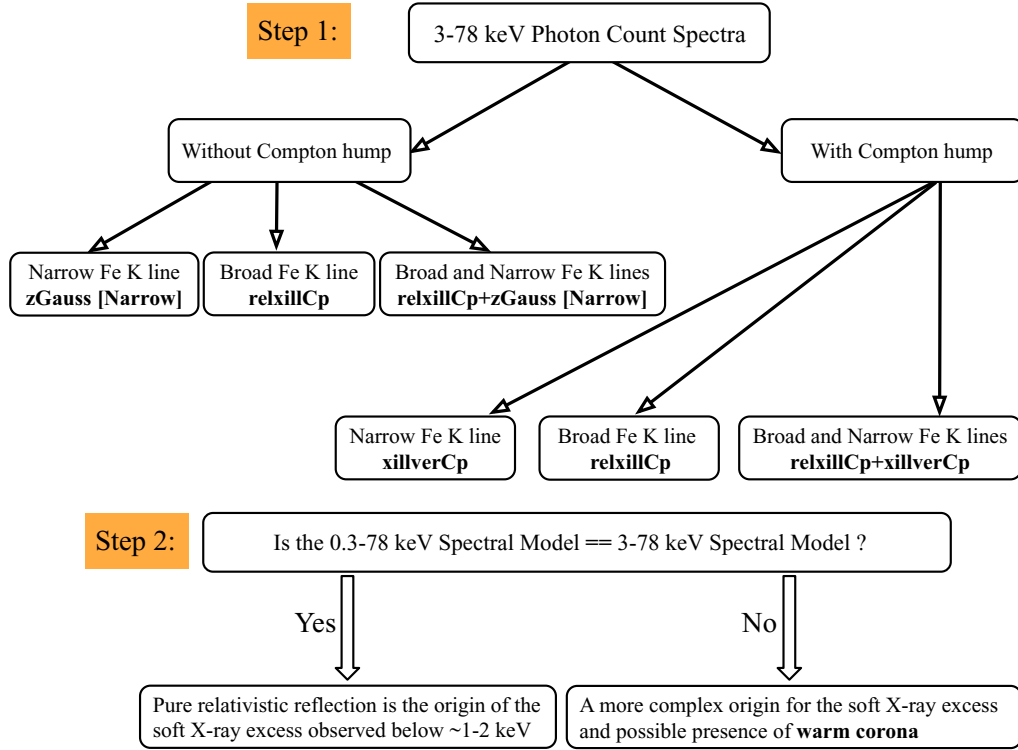


FIG. 2. The flowchart illustrates our systematic spectral fitting methodology, which unambiguously investigates the origin of soft x-ray excess in the AGN sample exhibiting diverse spectral features.

best-fit count spectral models and their corresponding components. We present the best-fit flux spectral model with components in Figs. A3 and A4. The best-fit broadband (0.3–78 keV) spectral model parameters for each source are presented in Table A6. In the Appendix, we discuss the hard-to-soft x-ray spectral fitting details for each source in the sample.

F. Testing pure warm coronal origin of soft x-ray excess without relativistic reflection

We test the pure warm coronal origin of soft x-ray excess by completely removing the `relxillCp` model from the broadband (0.3–78 keV) spectra and adding the `compTT` component for all sources in the sample to reconfirm the presence of relativistic reflection in the broadband spectra and our conclusions on the origin of soft x-ray excess. We present the source model expression including `compTT` and excluding `relxillCp`, along with the resulting fit statistic and DIC value, in columns (2), (4), and (7) of Table A7, respectively. The source model obtained in Sec. III.E, along with the corresponding fit statistic and DIC value, are shown in columns (3), (5), and (8) of Table A7, respectively. To quantitatively assess the significance of the source model obtained in Sec. III.E (labeled as Model 2 in Table A7) against the pure warm corona model (labeled as Model 1 in Table A7), we evaluate the significance of Model 2 over Model 1 through

the MLR test and calculate the difference between DIC values derived for Model 1 and Model 2. The confidence level computed by MLR and the Bayesian model selection metric agree with each other, reassuring the presence of relativistic disk reflection in the broadband spectra for all sources in the sample, as well as showing that a sole warm corona is insufficient to fit the entire soft x-ray excess without relativistic disk reflection.

G. MCMC analysis for parameter space exploration

To confirm that the parameters are not clustering at any local minima, we conduct an MCMC analysis on the best-fit model and explore the complete parameter space for each source. We draw the parameter distributions and determine confidence intervals for each free parameter from the converged MCMC chains. To run the MCMC chains, we use the algorithm of [82] implemented in XSPEC. We run MCMC chains with 100–300 walkers for $\sim 10^6$ – 10^7 iterations and burn the first $\sim [1 - 10]\%$ iterations until the chains were converged. We notice that the number of walkers needs to be at least four times greater than the number of free parameters in the model for faster convergence of the chains. We ensured that the chains converged with Gelman–Rubin’s MCMC convergence test, which resulted in the potential scale reduction factor being less than 1.1 for each parameter. The full posterior distributions of various model parameters and contour plots between each pair of parameters for the

best-fit spectral models of all sources in the sample are shown in Figs. A5 and A6. The dark, medium, and light blue contours represent 68.3%, 90%, and 95% confidence levels, respectively.

H. Relative contributions of relativistic and distant reflection in the Fe K band

We quantify the relative contributions of the `relxillCp` and distant reflection (either `zGauss_N` or `zGauss_N+zGauss2_N` or `xillverCp`) components in the 5–7 keV Fe K band to assess their respective strengths. Figure 3 demonstrates the relativistic disk reflected flux versus the nonrelativistic or distant reflection flux within the Fe K band. The 1:1 line denotes the point at which the relativistic and distant reflection components contribute equally to the Fe K band emission. From Fig. 3, we can see that relativistic reflection contributes more than the distant reflection in the Fe K band for all sources in the diagram except for some observations of Mrk 279. Additionally, we notice the variable nature of the relativistic disk reflected flux responsible for the broad Fe K emission whenever we have multiple flux measurements of a source. However, as expected, the distant reflection flux characterizing the narrow Fe K emission line(s) appears nonvariable or constant within error bars. To confirm this, we attempt to quantify the flux variability of both relativistic and distant

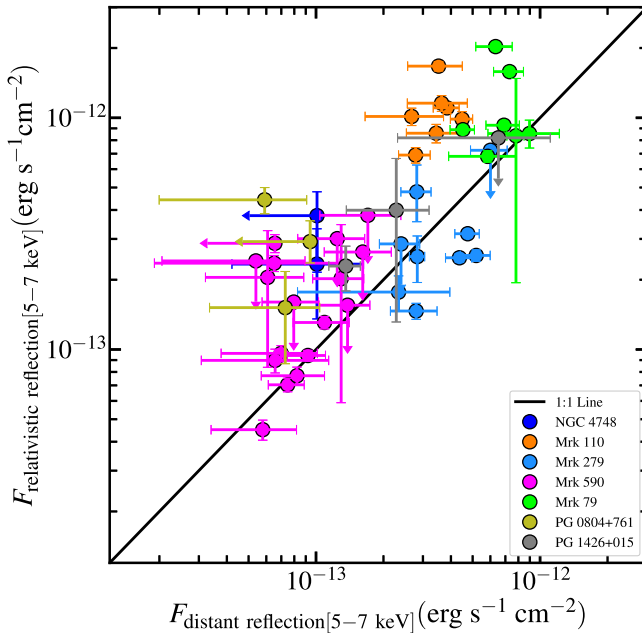


FIG. 3. The flux of the relativistic disk reflection model `relxillCp` versus nonrelativistic or distant reflection (either `zGauss_N` or `zGauss_N+zGauss2_N` or `xillverCp`) in the 5–7 keV band, demonstrating the relative contributions of these components to the Fe K band. The 1:1 line marks equal contributions from relativistic disk reflection and distant reflection components to the Fe K band.

reflection components by calculating their fractional root mean square (rms) variability using the formula provided in [83]. However, for any descriptive statistics analysis, at least ten data points are needed in order to obtain reliable results. The only source that satisfies this criterion is Mrk 590. The measured fractional rms variability of the relativistic disk reflection flux in the 5–7 keV band is $F_{\text{var}} = [28.6 \pm 23.6]\%$. However, we are unable to measure the fractional rms variability of the distant reflection flux because the mean squared error in the measured distant reflection flux is greater than its variance, resulting in a negative excess variance.⁴ This confirms the nonvariable nature of the observed distant reflection flux and is consistent with our visual inspection.

I. Relative contributions of relativistic reflection and warm Comptonization in the soft x-ray band

For sources requiring an additional warm Comptonization component to model the observed soft x-ray excess, we examine the relative contributions of the `relxillCp` and `compTT` components in the 0.3–2 keV soft x-ray band to evaluate their respective strengths. Figure 4 presents the relativistic disk reflected flux versus the warm coronal flux in this band. The 1:1 line indicates equal contributions from both relativistic disk reflection and warm Comptonization to the soft x-ray excess. As shown in Fig. 4, the warm corona contributes more to the soft x-ray excess in Mrk 110 (except for one observation), PG 0844 + 349, and PG 0804 + 761. The source for which relativistic disk reflection completely dominates over warm Comptonization is Mrk 79. For four AGN (Mrk 1310, NGC 4748, Mrk 279, and PG 1426 + 015), we find that relativistic disk reflected flux dominates the warm coronal flux in some observations, while in other observations of these four AGN, warm Comptonization dominates over relativistic disk reflection. Additionally, we notice that the warm coronal flux is unconstrained in several observations of some AGN (e.g. Mrk 1310, NGC 4748, and Mrk 79), likely due to low SNR or insufficient net source counts in those specific observations. Our results suggest a strong degree of independence between the warm corona and relativistic reflection components.

IV. RESULTS AND DISCUSSION

In this section, we discuss all the results derived from our broadband x-ray spectral modeling, the implications of the physical reflection model for the origin of the soft x-ray excess, the validity of the standard SS73 accretion disk theory, the first-time calculation of the disk-to-corona power transfer fraction, coronal (hot as well as warm) properties, and black hole spin population across mass scales ($\log M_{\text{BH}} \sim 5.5\text{--}9.0$). The details of broadband

⁴For detailed formulas, see Eqs. (1) and (2) in [84].

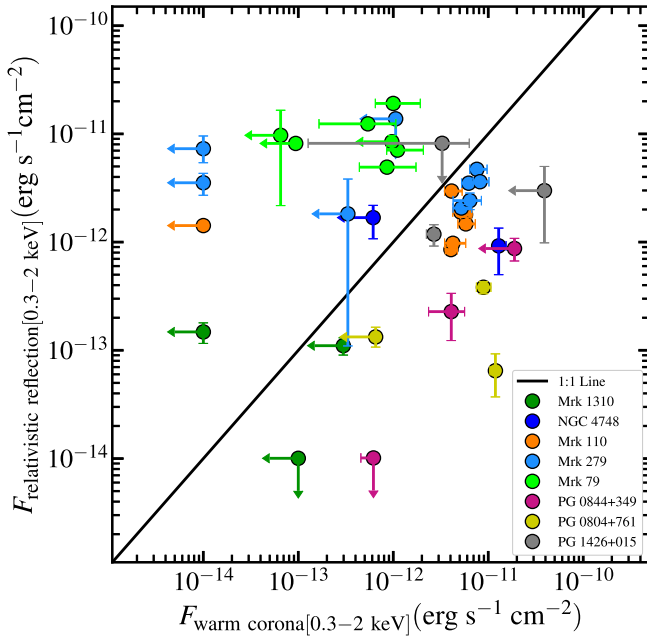


FIG. 4. The flux of *relxillCp* versus *compTT* in the 0.3–2 keV band, showing the contributions of the relativistic reflection and warm Comptonization components to the soft x-ray band. The 1:1 line indicates where relativistic disk reflection and warm coronal emission contribute equally to the observed soft x-ray excess.

spectral modeling for each source in the sample are presented in the Appendix. In Table A6, we report the best-fit source spectral model parameters and their 90% confidence intervals determined through MCMC parameter space exploration of the best-fit model. Figure A2 shows the broadband *XMM-Newton*/*NuSTAR* spectra, the best-fit count spectral model with components, and the corresponding residuals. The best-fit spectral energy flux models with components are presented in Figs. A3 and A4.

A. Physical origin of the soft x-ray excess emission: High-density disk reflection or warm Comptonization

Two models have been proposed to explain the observed soft x-ray excess. One model is the relativistic reflection or reprocessing of the incident hot coronal emission in the innermost part of the accretion disk [27,28,31]. The other model considers Compton upscattering of the optical/UV disk photons in a low-temperature ($kT \sim 0.1\text{--}2$ keV), optically thick ($\tau > 1$) Comptonizing medium or warm corona [46,85]. However, the relativistic reflection as the origin of the soft x-ray excess is a more consistent explanation, as it is the only model that can explain the broad Fe K emission line and Compton hump together with the soft x-ray excess. However, it was shown that the entire soft x-ray excess may not be well-fitted solely by relativistic reflection (e.g. Ark 120: [45,50]), especially when the disk density is low and fixed at $\log[n_e/\text{cm}^{-3}] = 15$.

Furthermore, fitting of soft x-ray excess with a fixed low-density relativistic disk reflection model resulted in unphysically high ($A_{\text{Fe}} > 10$) iron abundance in some sources (e.g. 1H 0707-495: [86]). To resolve these issues, [52] developed a new model where the density of the accretion disk is a free parameter varying in the range of $\log[n_e/\text{cm}^{-3}] = 15\text{--}20$, first implemented by [37]. When the innermost part of the disk becomes radiation pressure dominated, extra heating from free-free absorption increases the disk density because of its quadratic dependence on density, thus boosting the strength of the observed soft x-ray excess below 1 keV [52,87]. In Fig. 5 (left), we show the flux spectra calculated from the variable-density disk reflection model *relxillCp* for the disk density of $\log[n_e/\text{cm}^{-3}] = 15, 16, 17, 18, 19$, and 20. The parameters assumed for the model calculations shown in this plot are: $\Gamma = 2$, $kT_e = 300$ keV, $\xi = 500$ erg cm s $^{-1}$, $q_{\text{in}} = 8$, $a^* = 0.9$, $\theta = 45$ degree, $A_{\text{Fe}} = 1$, $\text{refl_frac} = 3$, and $\text{norm} = 1$. Even with the solar iron abundance, model flux is noticeably enhanced at the soft x-ray band when the disk density is higher than $n_e = 10^{15}$ cm $^{-3}$, and the difference from the canonical disk reflection model becomes more prominent for $n_e \geq 10^{17}$ cm $^{-3}$.

Disk ionization is also a crucial physical parameter in the *relxillCp* model and influences the strength of the observed soft x-ray excess. To illustrate how it affects the spectral components, especially the soft x-ray excess and broad Fe K emission line, we show the flux spectra for the disk ionization of $\xi/\text{erg cm s}^{-1} = 10, 100, 300, 500$, and 1000 in the right panel of Fig. 5. The other parameters considered for the model calculations shown in the plot are as follows: $\Gamma = 2$, $kT_e = 300$ keV, $\log[n_e/\text{cm}^{-3}] = 17$, $q_{\text{in}} = 8$, $a^* = 0.9$, $\theta = 45$ degree, $A_{\text{Fe}} = 1$, $\text{refl_frac} = 3$, and $\text{norm} = 1$. Evidently, as the disk becomes more ionized, the soft x-ray flux is enhanced with a broader Fe K emission line, even for the same electron density of the accretion disk.

The probability distribution of the disk density parameter for the sample is shown in Fig. 6 (left panel). We obtained disk density measurements higher than the canonical value of $\log[n_e/\text{cm}^{-3}] = 15$ with 90% confidence for four AGN: Mrk 110, Mrk 279, Mrk 590, and Mrk 79. Of these four AGN, three sources (Mrk 110, Mrk 279, and Mrk 79) exhibited strong relativistic reflection features (broad Fe K emission line, Compton hump), yet required a warm Comptonization component to account for the soft x-ray excess in addition to high-density relativistic disk reflection. For seven out of 11 AGN, the 90% lower limit of the density parameter has reached the canonical value of $n_e = 10^{15}$ cm $^{-3}$. Out of these seven AGN, an additional warm Comptonization component is required to describe the observed soft x-ray excess in five AGN.

Through joint *XMM-Newton*+*NuSTAR* broadband spectroscopy, we find that variable-density relativistic reflection can self-consistently explain both the broad Fe K emission

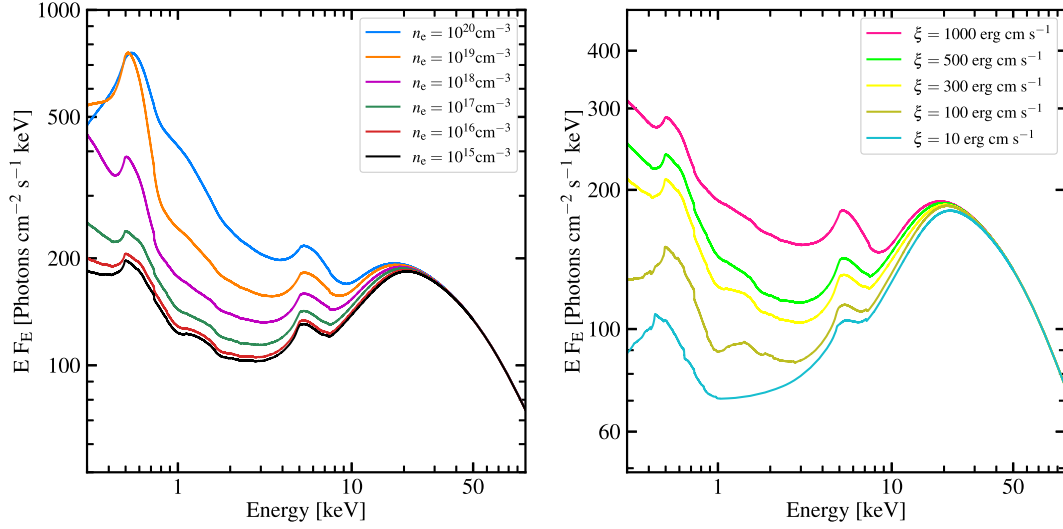


FIG. 5. The left panel shows the spectra calculated by the relativistic reflection model, `relxillCp`, for a range of disk densities, $\log[n_e/\text{cm}^{-3}] = 15, 16, 17, 18, 19$, and 20 . To make this plot, the model parameters are assumed to be $\Gamma = 2$, $kT_e = 300$ keV, $\xi = 500$ erg cm s $^{-1}$, $q_{\text{in}} = 8$, $a^* = 0.9$, $\theta = 45$ degree, $A_{\text{Fe}} = 1$, `refl_frac` = 3, and `norm` = 1. We can see a significant boost in the strength of the soft x-ray component from the disk density of $n_e = 10^{17}$ cm $^{-3}$ onward. The right panel depicts the spectra for various ionization states ($\xi/\text{erg cm s}^{-1} = 10, 100, 300, 500$, and 1000) of the accretion disk in the `relxillCp` model calculated for $\Gamma = 2$, $kT_e = 300$ keV, $\log[n_e/\text{cm}^{-3}] = 17$, $q_{\text{in}} = 8$, $a^* = 0.9$, $\theta = 45$ degree, $A_{\text{Fe}} = 1$, `refl_frac` = 3, and `norm` = 1. Even for the same disk density, the soft x-ray excess flux is enhanced with a broader Fe K emission line as the disk becomes more ionized up to $\log[\xi/\text{erg cm s}^{-1}] = 3$.

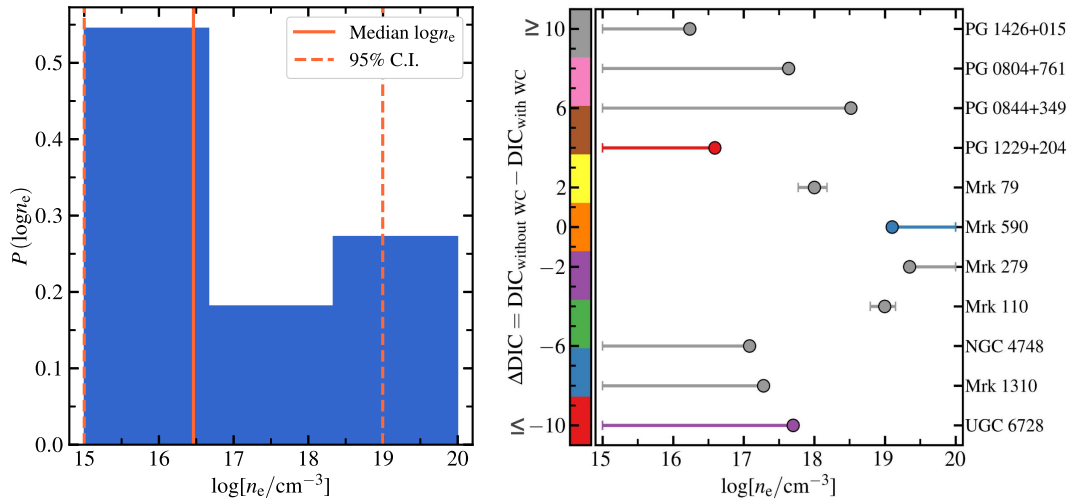


FIG. 6. The left panel shows the probability distribution of the measured electron density of the accretion disk for our sample. The color bar in the right panel depicts the difference between the Deviance Information Criteria, $\text{DIC}_{\text{without WC}}$ and $\text{DIC}_{\text{with WC}}$, for the variable-density disk reflection model without and with warm Comptonization, respectively. The Bayesian model selection metric supports the relevance of additional warm Comptonization for the origin of soft x-ray excess in eight AGN marked in gray color. As shown in column (9) of Table A5, the ΔDIC has an extensive range from -13.7 to 243.7 , so we cannot plot the full range in the colormap. Therefore, the upper and lower limits in the colormap are marked as ≥ 10 and ≤ -10 , respectively, which signify “very strong” positive and negative evidence of additional warm Comptonization over the variable-density disk reflection.

line and soft x-ray excess in three out of 11 AGN without requiring any additional warm coronal emission. The inner accretion disk is found to be ionized and dense, with the median ionization of $\xi \sim 10^{2.0}$ erg cm s $^{-1}$, median density of $n_e \sim 10^{16.5}$ cm $^{-3}$, and near-solar iron abundance of

$A_{\text{Fe}} \sim 2$ for the sample. The right panel in Fig. 6 shows $\Delta\text{DIC} = \text{DIC}_{\text{without WC}} - \text{DIC}_{\text{with WC}}$ versus disk density for the sample, confirming the relevance of an additional warm coronal emission for the observed soft x-ray excess in eight out of 11 AGN. We find that the temperature and optical

depth of the warm coronae for these eight AGN are in the range of $kT_{\text{wc}} \sim 0.2\text{--}2$ keV and $\tau_{\text{wc}} \sim 4\text{--}22$, respectively, which are consistent with the properties of warm coronae derived for a larger sample of AGN (see Fig. 5 of [46]). According to recent simulations, a warm corona can produce a featureless soft x-ray excess only if there is sufficient mechanical heating of the accretion disk [88–90]. The origin of such internal heating can be attributed to several physical processes, such as magnetic reconnection or viscous accretion within the corona [91,92]. The observed large optical depth of the warm corona supports the presence of intense magnetic pressure in the accretion disk [91].

A key simplification in all reflection models is the assumption that the density of the illuminated atmosphere remains constant. Therefore, our study investigates the effects of high density in x-ray illuminated disk atmospheres, assuming a constant-density vertical structure of the accretion disk (see Sec. 2 of [52] for more details), which may differ from reality. This simplification may explain why relativistic reflection alone cannot fully account for the soft x-ray excess without including additional warm Comptonized emission in most of the sources in the sample. However, developing a reflection model that incorporates variable density in the vertical direction of the accretion disk is beyond the scope of this paper.

B. Comparison with previous study of the sample with *XMM-Newton* only

The comparison of four key parameters, i.e., disk density, iron abundance, black hole spin, and disk inclination angle of the AGN sample, as determined from our broadband joint *XMM-Newton*+*NuSTAR* spectral modeling and from previous analyses by JJ19 using only 0.5–10 keV averaged *XMM-Newton* spectra, are presented in Fig. 7. However, a direct comparison of parameter uncertainties should be treated with caution, since our uncertainties are derived using MCMC, while JJ19 did not employ MCMC for error estimation. Moreover, the spectral modeling statistics and residuals reported in JJ19 are high for most sources, raising further concerns about the robustness of their parameter estimation.

As evident from the top left panel in Fig. 7, the disk density we measured using joint *XMM-Newton*/*NuSTAR* data is higher than that derived from *XMM-Newton* spectral fitting alone. This is because the additional spectral features across the broad bandpass let us more accurately disentangle the reflection from the continuum. Moreover, some sources (e.g. Mrk 79) in the previous *XMM-Newton* data did not reveal broad Fe K line emission due to short exposure. In this work, with more data from both *XMM-Newton* and *NuSTAR*, we prominently detected the broad Fe K emission feature, the modeling of which resulted in an enhanced contribution from the disk reflection component.

The iron abundances measured from our joint *XMM-Newton*+*NuSTAR* high-density reflection spectroscopy are

consistent with those obtained from high-density reflection modeling of previous *XMM-Newton* data for the sample (Fig. 7, top right). We measure solar or near-solar iron abundances for the sample, with a median value of $A_{\text{Fe}} \sim 2$. This is expected since the higher-density reflection model can reduce the inferred iron abundance by enhancing the continuum in the reflection component.

Comparison of our black hole spin measurements with those inferred from previous high-density spectral fitting of *XMM-Newton* data alone provides consistent results within the 90% confidence limits (Fig. 7, bottom left). In the earlier *XMM-Newton* spectral analysis by JJ19, the spin parameter was fixed at $a^* = 0.998$ for three sources (UGC 6728, Mrk 1310, and PG 1426 + 015) due to either the nondetection of a broad Fe K emission line in the *XMM-Newton* data or the lack of hard x-ray data above 10 keV. In this work, the inclusion of *NuSTAR* data, and therefore the Compton reflection hump, constrains the continuum emission better, which can potentially impact the determination of the red wing of the broad Fe K emission line and thus affect black hole spin measurements. As a result, our broadband spectroscopy improves both the accuracy and precision of black hole spin measurements for all sources in the sample. Notably, joint *XMM-Newton*/*NuSTAR* high-density relativistic reflection spectroscopy provides the first robust spin measurements for three AGN in our sample: UGC 6728, Mrk 1310, and PG 1426 + 015.

Previous measurements of the disk inclination angle using only *XMM-Newton* data provided extreme values for the sample, ranging from as low as five degrees to as high as 90 degrees, as illustrated in Fig. 7 (bottom right). However, through joint *XMM-Newton*/*NuSTAR* spectroscopy, we measure typical values for the disk inclination angle, clustering around 45 degrees. The median disk inclination for the sample is $\theta = 41 \pm 8$ degree at 2σ confidence. Accurate measurement of the inclination angle depends on the blue wing of the broad Fe K emission line, which is better characterized when both the continuum and Compton hump are well constrained. The inclusion of *NuSTAR* data not only enables tighter constraints on the broadband continuum but also improves the modeling accuracy of both the broad Fe K emission line and the Compton hump.

C. Disk density and disk-to-corona power transfer fraction

The electron density in the standard SS73 α -disk model [53], for a radiation pressure-dominated inner region and incorporating the transfer of a fraction of power from the disk to a corona, was derived by [54]:

$$n_e = \frac{1}{\sigma_{\text{T}} r_s} \frac{256\sqrt{2}}{27} \alpha^{-1} r^{3/2} \dot{m}^{-2} [1 - (r_{\text{in}}/r)^{1/2}]^{-2} (1 - f)^{-3}, \quad (1)$$

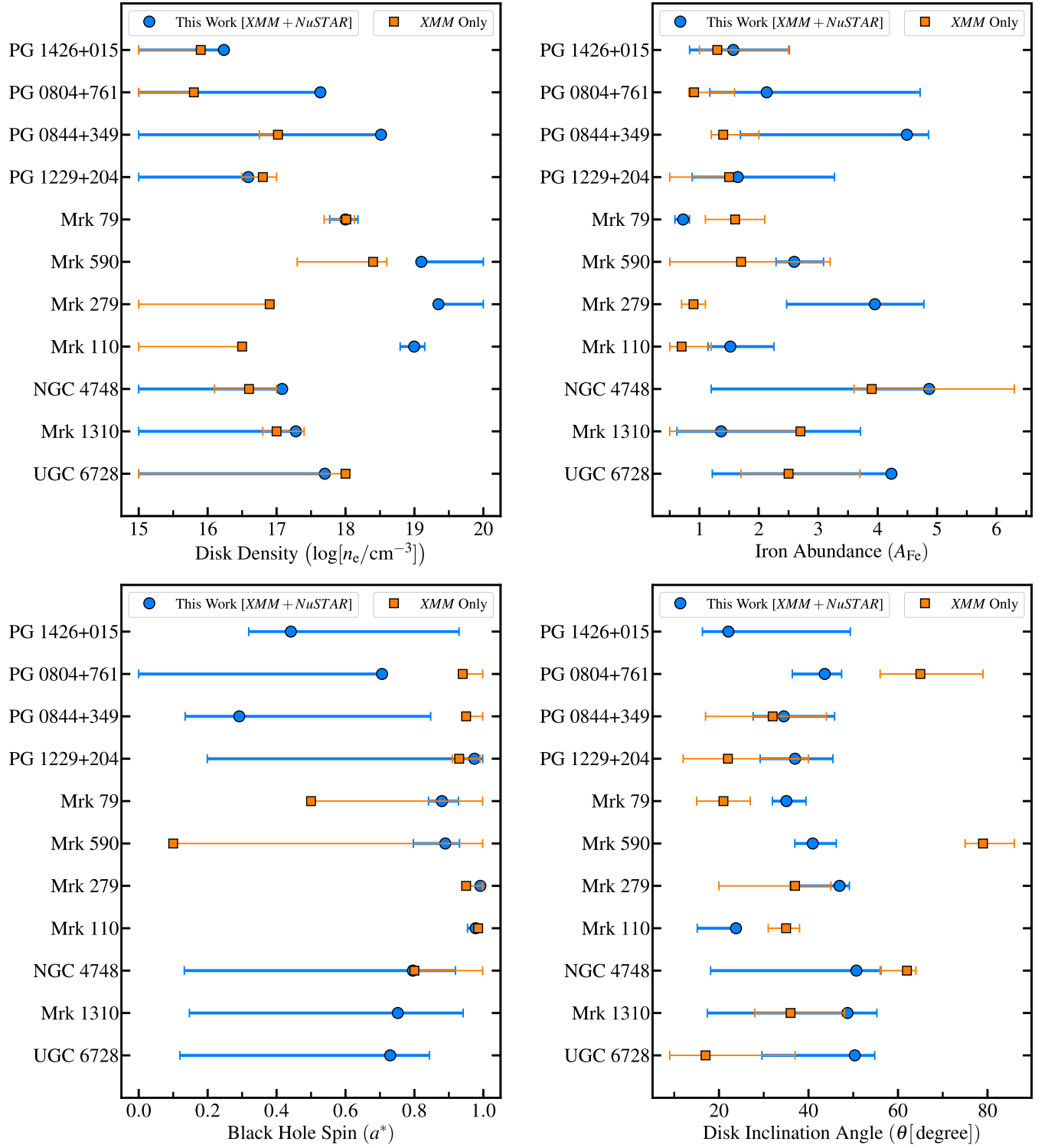


FIG. 7. Comparison of disk density, iron abundance, black hole spin, and disk inclination angle derived from our broadband (0.3–78 keV) joint *XMM-Newton*+*NuSTAR* spectral modeling with those obtained from the previous 0.5–10 keV *XMM-Newton* spectral fitting of the sample by JJ19. For UGC 6728, Mrk 1310, and PG 1426 + 015, JJ19 fixed the spin parameter at $a^* = 0.998$, precluding a direct comparison of spin measurements. As a result, orange squares are absent for these three sources in the bottom left panel. Likewise, no comparison of disk inclination angle is shown for PG 1426 + 015 in the bottom right panel, since this parameter was also fixed in JJ19. The error bars in all four panels represent the 90% confidence intervals.

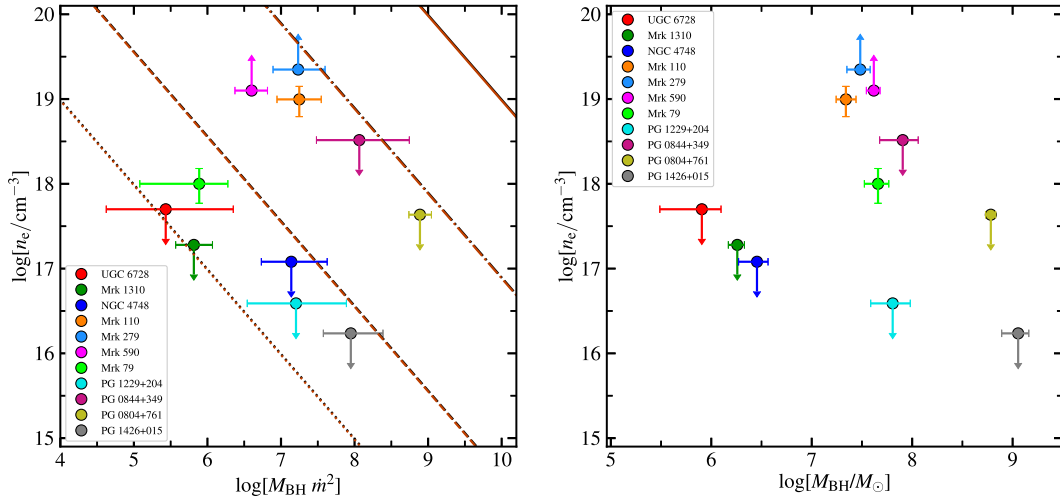


FIG. 8. Theoretically, the electron density (n_e) of the accretion disk depends on five parameters: M_{BH} , $\dot{m}$, f , r , and r_{in} . The left panel shows how disk density varies as a function of the black hole mass times the accretion rate squared, $M_{\text{BH}}\dot{m}^2$, plotted on a logarithmic scale. The density solutions for a radiation pressure-dominated disk at $r = 6r_s$ (in black) and $r = 10r_s$ (in brown) for $f = 0, 0.7, 0.95$, and 0.99 are depicted by the dotted, dashed, dash-dotted, and solid lines, respectively. The inner disk radius (r_{in}) is set at the sample median value of $r_{\text{isco}} = 2.8r_s$. The right panel shows the dependence of $\log n_e$ on $\log M_{\text{BH}}$ for the sample. In both panels, no significant correlation is observed between the parameters.

where f is the disk-to-corona power transfer fraction, representing the fraction of power released from the accretion disk into the hot corona. The range of the f -parameter is $f \in [0, 1)$. The $f = 0$ solution can provide a nonzero value of the electron density according to SZ94. However, the $f = 1$ solution is forbidden in the model. The viscosity parameter α is assumed to be 0.1 (see e.g. [93]). The Thomson scattering cross section is $\sigma_T = 6.64 \times 10^{-25} \text{ cm}^2$. The radius (r) of the accretion disk is in units of the Schwarzschild radius $r_s = \frac{2GM_{\text{BH}}}{c^2}$. The inner disk radius (r_{in}) is at the innermost stable circular orbit (r_{isco}) in the relativistic disk reflection model, the median of which is estimated to be $\sim 2.8r_s$ for the sample. $\dot{m} = \frac{\dot{M}}{M_E}$ denotes the dimensionless mass accretion rate.

Theoretically, the disk density depends on five parameters: M_{BH} , $\dot{m}$, r_{in} , r , and f . The black hole mass (M_{BH}) and dimensionless mass accretion rate ($\dot{m}$) for each source are obtained from the literature and listed in Table I. In the relativistic reflection model, the inner disk radius (r_{in}) is set to the innermost stable circular orbit (r_{isco}), which can be directly estimated from the black hole spin parameter, reported in Table A6. Thus, in the standard disk model, the two remaining unknown parameters for determining the disk density are the disk-to-corona power transfer fraction (f) and the disk radius (r).

To test the validity of the standard α -disk model, we first examine the dependence of disk density on the black hole mass and accretion rate for specific disk radius and f -parameter values. Figure 8 shows the measured disk density as a function of black hole mass times the accretion rate squared ($M_{\text{BH}}\dot{m}^2$) and black hole mass (M_{BH}), both

plotted on a logarithmic scale for our sample. The dotted, dashed, dash-dotted, and solid lines represent the theoretical density solutions for $f = 0, 0.7, 0.95$, and 0.99 , respectively, each computed at $r = 6r_s$ (black) and $r = 10r_s$ (brown). As shown in Fig. 8 (left panel), the impact of disk radius, r , on density is much less than that of the disk-to-corona power transfer fraction, f . If the standard α -disk theory holds and the intrinsic scatter in the f -parameter is negligible, we expect $\log n_e$ to anticorrelate with both $\log[M_{\text{BH}}\dot{m}^2]$ and $\log M_{\text{BH}}$, following $\log n_e \propto -\log[M_{\text{BH}}\dot{m}^2]$. However, we do not observe any such correlation, suggesting that the intrinsic scatter associated with the f -parameter is

TABLE II. The disk-to-corona power transfer fraction (f -parameter), measured at $r = 10r_s$ and expressed in percent for each source in the sample.

Source Name	f -parameter [percent]
UGC 6728	≤ 5.3
Mrk 1310	≤ 2.5
NGC 4748	≤ 58.9
Mrk 110	$91.3^{+1.0}_{-1.5}$
Mrk 279	≥ 93.3
Mrk 590	≥ 86.8
Mrk 79	$46.9^{+6.9}_{-10.2}$
PG 1229 + 204	≤ 42.9
PG 0844 + 349	≤ 93.3
PG 0804 + 761	≤ 93.0
PG 1426 + 015	≤ 57.8

substantial, and that f varies significantly between different sources in our sample. In principle, disk density is most sensitive to variations in f , with $n_e \propto (1 - f)^{-3}$ or $\log n_e \propto -3 \log(1 - f)$. Next, we evaluate the model density as a function of disk radius for various values of the f -parameter and compare the results with the measured disk densities for the sample, as illustrated in Fig. A7. The model density matches the measured density at different f -parameter values for individual sources. Notably, this agreement between the model and measured densities is observed at the common radius of $r = 10r_s$ across all sources. Therefore, we determine the f -values from the measured disk densities at $r = 10r_s$ for each source using Eq. (1) and report them in Table II. As it stands, we have taken care of all the model intrinsic scatters involved, and if the standard α -disk model is valid, we expect a positive correlation between f and $\log[M_{\text{BH}} \dot{m}^2]$. In Fig. 9 (left), we show the derived f -values for the sample as a function of $\log[M_{\text{BH}} \dot{m}^2]$. To evaluate the correlation between these parameters, we perform Spearman's rank correlation test. The calculated Spearman coefficient for the $f - \log[M_{\text{BH}} \dot{m}^2]$ correlation is $\rho_s = 0.74$ with a p-value of 0.01, which indicates a strong positive correlation between these parameters, thereby validating the standard α -disk model with a variable disk-to-corona power transfer fraction in the AGN sample.

Figure 9 (right) shows the distribution of the f -parameter for the sample, with a median value of $f = 0.68^{+0.25}_{-0.25}$ estimated at the 2σ confidence level. According to the density solution of SZ94 in Eq. (1), if all other input parameters are held constant, the black hole mass spanning $\log M_{\text{BH}} \sim 5.5\text{--}9.0$ for the sample can provide approximately 2 orders of magnitude variation in disk density. The dimensionless mass accretion rate ($\dot{m}$), ranging from

$\sim 0.1\text{--}2.5$ for the sample, can alone offer about 3 orders of magnitude variation in the measured density. However, the f -parameter varying in the range of $\sim 0.01\text{--}0.99$ can cause up to 6 orders of magnitude variation in the density parameter, thereby justifying the wide range of fitted disk densities for the sample.

D. Hot and warm coronal properties

The spectrum of the primary x-ray source or hot corona, which illuminates the accretion disk and produces reflection features in the `relxillCp` and `xillverCp` models, is described by the thermally Comptonized continuum model `nthComp`. The key parameters of the `nthComp` model are the photon index (Γ) of the primary continuum and the electron temperature (kT_e) of the hot corona, both of which are presented in Table A6. We first compare the hot coronal temperatures measured through our *XMM-Newton*+*NuSTAR* broadband spectral modeling of the sample with previous measurements reported in the literature [94–96], as shown in Fig. 10. For AGN when only a cutoff energy (E_c) estimate is available, we convert it to electron temperature using $kT_e = E_c/2.5$, since the cutoff energy of the primary continuum is typically estimated to be $\sim 2\text{--}3$ times the electron temperature [97]. No temperature comparison is possible for PG 0844 + 349 and PG 1426 + 015, as their coronal temperature or cutoff energy measurements are not available in the literature. Our broadband spectroscopy provides the first measurement of the coronal temperature for these two AGN. As evident from Fig. 10, the electron temperatures measured in this work are consistent with those reported in the literature within their 2σ confidence intervals. The left panel of Fig. 11 presents our measured temperature distribution

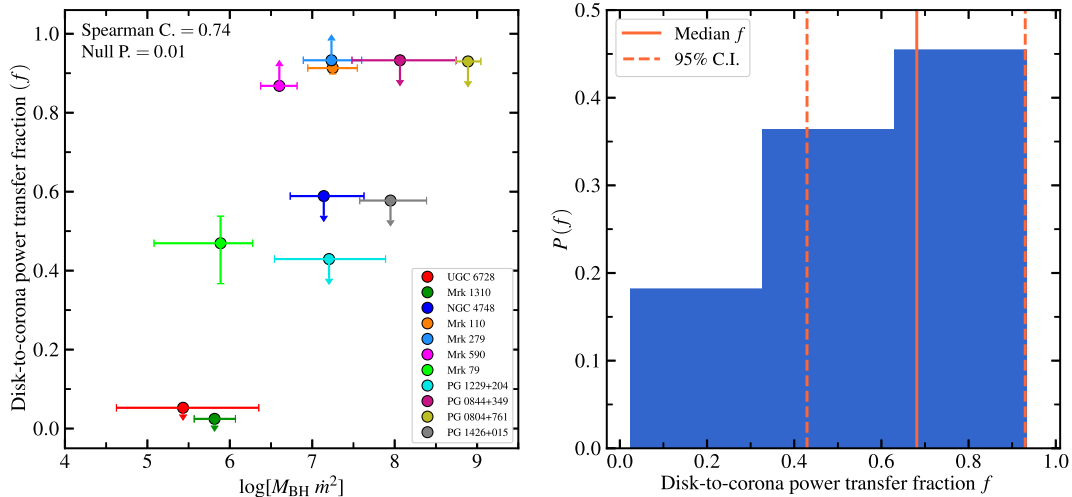


FIG. 9. The fraction (f) of power transferred from the disk into the corona, measured at $r = 10r_s$, is plotted as a function of $\log[M_{\text{BH}} \dot{m}^2]$ in the left panel. We find a strong positive correlation between f and $\log[M_{\text{BH}} \dot{m}^2]$ with a Spearman rank correlation coefficient of 0.74 [p-value = 0.01]. The right panel shows the distribution of the f -parameter, which has a median value of $f = 0.68^{+0.25}_{-0.25}$.

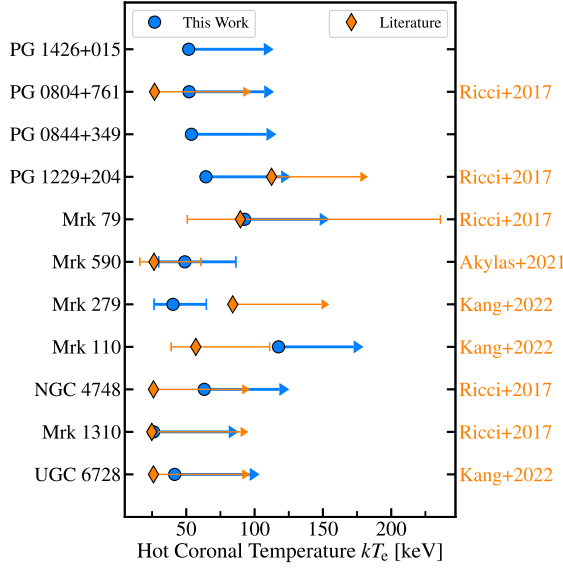


FIG. 10. Comparison of the electron temperature (blue circles) of hot corona obtained from our broadband *XMM-Newton*+*NuSTAR* spectral modeling of the sample and their previous measurements (orange diamonds) from the literature [94–96], labeled on the right-hand side of the Y-axis. The errors are reported at the 90% confidence level. Orange diamonds are absent for PG 0844 + 349 and PG 1426 + 015 since no estimate of their coronal temperature is available in the literature.

of the hot corona for the sample, which has a median value of 54^{+11}_{-12} keV at the 2σ confidence level.

Another physical parameter that characterizes the corona is its optical depth. However, optical depth is not a free parameter in the *nthComp* model. Therefore, we need to estimate the optical depth (τ_e) of the hot

corona, which is related to the electron temperature (kT_e) and photon index (Γ) of the *nthComp* model via the formula [23]:

$$\tau_e = \sqrt{2.25 + \frac{3}{\frac{kT_e}{m_e c^2} \times [(\Gamma + 0.5)^2 - 2.25]}} - 1.5. \quad (2)$$

An anticorrelation between kT_e and τ_e is expected from the above equation itself. Therefore, we did not conduct a correlation analysis between these two parameters. Such an analysis is only meaningful when τ_e is independently measured via models, as demonstrated by [98]. The calculated optical depth (τ_e) of the hot corona for our sample lies in the range of ~ 0.15 – 1.86 , with a mean value of ~ 1.0 . This range aligns well with the optical depth range of ~ 0.19 – 1.78 inferred for the AGN sample in [99]. The distribution of hot coronal optical depths for our AGN sample is presented in Fig. 11 (right), showing a median value of $0.98^{+0.22}_{-0.28}$ at 2σ confidence. The optical depth range inferred for the hot corona in our study is reasonable, as demonstrated by the numerical simulations of [24]. Furthermore, to clearly observe the relativistic reflection features from the inner disk, the coronal optical depth is required to be less than or close to unity, as argued by [100].

The physical origin of the warm corona remains an open question. It may correspond to an anomalously heated surface layer of the accretion disk, as suggested by [90]. In the *compTT* model, optical/UV seed photons from the accretion disk are Compton upscattered in an optically thick, low-temperature corona, producing excess emission in the soft x-ray band of AGN. The key parameters of the *compTT* model are the temperature (kT_{wc}) and optical

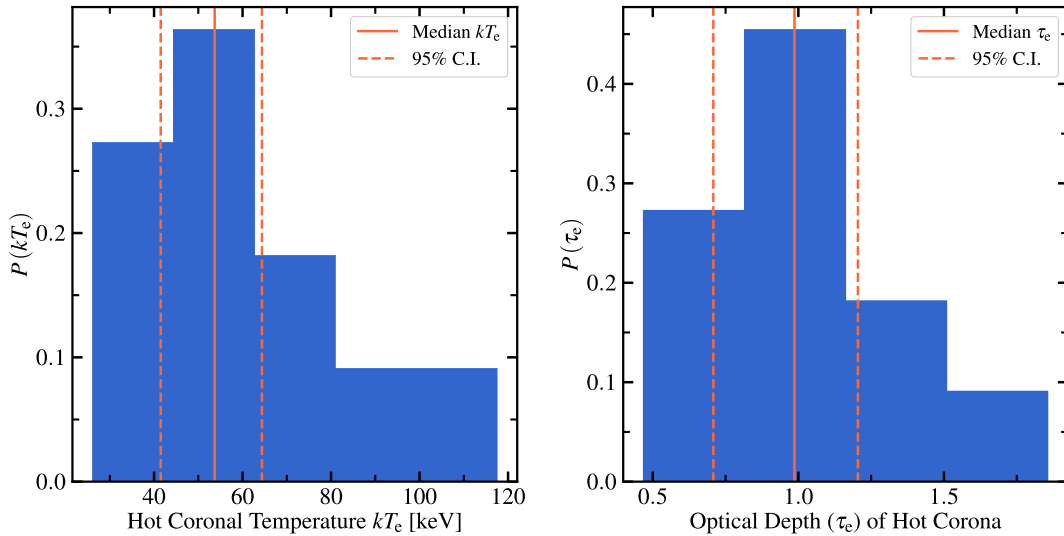


FIG. 11. Probability distribution of measured temperature (kT_e) and inferred optical depth (τ_e) of the hot corona for 11 AGN from this work. The median values of hot coronal temperature and optical depth for the sample are $kT_e = 54^{+11}_{-12}$ keV and $\tau_e = 0.98^{+0.22}_{-0.28}$, respectively.

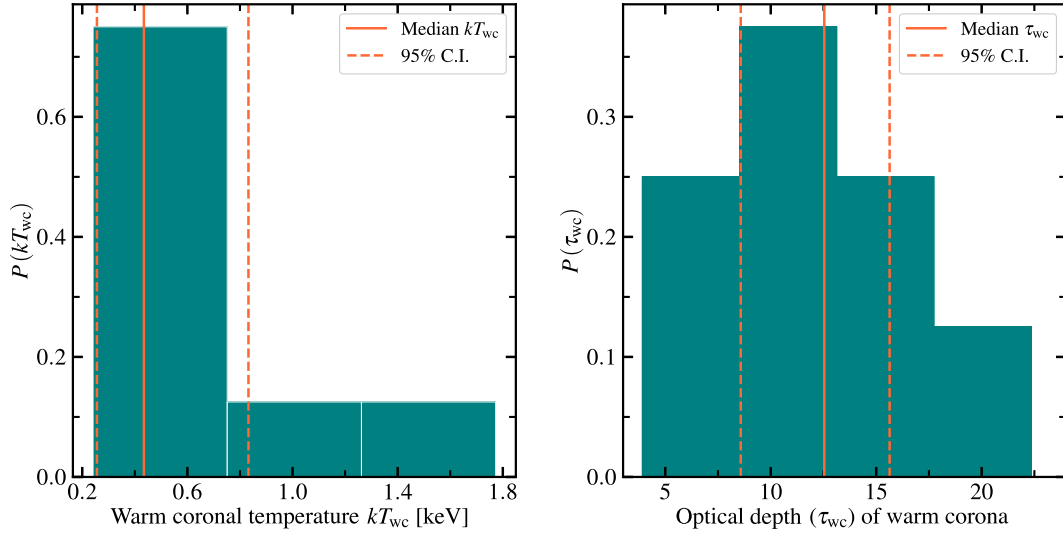


FIG. 12. Probability distribution of measured temperature and optical depth of the warm corona for eight AGN in our sample that require an additional warm Comptonization to describe the observed soft x-ray excess. The sample median values of the warm coronal temperature and optical depth are $kT_{\text{wc}} = 0.43^{+0.40}_{-0.18}$ keV and $\tau_{\text{wc}} = 12.5^{+3.1}_{-3.9}$, respectively.

depth (τ_{wc}) of the warm corona, which are listed in Table A6 for the eight AGN in our sample that require an additional warm Comptonization component to model the observed soft x-ray excess. The distribution of the measured temperature of the warm corona for these eight AGN is presented in Fig. 12 (left), revealing a median value of $kT_{\text{wc}} = 0.43^{+0.40}_{-0.18}$ keV at the 2σ confidence level. The right panel of Fig. 12 shows the distribution of the measured optical depth for the warm corona in these eight AGN, which has a median value of $\tau_{\text{wc}} = 12.5^{+3.1}_{-3.9}$ at 2σ confidence. Our sample spans almost the full range of temperature and optical depth allowed for the warm corona.

E. Black hole spin measurements from the relativistic reflection features

We measure black hole spin by modeling relativistic disk reflection features, primarily the broad Fe K emission line and soft x-ray excess, using the `relxillCp` model under the assumption that emission originates from the innermost stable circular orbit. The accuracy of spin measurements mainly depends on the red wing of the broad Fe K emission line. Since the soft x-ray band is contaminated by warm coronal emission in most sources, we first estimate the spin parameter by modeling the hard x-ray (3–78 keV) spectra, which contain the broad Fe K emission feature, for all

TABLE III. The best-fit source spectral models and spin measurements obtained from both hard x-ray (3–78 keV) and broadband (0.3–78 keV) relativistic reflection spectroscopy of the sample. Columns (2) and (3) list the best-fit hard x-ray source spectral model and the corresponding spin parameter obtained from the hard x-ray spectral modeling. Columns (4) and (5) display the broadband best-fit source spectral model and the spin parameter determined from the broadband spectral modeling for each source.

Source (1)	3–78 keV Best-fit model (2)	$a_{\text{3–78 keV}}^*$ (3)	0.3–78 keV Best-fit model (4)	$a_{\text{0.3–78 keV}}^*$ (5)
UGC 6728	relxillCp+nthComp	$0.658^{+0.249}_{-0.518}$	relxillCp+nthComp	$0.730^{+0.114}_{-0.610}$
Mrk 1310	relxillCp+nthComp	$0.736^{+0.219}_{-0.606}$	compTT+relxillCp+nthComp	$0.752^{+0.190}_{-0.605}$
NGC 4748	relxillCp+xillverCp+nthComp	$0.705^{+0.240}_{-0.573}$	compTT+relxillCp+xillverCp+nthComp	$0.796^{+0.123}_{-0.663}$
Mrk 110	relxillCp+xillverCp+nthComp	≥ 0.970	compTT+relxillCp+xillverCp+nthComp	$0.978^{+0.016}_{-0.024}$
Mrk 279	relxillCp+xillverCp+nthComp	≥ 0.968	compTT+relxillCp+xillverCp+nthComp	≥ 0.991
Mrk 590	relxillCp+zGauss_N+zGauss2_N+nthComp	$0.901^{+0.048}_{-0.130}$	relxillCp+zGauss_N+zGauss2_N+nthComp	$0.890^{+0.041}_{-0.093}$
Mrk 79	relxillCp+xillverCp+nthComp	$0.944^{+0.038}_{-0.762}$	compTT+relxillCp+xillverCp+nthComp	$0.880^{+0.048}_{-0.039}$
PG 1229 + 204	relxillCp+nthComp	$0.676^{+0.226}_{-0.553}$	relxillCp+nthComp	$0.974^{+0.024}_{-0.775}$
PG 0844 + 349	relxillCp+nthComp	≤ 0.968	compTT+relxillCp+nthComp	$0.292^{+0.555}_{-0.157}$
PG 0804 + 761	relxillCp+zGauss_N+nthComp	≤ 0.931	compTT+relxillCp+zGauss_N+nthComp	≤ 0.706
PG 1426 + 015	relxillCp+xillverCp+nthComp	$0.460^{+0.463}_{-0.297}$	compTT+relxillCp+xillverCp+nthComp	$0.441^{+0.488}_{-0.122}$

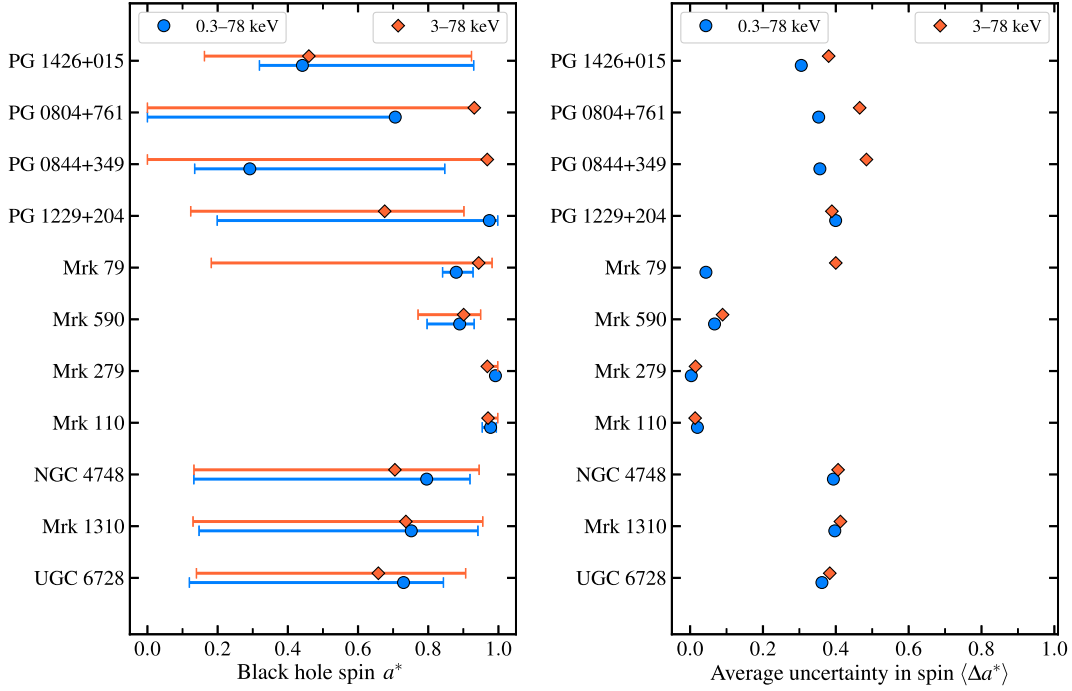


FIG. 13. The left panel presents black hole spins measured using hard x-ray (3–78 keV; orange diamonds) and broadband (0.3–78 keV; blue circles) relativistic reflection spectroscopy. All errors are measured through MCMC at the 90% confidence level. Details of the best-fit spectral models for both energy ranges are provided in Table III. The right panel shows the average uncertainties for spin measurements in each case.

sources in the sample, as quantitatively demonstrated in Table A2. Additionally, with the use of *NuSTAR* data above 10 keV, we obtain better constraints on the hard x-ray continuum and detect Compton reflection hump, which can provide a more accurate determination of the red wing of the broad Fe K emission line. The black hole spin values obtained from the best-fit hard x-ray (3–78 keV) spectra and their 90% credible intervals determined through MCMC are presented in Table III and shown as orange diamonds in the left panel of Fig. 13.

Although the warm Comptonization model `compTT` dominates the soft x-ray (0.3–2 keV) band for some sources (e.g. PG 1426 + 015), the relativistic disk reflection model `relxillCp` also contributes to this excess emission (see Fig. 4). The best-fit spin values measured through broadband (0.3–78 keV) spectral modeling and corresponding 90% credible intervals obtained from MCMC analyses of the sample are presented in Table III and depicted by blue circles in the left panel of Fig. 13. The average uncertainties in the black hole spin parameter, as measured from the hard x-ray and broadband spectral modeling of the sample, are shown in the right panel of Fig. 13. Overall, the inclusion of soft x-ray spectra provides tighter constraints on the spin parameter for the sample, notably for the four AGN: Mrk 79, PG 0844 + 349, PG 0804 + 761, and PG 1426 + 015. In these four AGN, warm coronal emission was present in the soft x-ray band in addition to the relativistic disk reflection. However, Mrk 79 exhibits relativistic disk reflection-dominated soft x-ray spectra (Fig. 4), which substantially

reduces spin uncertainties and significantly refines the measured spin through broadband spectroscopy (Fig. 13).

In Fig. 14, we present the dimensionless black hole spin parameter as a function of black hole mass for 11 AGN measured using our broadband spectral modeling, together with the 13 low-mass dwarf AGN from [37] and 36 AGN with the most recent reliable spin-mass measurements. With the addition of 11 new, refined spin measurements through broadband *XMM-Newton*+*NuSTAR* relativistic reflection spectroscopy, the total spin sample size has reached $N = 60$. Thus, this work is increasing or refining the available spin measurements in the AGN population by approximately 20%. The distribution of black hole spins for our sample is shown in Fig. 15 (left panel), which has a median of $0.81^{+0.16}_{-0.11}$ at the 2σ confidence level. The right panel of Fig. 15 shows the spin distribution for all 60 AGN, where the median is $0.87^{+0.04}_{-0.07}$ at 2σ confidence.

The measurement of high spins is not the shortcoming of the relativistic reflection model. We have previously measured the black hole spin parameter in faint dwarf AGN using the same relativistic reflection spectroscopy and found a range of spin values, from low and moderate to high [37]. A fundamental parameter governed by black hole spin is the radiative efficiency (η) of the accretion flow, given by

$$\eta = 1 - \left(1 - \frac{2r_g}{3r_{\text{isco}}}\right)^{1/2}. \quad (3)$$

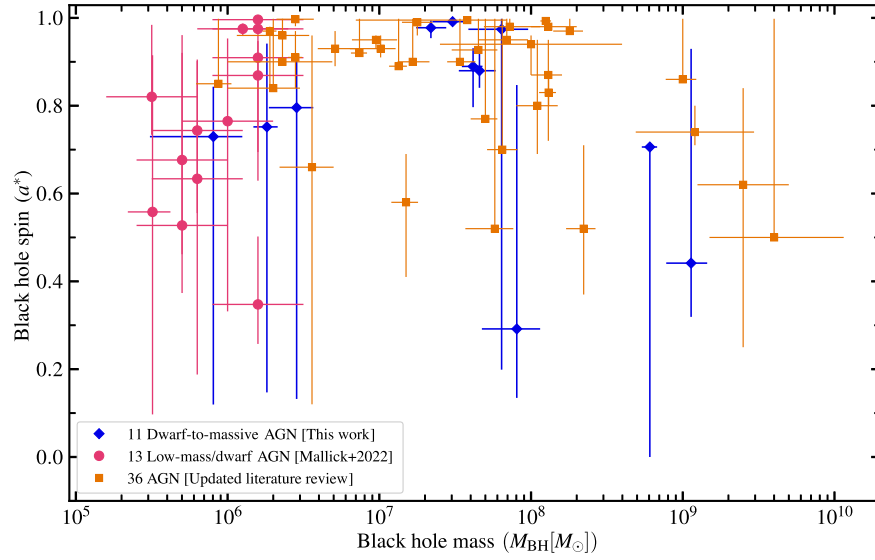


FIG. 14. Evolution of dimensionless black hole spin as a function of black hole mass constructed using the most updated spin and mass measurements. The blue diamonds show the spin measurements of the 11 AGN from this work. The pink circles represent spin measurements of low-mass AGN (including dwarf AGN) from [37], while the orange squares denote spin measurements compiled from the updated literature review. The error bars on the spin values indicate the 90% confidence intervals.

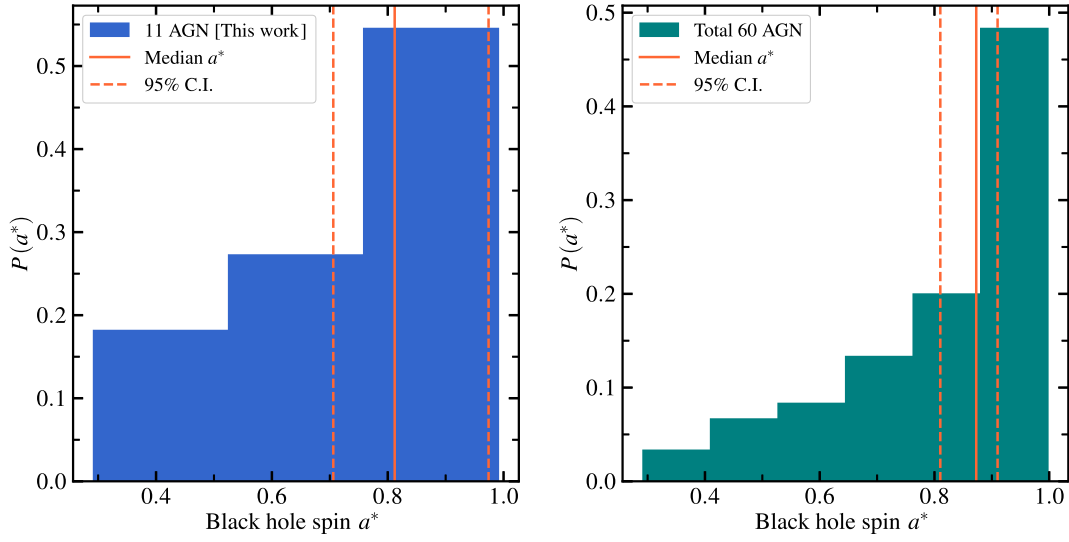


FIG. 15. Probability distribution of black hole spin for 11 AGN from this work and all 60 AGN with updated spin measurements, including those 11 AGN. Our spin measurements are increasing or refining the available spin population of AGN by $\sim 20\%$. The sample medians of the spin parameter for the 11 AGN in our sample and the total 60 AGN are $0.81^{+0.16}_{-0.11}$ and $0.87^{+0.04}_{-0.07}$, respectively, at the 95% confidence level.

For prograde orbits restricted to $\theta = \pi/2$ plane, the formula for r_{isco} as derived by [101] is:

$$r_{\text{isco}} = (3 + Z_2 - [(3 - Z_1)(3 + Z_1 + 2Z_2)]^{1/2})r_g, \quad (4)$$

where

$$Z_1 = 1 + (1 - a^{*2})^{1/3}[(1 + a^*)^{1/3} + (1 - a^*)^{1/3}], \quad (5)$$

$$Z_2 = (3a^{*2} + Z_1^2)^{1/2}. \quad (6)$$

As evident from the above equations, radiative efficiency is purely a function of black hole spin. AGN with high spins ($a^* > 0.9$) exhibit high radiative efficiency ($\eta > 0.2$). In addition to spin, radiative efficiency directly influences the luminosity of an accreting object. For a steady-state accretion flow, the total radiated luminosity is $L = \eta \dot{M} c^2$, where

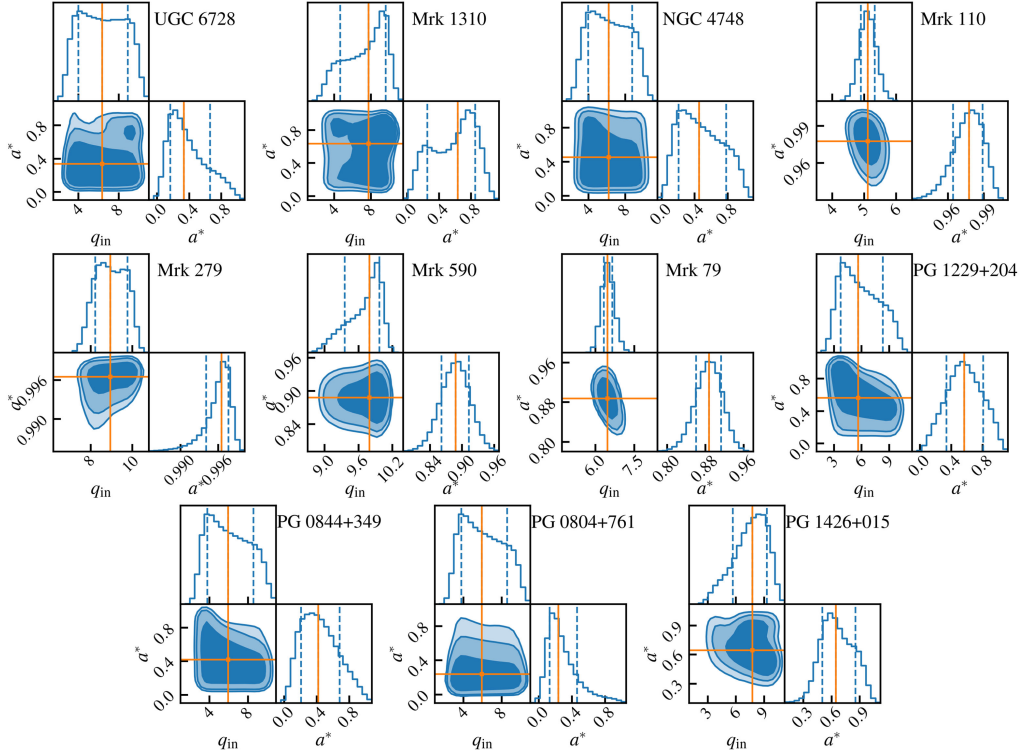


FIG. 16. Corner plots between dimensionless black hole spin (a^*) and inner emissivity index (q_{in}) of the sample. Vertical lines show the median and 68.3% confidence intervals. The dark, medium, and light blue areas represent 68.3%, 90%, and 95% confidence levels, respectively, with square symbols indicating the parameter medians.

$\dot{M}$ is the accretion rate. Highly spinning black holes have high radiative efficiency, which makes them more luminous even if they have a comparable accretion rate. Therefore, accreting black holes with high spins are more likely to dominate flux-limited samples (e.g. [102]).

As a final note, we emphasize that in the relativistic reflection method for spin estimation, high-spin measurements are likely to be more reliable than low-spin measurements. This is because the method requires blurred reflection features, ideally the broadened Fe K line, which can only be produced when the inner edge of the disk is very close to the black hole and the corona is compact or located near the black hole. If the Fe K line is not very broadened, the black hole may not have a high spin even if the corona is compact. Alternatively, the corona may not be compact or located near the black hole, even if the black hole has a high-spin value. In essence, spin measurements are influenced by the inner edge of the illuminated part of the disk, while intrinsic spin governs the true inner edge of the disk. Therefore, a correlation between measured spin and coronal compactness may indicate that we are actually sampling the illuminated inner edges of the disks rather than their true ISCOs. To evaluate the reliability of our spin measurements, we search for any correlations between black hole spin and the inner emissivity index, which serves as a proxy for the corona's compactness or proximity to the black hole. In Fig. 16, we present contour plots of spin

versus the inner emissivity index for our sample, which show no significant correlation or degeneracy between the two parameters. This finding demonstrates that our spin measurements accurately probe the true ISCOs of the central SMBHs in these 11 AGN.

V. SUMMARY AND CONCLUSIONS

In this paper, we apply the updated higher-density disk reflection model `relxillCp` to joint *XMM-Newton* + *NuSTAR* broadband spectra of a sample of type-1 AGN spanning nearly the complete range of central black hole masses from $M_{\text{BH}} \sim 10^{5.5} M_{\odot}$ to $10^9 M_{\odot}$. We systematically investigate the origin of both hard and soft x-ray excess in all sources of the sample, and assess the relevance of both high-density disk reflection and warm Comptonization for the observed soft x-ray excess using a Bayesian framework. Additionally, we compute the disk-to-corona power transfer fraction for the first time in any accreting system. Finally, we constrain the SMBH spin parameter across mass scales using both hard x-ray (3–78 keV) and broadband (0.3–78 keV) relativistic reflection spectroscopy. The main results and conclusions of our work are summarized below:

- (1) The relativistic reflection model, incorporating a variable-density accretion disk with a broken power-law emissivity profile, can describe the soft x-ray excess,

together with the broad Fe K emission line and Compton hump in three out of 11 AGN, namely UGC 6728, Mrk 590, and PG 1229 + 204. For the remaining eight sources, an additional low-temperature or warm Comptonization component is necessary to accurately model the soft x-ray excess, which suggests its hybrid origin.

- (2) The measured temperature and optical depth of the warm corona for these eight AGN in our sample are $kT_{\text{wc}} \sim 0.2\text{--}2$ keV and $\tau_{\text{wc}} \sim 4\text{--}22$, respectively, spanning nearly the entire theoretically allowed range for the warm corona. The median values are $kT_{\text{wc}} = 0.43^{+0.40}_{-0.18}$ keV and $\tau_{\text{wc}} = 12.5^{+3.1}_{-3.9}$, respectively. Among these eight AGN, the measured disk density is significantly higher than the canonical value of $n_e = 10^{15} \text{ cm}^{-3}$, even with the presence of a warm corona in three AGN (Mrk 110, Mrk 279, and Mrk 79). For the remaining five AGN (Mrk 1310, NGC 4748, PG 0844 + 349, PG 0804 + 761, and PG 1426 + 015), the lower limit of the disk density is consistent with the canonical value.
- (3) The inner accretion disk of the AGN sample is found to be ionized and dense with a median ionization of $\xi \sim 10^2 \text{ erg cm}^{-2} \text{ s}^{-1}$ and a median density of $n_e \sim 10^{16.5} \text{ cm}^{-3}$ without requiring a very high super-solar iron abundance. The iron abundance of the accretion disk is near solar for the sample, with a median value of ~ 2 relative to the solar abundance.
- (4) We did not find any anticorrelation between disk density and either black hole mass times the accretion rate squared or black hole mass alone. This indicates that the intrinsic scatter in the fraction of power transferred from the accretion disk to the corona is substantial, and our sample spans a broad range of disk-to-corona power transfer fractions.
- (5) For the first time, we calculate the disk-to-corona power transfer fraction for each source, finding a sample median of $f = 0.68^{+0.25}_{-0.25}$ at 2σ confidence. Importantly, we observe a strong positive correlation between the f -parameter and black hole mass times the accretion rate squared, $\log(M_{\text{BH}}\dot{m}^2)$, which is consistent with the theoretical prediction of a radiation pressure-supported accretion disk.
- (6) The coupling between the accretion disk and hot corona is directly evident from the fraction of the disk power transferred into the hot corona. The measured electron temperature and optical depth of the hot corona have medians of $kT_e = 54^{+11}_{-12}$ keV and $\tau_e = 0.98^{+0.22}_{-0.28}$, respectively, for the sample. The hot corona has a reasonably low optical depth to make the relativistic reflection from the inner accretion disk observable.
- (7) By utilizing joint *XMM-Newton*+*NuSTAR* relativistic reflection spectroscopy, we are increasing or

refining the AGN population for which a spin measurement is available by around 20% across the mass range of $\log M_{\text{BH}} \sim 5.5\text{--}9.0$. This allows us to achieve a total spin sample of 60 AGN. The median spins for 11 AGN in our sample and all 60 AGN are $0.81^{+0.16}_{-0.11}$ and $0.87^{+0.04}_{-0.07}$, respectively, at 2σ confidence.

VI. FUTURE PROSPECTS

The RELXILL family of reflection models requires further development to provide a more robust determination of the parameter space and the origin of the soft x-ray excess. Currently, the *relxillCp* model assumes a broken power-law emissivity profile for the coronal illumination of the accretion disk. However, this assumption has not yet been fully validated due to uncertainties regarding the coronal geometry. Future observations with *IXPE* [103] will be crucial for constraining the geometry of the AGN corona in this context.

The other noteworthy simplification in current reflection models is the assumption that the density and ionization structure remain uniform across the entire disk. In reality, accretion disks are likely to exhibit variations in density and ionization structure, both radially and vertically. Thus, determining the origin of the soft x-ray excess will require the development of a disk reflection model that accounts for radial and vertical variations in disk density and ionization state. In addition to refining these models, we must design advanced x-ray telescopes capable of detecting the features in x-ray spectra predicted by these more realistic models.

Next-generation x-ray observatories, such as *NewAthena* [104], *Colibrí* [105], and *HEROIX* (Thomas *et al.*, in prep) will significantly advance the scope of this study in multiple ways. Overall, these future missions will allow similar analyses to be performed at much higher redshifts, with better accuracy in parameter estimation and faster processing.

NewAthena with its advanced sensitivity in soft x-rays and high spatial resolution will extend the results of this study to fainter AGN populations, thereby probing the disk-to-corona power transfer in lower luminosity regimes, even for non-central, gas-starved supermassive black holes (see e.g. [106]). *NewAthena* and *HEROIX* will jointly provide tighter constraints on the temperatures of warm and hot coronae by observing a larger and more diverse sample of AGN across a wide range of redshifts and mass scales. They will play a pivotal role in refining spin measurements for a significantly larger number of supermassive black holes (e.g. [107]), especially for high(er)-redshift AGN, which remain under-represented in current spin demographic studies. Those constraints will also be crucial to improving our knowledge of the population of black hole seeds (e.g. [108]), typically formed at redshifts $z = 20\text{--}30$ [109].

Both *NewAthena* and *Colibrí* will be instrumental in testing competing models for the soft x-ray excess (high-density disk

reflection versus warm Comptonization) due to their superior capability to resolve these components spectroscopically. The determination of the x-ray spectral energy distribution of the faint AGN population recently discovered by *JWST* (see e.g. [110]) will also open up new possibilities to investigate peculiar spectral shapes and assess the spectral impact of super-Eddington accretors as recently investigated by several studies (see e.g. [111]).

In summary, several future x-ray facilities will offer immense possibilities for enhancing and expanding the work performed in this paper.

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DATA AVAILABILITY

All data used in this study are publicly available from the NASA High Energy Astrophysics Science Archive Research Center (HEASARC [112]).

APPENDIX: MODELING DETAILS OF INDIVIDUAL AGN

Here, we discuss the hard-to-soft x-ray spectral fitting details for individual AGN in the sample.

1. UGC 6728

The source exhibits soft x-ray excess below ~ 2 keV in the *XMM-Newton* spectra and a broad Fe K emission line with a width of ~ 0.13 keV in the 6–7 keV range of the *NuSTAR* spectra (Fig. A1). We noticed only a weak Compton hump in the 15–30 keV range. The confidence level of `relxillCp` for fitting the broad Fe K emission line is 99.95% with $\Delta\text{DIC} = 12.6 > 10$. The fitting of the Compton reflection hump with the `xillverCp` model is found to be insignificant, with a confidence level of 86.4% and $\Delta\text{DIC} = -4.3 < 0$. However, the Compton hump is well modeled by the `relxillCp` model with 95.72% confidence and $\Delta\text{DIC} = 14.2 > 10$. Therefore, the hard x-ray (3–78 keV) spectra require mainly the relativistic disk reflection model `relxillCp` to fit the broad Fe K emission together with the Compton hump, and the model `TBabs*(relxillCp + nthComp)` best fits the hard x-ray spectra with $\chi^2/\text{dof} = 375.8/360$. Once we extrapolate the hard x-ray best-fit model down to 0.3 keV, we find that the `relxillCp` model can self-consistently fit the soft x-ray excess emission, yielding a very good fit with $\chi^2/\text{dof} = 514.7/460$. No structural residuals are seen in the entire energy band. To test the relevance of the warm coronal emission for the origin of soft x-ray excess, we added the warm Comptonization model, `compTT`, which provided $\text{DIC}_{\text{with WC}} = 545.2$ through Bayesian analysis. The confidence level of the warm coronal emission as evaluated by the MLR test is only 77.18%. The difference between the Deviance Information Criteria without and with the warm Comptonization component is $\Delta\text{DIC} = \text{DIC}_{\text{without WC}} - \text{DIC}_{\text{with WC}} = 543.9 - 545.2 = -1.3 < 0$, implying that an extra warm Comptonization is not required to fit the observed soft x-ray excess in UGC 6728. Considering the LOS Galactic absorption (`TBabs`), the broadband (0.3–78 keV) best-fit model expression is

$$\text{TBabs} * (\text{relxillCp} + \text{nthComp}).$$

Figure A2 shows the broadband *XMM-Newton*/*NuSTAR* spectra, the best-fit count spectral model with components, and the corresponding residuals. We plot the best-fit averaged spectral energy flux model with `relxillCp` and `nthComp` components in Fig. A3. The best-fit spectral model parameters and their 90% confidence intervals determined through MCMC parameter exploration are presented in Table A6. The best-fit values for the disk density, iron abundance, black hole spin, and disk inclination angle are $\log[n_e/\text{cm}^{-3}] \leq 17.7$, $A_{\text{Fe}} \leq 4.3$, $a^* = 0.73^{+0.114}_{-0.61}$, and $\theta = 50^{+4}_{-21}$ degree, respectively. The temperature of the hot corona is measured to be $kT_e \geq 42$ keV. Bayesian analysis suggested no difference between the high-density and canonical disk reflection models, which is supported by the disk density parameter reaching its lower limit of $\log[n_e/\text{cm}^{-3}] = 15$.

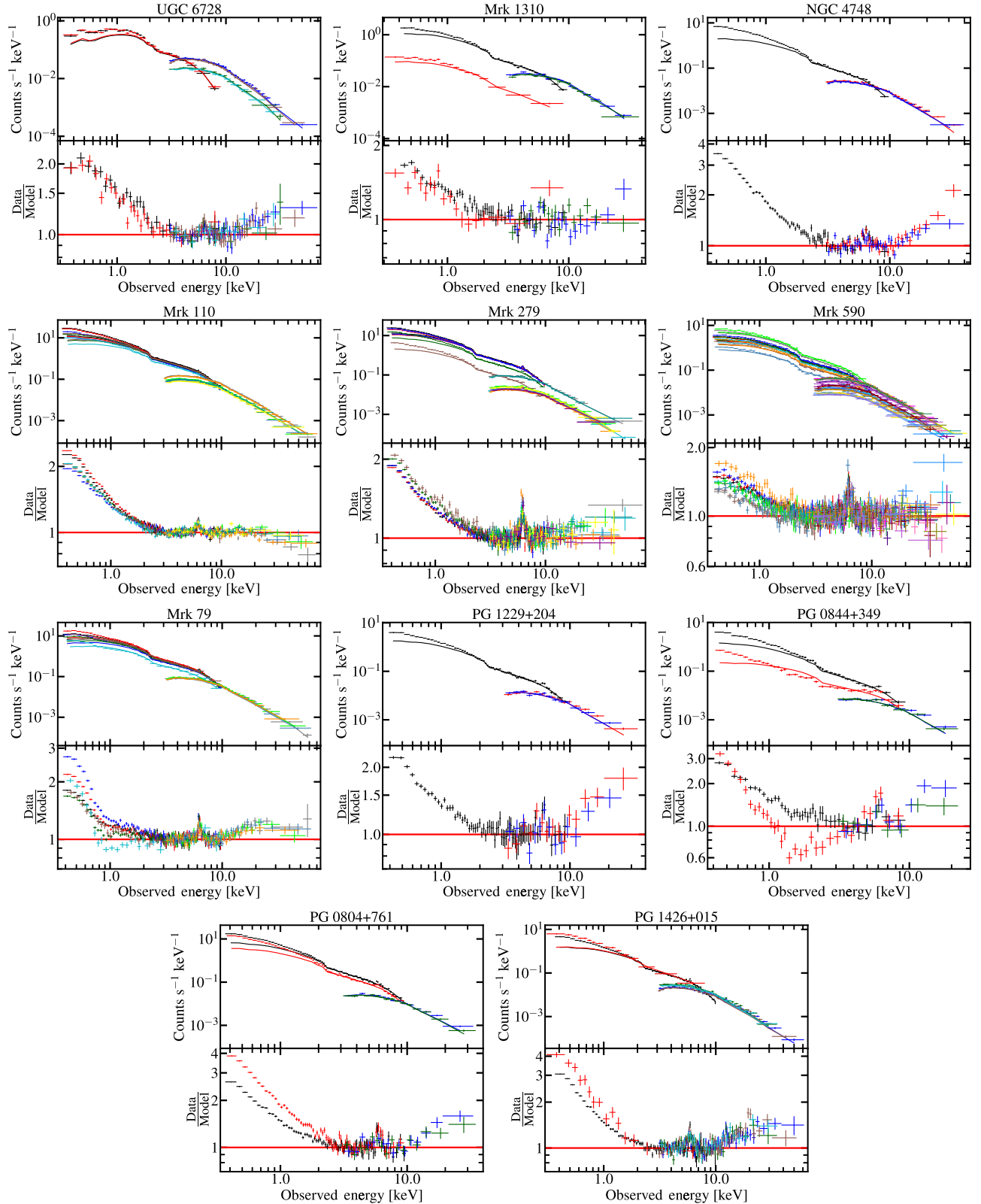


FIG. A1. The top panels show the *XMM-Newton* EPIC and *NuSTAR* FPM spectral data fitted by the Galactic absorption corrected power-law model (TBabs**zpower1w*) in the range 3–5 keV and 7–10 keV and then extrapolated over the complete energy range for each source. The bottom panel is the corresponding data-to-model ratio. The ratio plot shows soft x-ray excess below around 1–2 keV and Fe K emission in the 6–7 keV range for each source, along with a hard x-ray excess in the 15–30 keV range for some sources. The spectra are rebinned to visualize and demonstrate the shape of the residuals more clearly.

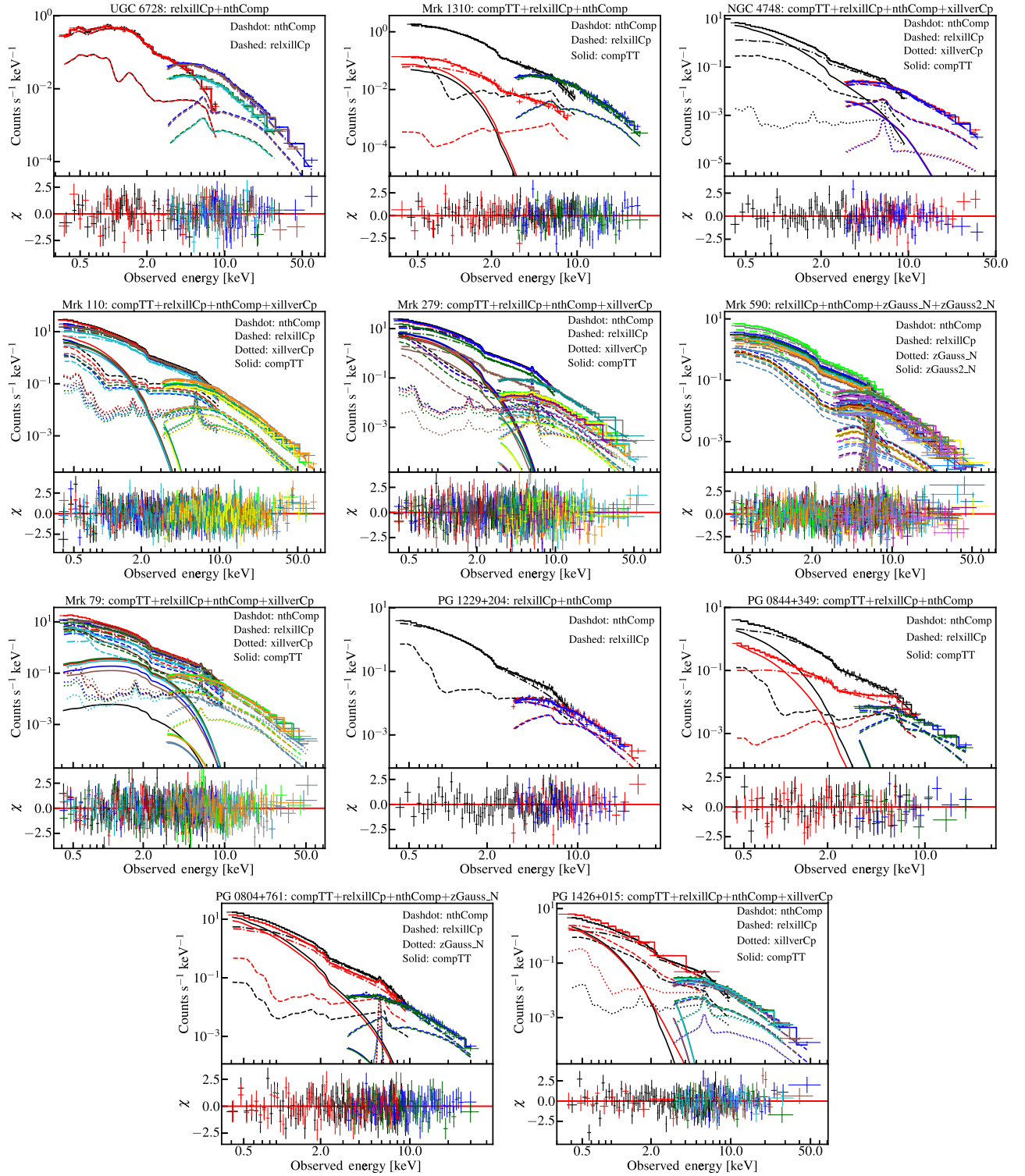


FIG. A2. The *XMM-Newton*/EPIC, *NuSTAR*/FPMA, and FPMB count spectra with the best-fit models and residuals as a function of energy. Five sets of best-fit models have been obtained for the sample. The spectra are binned up only for plotting purposes.

2. Mrk 1310

The source revealed soft x-ray excess below ~ 1.5 keV in the *XMM-Newton* spectra and a broad Fe K emission line of width ~ 0.14 keV in the 6–7 keV band of the *NuSTAR*

spectra (Fig. A1). The Compton hump was not detected in the *NuSTAR* spectra. The presence of a Compton hump, as modeled by either *xillverCp* or *relxillCp*, is found to be insignificant with confidence levels of only 35.54%

and 10.94%, and $\Delta\text{DIC} = -1.1$ and -1.8 , respectively. The confidence level of the detected broad Fe K emission feature fitted by the `relxillCp` model is 99.92% with $\Delta\text{DIC} = 6.1 > 2$. Therefore, the hard x-ray spectra of Mrk 1310 are best described by the model, `TBabs*(relxillCp+nthComp)`, with $\chi^2/\text{dof} = 193.2/158$. The extrapolation of the hard x-ray best-fit model down to 0.3 keV can describe the broadband spectra well with $\chi^2/\text{dof} = 277.1/228$. No significant features are seen in the residual plot. However, we still test for the presence of a warm corona by adding the warm Comptonization model, `compTT`, which provides $\chi^2/\text{dof} = 263.4/223$ and $\text{DIC}_{\text{with WC}} = 297.9$. The warm coronal emission has a confidence level of 98.24% as calculated by the MLR test. The measured difference between the Deviance Information Criteria without and with the warm Comptonization component is $\Delta\text{DIC} = \text{DIC}_{\text{without WC}} - \text{DIC}_{\text{with WC}} = 308.9 - 297.9 = 11.0 > 10$. Both the MLR test and Bayesian model selection metric imply that the warm corona contributes to the observed soft x-ray excess in Mrk 1310. Therefore, the broadband best-fit model, including `TBabs`, is

$$\text{TBabs} * (\text{compTT} + \text{relxillCp} + \text{nthComp}).$$

In Fig. A2, we show the broadband *XMM-Newton*/*NuSTAR* spectra, the best-fit count spectral model with components, and the corresponding residual plot. Figure A3 shows the best-fit averaged spectral energy flux model with `compTT`, `relxillCp`, and `nthComp` components. In Table A6, we present the best-fit spectral model parameters and their 90% confidence intervals obtained from the MCMC calculation. The best-fit values of disk density, iron abundance, black hole spin, and disk inclination angle are $\log[n_e/\text{cm}^{-3}] \leq 17.3$, $A_{\text{Fe}} = 1.4^{+2.3}_{-0.7}$, $a^* = 0.752^{+0.19}_{-0.605}$, and $\theta = 49^{+7}_{-31}$ degree, respectively. The lower limit of the hot coronal temperature is estimated to be 26 keV. The Bayesian analysis did not find any difference between the high-density and canonical disk reflection models since the density parameter is pegged at the lower bound of $\log[n_e/\text{cm}^{-3}] = 15$.

3. NGC 4748

The *XMM-Newton*/*NuSTAR* spectra of NGC 4748 reveal a narrow Fe K_α core at ~ 6.4 keV with a broad Fe K emission feature of width ~ 0.35 keV in the 6–7 keV band, a Compton hump above 15 keV, and soft x-ray excess below ~ 2 keV (Fig. A1). The broad Fe K emission line is modeled by `relxillCp` and has a confidence level of 98.8% with $\Delta\text{DIC} = 9.7 > 2$. The confidence levels of the Compton hump fitted by `xillverCp` or `relxillCp` are 99.11% and 97.19%, with $\Delta\text{DIC} = 5.3$ and 4.2, respectively. Therefore, the model, `TBabs*(relxillCp+xillverCp+nthComp)`, best explains the hard x-ray

(3–78 keV) spectra with $\chi^2/\text{dof} = 207.2/197$, where the narrow 6.4 keV Fe K_α line and part of the Compton hump are modeled by `xillverCp`, while the broad Fe K emission line and most of the Compton hump are modeled by `relxillCp`. The extrapolation of the hard x-ray best-fit model can fit the broadband (0.3–78 keV) x-ray spectra with $\chi^2/\text{dof} = 281.0/242$, where the soft x-ray excess is modeled by `relxillCp`. However, we notice some excess emission in the hard x-ray band, which means `relxillCp` cannot explain both the soft and hard x-ray excess emission self-consistently. We then add the `compTT` model and find that the spectral model, `TBabs*(compTT+relxillCp+xillverCp+nthComp)`, explains the broadband spectra the best with $\chi^2/\text{dof} = 258.3/238$. There are no structural residuals in the entire energy band. The Bayesian model selection metric very strongly prefers warm Comptonization over high-density disk reflection for fitting of soft x-ray excess with $\Delta\text{DIC} = \text{DIC}_{\text{without WC}} - \text{DIC}_{\text{with WC}} = 306.7 - 290.6 = 16.1 > 10$. The MLR test finds that the confidence level of the warm coronal emission is 99.99%. Figure A2 shows the broadband *XMM-Newton*/*NuSTAR* spectra, the best-fit count spectral model with components, and the corresponding residual plot. We plot the best-fit averaged spectral energy flux model with `compTT`, `relxillCp`, `xillverCp`, and `nthComp` components in Fig. A3. The expression for the broadband best-fit model considering `TBabs`,

$$\text{TBabs} * (\text{compTT} + \text{relxillCp} + \text{xillverCp} + \text{nthComp}).$$

The best-fit source spectral model parameters and their 90% confidence intervals derived through MCMC computation are presented in Table A6. The best-fit values of disk density, iron abundance, black hole spin, and disk inclination angle are estimated to be $\log[n_e/\text{cm}^{-3}] \leq 17.1$, $A_{\text{Fe}} \leq 4.9$, $a^* = 0.796^{+0.123}_{-0.663}$, and $\theta = 51^{+5}_{-33}$ degree, respectively. We find the lower limit for the hot coronal temperature to be 63 keV. Our joint *XMM-Newton*/*NuSTAR* spectral modeling obtained stringent constraints on the spin parameter, which was previously unconstrained by JJ19's *XMM-Newton* spectroscopy alone.

4. Mrk 110

The source exhibits soft x-ray excess below ~ 2 keV and a narrow Fe K_α core at ~ 6.37 keV along with a broad Fe K emission feature of width ~ 0.15 keV in the 6–7 keV band, and a Compton hump like structure in the 15–30 keV band (Fig. A1). The modeling of the Compton hump with `xillverCp` or `relxillCp` is found to be significant with $>99.99\%$ confidence and $\Delta\text{DIC} = 28.0$ and 20.1, respectively. The broad Fe K emission line fitted by the `relxillCp` model has a significance of $>99.99\%$ with $\Delta\text{DIC} = 38.4 > 10$. Therefore, the model `TBabs*`

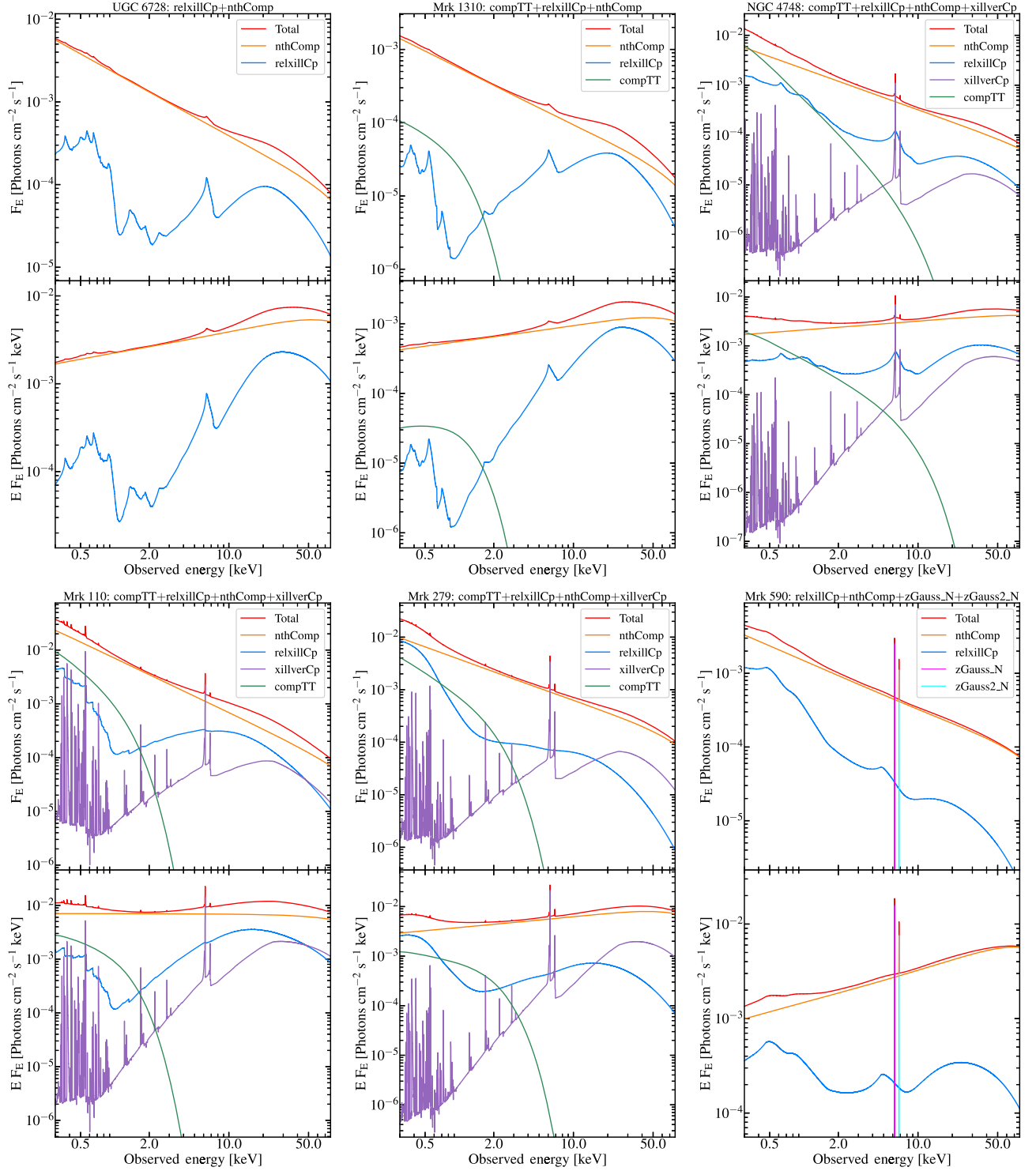


FIG. A3. The best-fit averaged spectral energy flux model (in red) with various model components for each source in the sample. The orange, blue, purple, green, and magenta/cyan solid lines represent the primary coronal emission (nthComp), relativistic disk reflection (relxillCp), distant reflection (xillverCp), warm coronal emission (compTT), and narrow Fe K_α/K_β emission cores (zGauss_N, zGauss2_N), respectively.

(relxillCp+xillverCp+nthComp) best represents the hard x-ray (3–78 keV) spectra of Mrk 110 with $\chi^2/\text{dof} = 1233.7/1155$, where the xillverCp component fits the narrow Fe K_α core and most of the Compton

hump, while the relxillCp component fits the broad Fe K emission line and part of the Compton hump. By extrapolating the hard x-ray best-fit model down to 0.3 keV, we find that the model is unable to fit the

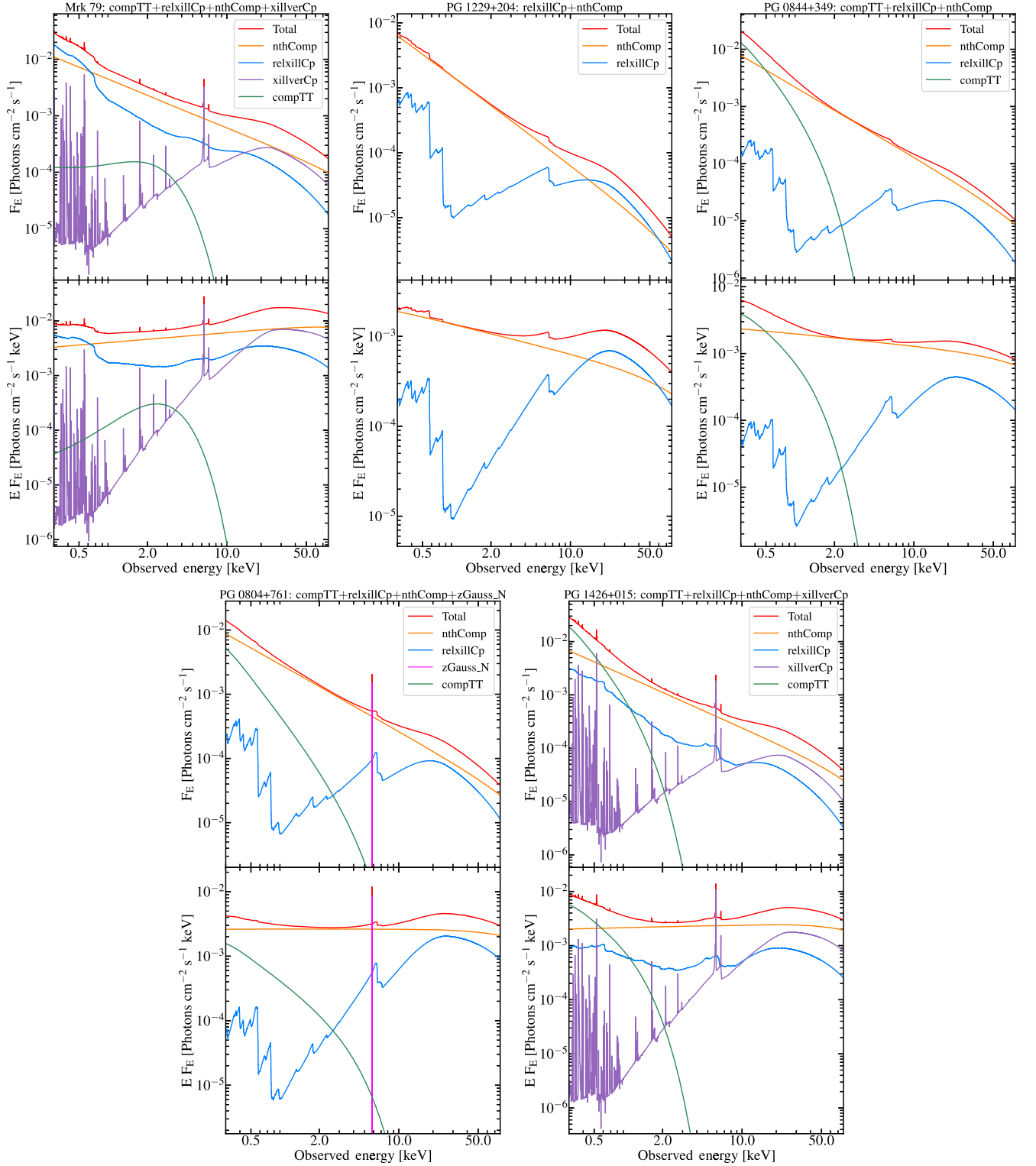


FIG. A4. Continued from previous page.

broadband (0.3–78 keV) spectra satisfactorily with significant residuals observed in the hard x-ray band, providing $\chi^2/\text{dof} = 1904.4/1434$. This suggests that the relativistic reflection model `relxillCp` cannot self-consistently fit both the soft and hard x-ray excess emission, and an extra

warm Comptonization for soft x-ray excess is perhaps needed, as found by [51]. Hence, we add the warm Comptonization model `compTT` and find that the model `TBabs* (compTT + relxillCp + xillverCp + nthComp)` represents the broadband spectra well with $\chi^2/\text{dof} = 1647/1425$, where

the soft x-ray excess is fitted by `compTT`, and `relxillCp` describes the broad Fe K emission feature and part of the Compton hump. The Bayesian analysis strongly prefers the warm coronal origin of soft x-ray excess over high-density disk reflection with $\Delta\text{DIC} = \text{DIC}_{\text{without WC}} - \text{DIC}_{\text{with WC}} = 1996.5 - 1752.8 = 243.7 \gg 10$. The significance of the warm coronal emission as computed by the MLR test is $>99.99\%$. In Fig. A2, we show the broadband *XMM-Newton*/*NuSTAR* spectra, the best-fit count spectral model with components, and the corresponding residuals. Figure A3 presents the best-fit averaged spectral energy flux model with `compTT`, `relxillCp`, `xillverCp`, and `nthComp` components. With `TBabs`, we can write the broadband best-fit model as

$$\text{TBabs} * (\text{compTT} + \text{relxillCp} \\ + \text{xillverCp} + \text{nthComp}).$$

Table A6 presents the best-fit spectral model parameters and their 90% confidence intervals, obtained by exploring the complete parameter space through MCMC. The best-fit values of disk density, iron abundance, black hole spin, and disk inclination angle are $\log[n_e/\text{cm}^{-3}] = 19.0^{+0.2}_{-0.2}$, $A_{\text{Fe}} = 1.5^{+0.7}_{-0.4}$, $a^* = 0.978^{+0.016}_{-0.024}$, and $\theta \leq 24$ degree, respectively. The lower limit of the hot coronal temperature is measured to be 118 keV. Through joint *XMM-Newton*/*NuSTAR* spectroscopy, we can constrain the disk density for which only an upper limit was calculated by JJ19's *XMM-Newton* spectral modeling alone. The Bayesian analysis strongly supports the higher-density disk against the canonical disk reflection model with $\Delta\text{DIC} > 10$.

5. Mrk 279

The *XMM-Newton*/*NuSTAR* spectra of Mrk 279 show two narrow lines at ~ 6.38 keV and ~ 6.98 keV corresponding to Fe K_α and Fe K_β emission cores, respectively, one broad Fe K emission line of width ~ 0.24 keV in the 6–7 keV band, a Compton hump above 15 keV, and soft x-ray excess below ~ 2 keV (Fig. A1). The broad Fe K emission line is modeled by `relxillCp`, which has a significance of $>99.99\%$ with $\Delta\text{DIC} = 44.0 > 10$. The modeling of the Compton hump by `xillverCp` or `relxillCp` is found to be significant with $\Delta\text{DIC} = 70.6$ and 68.4 , respectively, and has a confidence level of $>99.99\%$. Therefore, to fit the hard x-ray (3–78 keV) spectra of the source, we employ the model `TBabs*(relxillCp+xillverCp+nthComp)`, which provided the best-fit with $\chi^2/\text{dof} = 1031.3/956$. Here, the two narrow Fe K emission line cores and most of the Compton hump are modeled by the `xillverCp` model, and the `relxillCp` model fits the broad Fe K emission and part of the Compton hump. The extrapolation of the hard x-ray best-fit model can fit the broadband (0.3–78 keV) x-ray spectra well with $\chi^2/\text{dof} = 1376.9/1181$. However, there are some excess

residuals in the soft x-ray band, which may indicate the need for an extra soft x-ray component. We then add the `compTT` model and find that the model `TBabs*(compTT+relxillCp+xillverCp+nthComp)` best fits the broadband spectra of the source with $\chi^2/\text{dof} = 1307.6/1170$ and $\text{DIC}_{\text{with WC}} = 1413.0$, where both `compTT` and `relxillCp` contribute to the soft x-ray excess. The presence of warm Comptonization is significant with a confidence level of $>99.99\%$, which is further supported by our Bayesian analysis with the difference between the Deviance Information Criteria without and with warm Comptonization is $\Delta\text{DIC} = \text{DIC}_{\text{without WC}} - \text{DIC}_{\text{with WC}} = 1463.2 - 1413.0 = 50.2 > 10$. The broadband *XMM-Newton*/*NuSTAR* spectra, the best-fit count spectral model along with all components, and the residual plot are shown in Fig. A2. We plot the best-fit averaged spectral energy flux model with `compTT`, `relxillCp`, `xillverCp`, and `nthComp` components in Fig. A3. Including `TBabs`, the broadband best-fit model expression is

$$\text{TBabs} * (\text{compTT} + \text{relxillCp} \\ + \text{xillverCp} + \text{nthComp}).$$

We explore the complete parameter space through the MCMC method and list the best-fit source spectral model parameters and their 90% confidence intervals in Table A6. The best-fit values of disk density, iron abundance, black hole spin, and disk inclination angle are $\log[n_e/\text{cm}^{-3}] \geq 19.3$, $A_{\text{Fe}} = 4.0^{+0.8}_{-1.5}$, $a^* \geq 0.991$, and $\theta = 47^{+2}_{-11}$ degree, respectively. The temperature of the hot corona is measured to be $kT_e = 40^{+24}_{-14}$ keV. The Bayesian analysis strongly preferred the higher-density disk over the canonical disk reflection model with $\Delta\text{DIC} > 10$.

6. Mrk 590

In the *XMM-Newton*/*NuSTAR* spectra of the source, we find narrow Fe K_α and Fe K_β emission line cores at ~ 6.38 keV and ~ 7.0 keV, respectively along with a broad Fe K emission feature of width ~ 0.3 keV in the 6–7 keV band, and soft x-ray excess below ~ 1 keV (Fig. A1). There is no strong Compton hump observed in the *NuSTAR* spectra. The confidence levels of the Compton hump as modeled by either `xillverCp` or `relxillCp` are only 89.06% with $\Delta\text{DIC} = 0.6$, and 87.3% with $\Delta\text{DIC} = -8$, respectively. Since the Compton hump is not significant, to fit the Fe K_α and Fe K_β emission cores, we used two narrow ($\sigma = 10$ eV) Gaussian emission lines, `zGauss_N` and `zGauss2_N`, respectively. The broad Fe K emission feature is modeled by `relxillCp` and has a significance of 97.27% with $\Delta\text{DIC} = 5.4 > 2$. Therefore, the hard x-ray (3–78 keV) spectra of the source are best described by the model `TBabs*(relxillCp+zGauss_N+zGauss2_N+nthComp)` with $\chi^2/\text{dof} = 1640.5/1623$. Once we extrapolate the hard x-ray best-fit model down to 0.3 keV,

the same model can fit the broadband (0.3–78 keV) spectra well, providing a reasonable fit with $\chi^2/\text{dof} = 2030.2/2028$. Self-consistently, the `relxillCp` model explains the observed soft x-ray excess. We still incorporate the `compTT` model and test its relevance for the soft x-ray excess using both the MLR test and Bayesian analysis. The confidence level of warm Comptonization as evaluated by the MLR test is 89.53%. The estimated difference between the Deviance Information Criteria without and with warm Comptonization is $\Delta\text{DIC} = \text{DIC}_{\text{without WC}} - \text{DIC}_{\text{with WC}} = 2229.5 - 2236.9 = -7.4 < 0$, confirming that high-density disk reflection is sufficient enough to fit the observed soft x-ray excess and an extra warm Comptonization is not required. Considering `TBabs`, the broadband best-fit model expression can be written as

$$\text{TBabs} * (\text{relxillCp} + \text{zGauss_N} \\ + \text{zGauss2_N} + \text{nthComp}).$$

We show the broadband *XMM-Newton*/*NuSTAR* spectra, the best-fit count spectral model with components, and the corresponding residuals in Fig. A2. The best-fit averaged spectral energy flux model with `relxillCp`, `zGauss_N`, `zGauss2_N`, and `nthComp` components are presented in Fig. A3. Table A6 shows the best-fit spectral model parameters and their 90% confidence intervals, where the complete parameter space is explored through MCMC. The best-fit values of disk density, iron abundance, black hole spin, and disk inclination angle are $\log[n_e/\text{cm}^{-3}] \geq 19.1$, $A_{\text{Fe}} = 2.6_{-0.3}^{+0.5}$, $a^* = 0.89_{-0.093}^{+0.041}$, and $\theta = 41_{-4}^{+5}$ degree, respectively. The measured electron temperature of the hot corona is $kT_e = 49_{-19}^{+37}$ keV. The joint *XMM-Newton*+*NuSTAR* relativistic reflection spectroscopy provided more stringent constraints on the black hole spin parameter, which was previously unconstrained by JJ19's *XMM-Newton* spectral fitting alone. Through Bayesian analysis, we verify that the higher-density disk is preferred over the canonical disk reflection model with $\Delta\text{DIC} = 8.7 > 2$.

7. Mrk 79

The source Mrk 79 revealed narrow Fe K_α and Fe K_β emission line cores at ~ 6.38 keV and ~ 6.95 keV, respectively, together with a broad Fe K emission feature of width ~ 0.3 keV in the 6–7 keV band, a Compton hump in the 15–40 keV range, and variable soft x-ray excess emission below ~ 2 keV (Fig. A1). The broad Fe K emission feature is modeled by `relxillCp`, which has a confidence level of 99.9% and $\Delta\text{DIC} = 13.6 > 10$. The Compton hump modeled by `xillverCp` or `relxillCp` has confidence levels of 90.72% and 91.43%, and $\Delta\text{DIC} = 2.3$ and $5.3 > 2$, respectively. Therefore, the model `TBabs*(relxillCp+xillverCp+nthComp)` best describes the hard x-ray (3–78 keV) spectra of the source, providing

$\chi^2/\text{dof} = 716/718$, where `xillverCp` models Fe K_α/K_β emission line cores and part of the Compton hump, while the broad Fe K emission and most of the Compton hump are modeled by `relxillCp`. When extrapolated down to 0.3 keV, the hard x-ray best-fit model can fit the broadband spectra well with $\chi^2/\text{dof} = 1138.6/972$. However, we notice some positive residuals in the soft x-ray band, which could indicate the presence of an additional soft x-ray component. Therefore, we add the warm Comptonization model `compTT` to test for the presence of a warm corona and find that the broadband spectra of the source are best described by the model `TBabs*(compTT+relxillCp+xillverCp+nthComp)` with $\chi^2/\text{dof} = 1075.0/963$ and $\text{DIC}_{\text{with WC}} = 1196.3$, where soft x-ray excess is modeled by both `compTT` and `relxillCp`. However, relativistic disk reflection contributes more to the soft x-ray excess than warm Comptonization. The MLR test finds that warm Comptonization has a confidence level of 99%, which is further supported by our Bayesian analysis, revealing a difference of $\Delta\text{DIC} = \text{DIC}_{\text{without WC}} - \text{DIC}_{\text{with WC}} = 1211.3 - 1196.3 = 15.0 > 10$ in the Deviance Information Criteria between models without and with warm Comptonization. Therefore, we can write the broadband best-fit model expression as

$$\text{TBabs} * (\text{compTT} + \text{relxillCp} \\ + \text{xillverCp} + \text{nthComp}).$$

Figure A2 shows the broadband *XMM-Newton*/*NuSTAR* spectra, the best-fit count spectral model with components, and the corresponding residuals. We present the best-fit averaged spectral energy flux model with `compTT`, `relxillCp`, `xillverCp`, and `nthComp` components in Fig. A4. The best-fit spectral model parameters and their 90% confidence intervals derived through MCMC are presented in Table A6. Our MCMC analysis confirms that all the reflection model parameters are constrained with the best-fit values of disk density, iron abundance, black hole spin, and disk inclination angle are $\log[n_e/\text{cm}^{-3}] = 18.0_{-0.2}^{+0.2}$, $A_{\text{Fe}} = 0.7_{-0.1}^{+0.1}$, $a^* = 0.88_{-0.039}^{+0.048}$, and $\theta = 35_{-3}^{+4}$ degree, respectively. We obtained a more precise measurement of the black hole spin parameter, which was not constrained by the relativistic reflection spectroscopy of only *XMM-Newton* spectra (see Fig. 7). The Bayesian model selection metric strongly supports the higher-density disk reflection model over the canonical one with $\Delta\text{DIC} > 10$.

8. PG 1229+204

The *XMM-Newton*/*NuSTAR* spectra of PG 1229+204 unveiled a strong soft x-ray excess below ~ 2 keV, a broad Fe K emission line of width ~ 0.2 keV in the 6–7 keV band, and an excess emission above ~ 10 keV indicating a

Compton hump (Fig. A1). However, the detection significance of the Compton hump modeled by either `xillverCp` or `relxillCp` is less than 1σ with $\Delta\text{DIC} = -6.0$ and $-1.8 < 0$, respectively. The broad Fe K emission line is modeled by `relxillCp` and has a confidence level of 94.99% with $\Delta\text{DIC} = 9.7 > 2$. Therefore, the hard x-ray (3–78 keV) spectra of the source are best modeled by `TBabs*(relxillCp+nthComp)` with $\chi^2/\text{dof} = 123.1/113$. Once we extrapolate the hard x-ray best-fit model down to 0.3 keV, we can fit the complete 0.3–78 keV energy band by the same model, providing a reasonable fit with $\chi^2/\text{dof} = 162.0/154$, where the `relxillCp` model explains the soft x-ray excess emission self-consistently. However, we still tested the relevance of warm corona for soft x-ray excess and added the warm Comptonization model `compTT`. The significance of warm Comptonization as evaluated by the MLR test is only 56.63%. Additionally, our Bayesian analysis finds that the difference between the Deviance Information Criteria without and with warm Comptonization is $\Delta\text{DIC} = \text{DIC}_{\text{without WC}} - \text{DIC}_{\text{with WC}} = 188.2 - 201.9 = -13.7 < 0$, confirming that the relativistic disk reflection is sufficient for the modeling of soft x-ray excess and an additional warm Comptonization component is not required. Considering `TBabs`, the broadband best-fit model can be written as

$$\text{TBabs} * (\text{relxillCp} + \text{nthComp}).$$

The broadband *XMM-Newton/NuSTAR* spectra, the best-fit count spectral model with components, and the corresponding residual plot are shown in Fig. A2. We plot the best-fit averaged spectral energy flux model with `relxillCp` and `nthComp` components in Fig. A4. The complete parameter space of the best-fit model was explored using MCMC computation. In Table A6, we present the best-fit spectral model parameters and their 90% confidence intervals. The best-fit values of disk density, iron abundance, black hole spin, and disk inclination angle are $\log[n_e/\text{cm}^{-3}] \leq 16.6$, $A_{\text{Fe}} = 1.6_{-0.8}^{+1.6}$, $a^* = 0.974_{-0.775}^{+0.024}$, and $\theta = 37_{-8}^{+8}$ degree, respectively. The temperature of the hot corona is measured to be $kT_e \geq 64$ keV.

9. PG 0844 + 349

The *XMM-Newton/NuSTAR* spectra of PG 0844 + 349 show a variable soft x-ray excess below around 1–2 keV, a broad Fe K emission line of width ~ 0.6 keV in the 5–7 keV band, and a possible presence of Compton hump above 10 keV (Fig. A1). The broad Fe K emission line modeled by `relxillCp` has a confidence level of $> 99.99\%$ with $\Delta\text{DIC} = 30.2 > 10$. However, the Compton hump modeled by either `xillverCp` or `relxillCp` has significance levels below 1σ with $\Delta\text{DIC} = -4.1$ and $-7.7 < 0$, respectively. Therefore, we model the hard x-ray (3–78 keV) spectra using `TBabs*`

(`relxillCp+nthComp`), which provided the best-fit with $\chi^2/\text{dof} = 88/86$. When extrapolated down to 0.3 keV, the hard x-ray best-fit model cannot fit the broadband spectra well, resulting in significant residuals in the hard x-ray band, with $\chi^2/\text{dof} = 192.0/155$. This implies that the `relxillCp` model is incapable of explaining both soft and hard x-ray excess emission self-consistently. By incorporating `compTT`, we find that the model `TBabs*(compTT+relxillCp+nthComp)` successfully fits the broadband spectra with $\chi^2/\text{dof} = 158.7/150$ and $\text{DIC}_{\text{with WC}} = 191.4$, where both `compTT` and `relxillCp` contribute to the observed soft x-ray excess. The MLR test indicates that the warm corona has a significance level of $> 99.99\%$. This is further supported by our Bayesian analysis, which finds that the difference between the Deviance Information Criteria without and with warm Comptonization is $\Delta\text{DIC} = \text{DIC}_{\text{without WC}} - \text{DIC}_{\text{with WC}} = 236.7 - 191.4 = 45.3 > 10$. Therefore, the broadband best-fit model containing `TBabs` can be expressed as

$$\text{TBabs} * (\text{compTT} + \text{relxillCp} + \text{nthComp}).$$

Figure A2 shows the broadband *XMM-Newton/NuSTAR* spectra, the best-fit count spectral model with components, and the corresponding residuals. In Fig. A4, we present the best-fit averaged spectral energy flux model with `compTT`, `relxillCp`, and `nthComp` components. The best-fit spectral model parameters and their 90% confidence intervals are presented in Table A6, where we explore the complete parameter space and calculate parameter uncertainties using the MCMC method. The best-fit values of disk density, iron abundance, black hole spin, and disk inclination angle are $\log[n_e/\text{cm}^{-3}] \leq 18.5$, $A_{\text{Fe}} = 4.5_{-2.8}^{+0.4}$, $a^* = 0.292_{-0.157}^{+0.555}$, and $\theta = 34_{-7}^{+11}$ degree, respectively. The lower limit of the hot coronal temperature is found to be 54 keV. Our broadband relativistic reflection spectroscopy constrains the black hole spin parameter, which was previously unconstrained through *XMM-Newton* reflection spectroscopy alone, as shown in Fig. 7.

10. PG 0804 + 761

The source revealed a soft x-ray excess below ~ 2 keV, a narrow Fe K_α emission line core at ~ 6.4 keV along with a broad Fe K emission feature of width ~ 0.3 keV in the 6–7 keV band, and a possible Compton hump above 10 keV, as shown in Fig. A1. However, the significance levels of the Compton hump as modeled by `xillverCp` or `relxillCp` are only 81.53% and 78.1%, with $\Delta\text{DIC} = -1.8$, and $-4.0 < 0$, respectively. Since the Compton hump is not statistically significant, we fit the Fe K_α emission core using a narrow ($\sigma = 10$ eV) Gaussian emission (`zGauss_N`) line. The `relxillCp` model fits the broad Fe K emission line with a confidence level of 97.72% and $\Delta\text{DIC} = 5.1 > 2$. Thus, the hard x-ray (3–78 keV) spectra of the source are

best represented by the model $\text{TBabs}^*(\text{relxillCp} + \text{zGauss_N} + \text{nthComp})$ with $\chi^2/\text{dof} = 159.5/191$. We then extrapolated the best-fit hard x-ray model down to 0.3 keV to fit the broadband (0.3–78 keV) spectra, which resulted in $\chi^2/\text{dof} = 301.6/277$. However, we noticed excess residuals in the hard x-ray band, suggesting that the relxillCp model cannot self-consistently explain both the soft and hard x-ray excess in the source. Therefore, we include the warm Comptonization model compTT for the soft x-ray excess, and the model $\text{TBabs}^*(\text{compTT} + \text{relxillCp} + \text{zGauss_N} + \text{nthComp})$ provides the best fit to the broadband spectra with $\chi^2/\text{dof} = 252.2/272$ and $\text{DIC}_{\text{with WC}} = 310.1$, where both compTT and relxillCp contribute to the observed soft x-ray excess. The MLR test shows that the warm Comptonization component is significant at the $>99.99\%$ level, which is further supported by our Bayesian analysis, where the difference between the Deviance Information Criteria without and with warm Comptonization is $\Delta\text{DIC} = \text{DIC}_{\text{without WC}} - \text{DIC}_{\text{with WC}} = 348.5 - 310.1 = 38.4 > 10$. The broadband best-fit model expression incorporating TBabs is

$$\text{TBabs} * (\text{compTT} + \text{relxillCp} + \text{zGaussN} + \text{nthComp}).$$

In Fig. A2, we show the broadband *XMM-Newton*/*NuSTAR* spectra, the best-fit count spectral model with components, and the corresponding residual plot. The best-fit averaged spectral energy flux model with components is plotted in Fig. A4. In Table A6, we present the best-fit spectral model parameters and their 90% confidence intervals obtained through MCMC parameter space exploration of the best-fit model. The best-fit values of disk density, iron abundance, black hole spin, and disk inclination angle are estimated to be $\log[n_e/\text{cm}^{-3}] \leq 17.6$, $A_{\text{Fe}} = 2.1^{+2.6}_{-1.0}$, $a^* \leq 0.706$, and $\theta = 44^{+4}_{-7}$ degree, respectively. The lower limit measured for the electron temperature of the hot corona is 52 keV.

11. PG 1426 + 015

The *XMM-Newton*/*NuSTAR* spectra of the source revealed a soft x-ray excess below ~ 2 keV, a narrow Fe K_α emission line core at ~ 6.38 keV together with a broad Fe K emission feature of width ~ 0.5 keV in the 6–7 keV band, and a strong Compton hump in the 15–40 keV range (Fig. A1). The Compton hump modeled using either xillverCp or relxillCp has significance levels of 96.49% and 96.60%, with $\Delta\text{DIC} = 8.1(> 6)$ and $13.7(> 10)$, respectively. The broad Fe K emission line is modeled by relativistic disk reflection relxillCp and is significant at the 99.95%

level with $\Delta\text{DIC} = 11.4 > 10$. Therefore, the hard x-ray (3–78 keV) spectra of the source are best modeled by $\text{TBabs}^*(\text{relxillCp} + \text{xillverCp} + \text{nthComp})$ with $\chi^2/\text{dof} = 373.8/350$, where the xillverCp component fits the narrow Fe K_α emission line core and most of the Compton hump, while the broad Fe K emission line and part of the Compton hump are explained by relxillCp . However, the extrapolation of the hard x-ray best-fit model is unable to explain the broadband (0.3–78 keV) x-ray spectra well, yielding $\chi^2/\text{dof} = 524.9/420$ with significant residuals in the hard x-ray band, which implies that the relxillCp model cannot fit both soft and hard x-ray excess self-consistently. Therefore, we add the compTT model for soft x-ray excess and find that the model $\text{TBabs}^*(\text{compTT} + \text{relxillCp} + \text{xillverCp} + \text{nthComp})$ fits the broadband spectra well with $\chi^2/\text{dof} = 477.8/415$. No structures or features are seen in the residual spectra. The significance of the warm Comptonization component is $> 99.99\%$ as determined by the MLR test. The Bayesian model selection metric confirms the relevance of warm Comptonization over high-density disk reflection for soft x-ray excess, since the difference between the Deviance Information Criteria without and with the warm Comptonization component is $\Delta\text{DIC} = \text{DIC}_{\text{without WC}} - \text{DIC}_{\text{with WC}} = 575.8 - 534.6 = 41.2 > 10$. We can write the broadband best-fit model considering TBabs as

$$\text{TBabs} * (\text{compTT} + \text{relxillCp} + \text{xillverCp} + \text{nthComp}).$$

Figure A2 shows the broadband *XMM-Newton*/*NuSTAR* spectra, the best-fit count spectral model with components, and the corresponding residuals. We plot the best-fit averaged spectral energy flux model with all four (compTT , relxillCp , xillverCp , and nthComp) components in Fig. A4. Our MCMC analysis explored the complete parameter space of the best-fit model. The resulting best-fit spectral model parameters and their 90% confidence intervals are presented in Table A6. The best-fit values of disk density, iron abundance, black hole spin, and disk inclination angle are $\log[n_e/\text{cm}^{-3}] \leq 16.2$, $A_{\text{Fe}} = 1.6^{+0.9}_{-0.7}$, $a^* = 0.442^{+0.488}_{-0.122}$, and $\theta = 22^{+27}_{-6}$ degree, respectively. We measured the electron temperature of the hot corona to be $kT_e \geq 52$ keV. While performing the *XMM-Newton* spectral modeling of the source, JJ19 fixed the spin parameter at 0.998. Through joint *XMM-Newton*/*NuSTAR* relativistic reflection spectroscopy, we provide the first measurement of the black hole spin parameter for PG 1426 + 015.

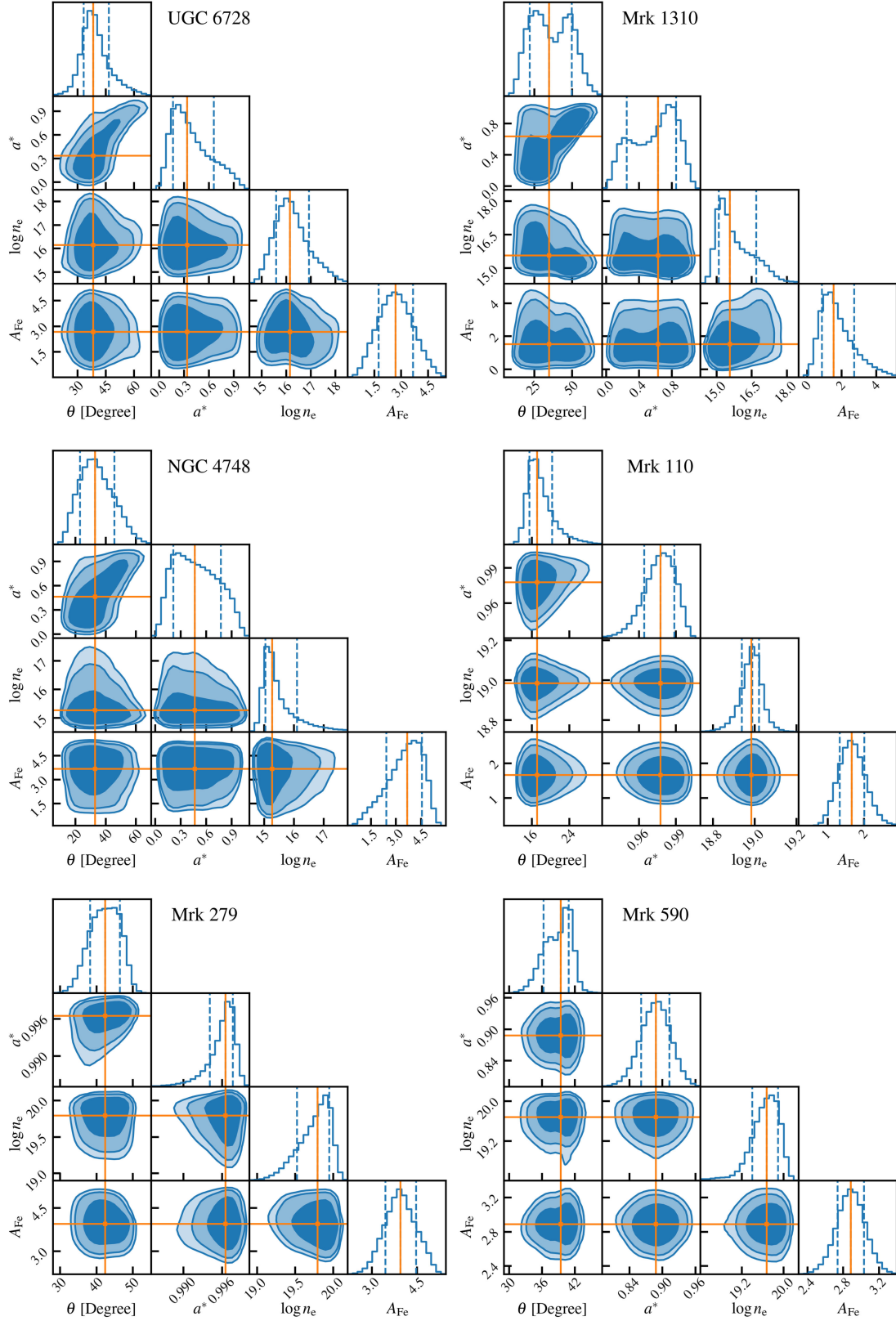


FIG. A5. Corner plots for black hole spin, disk density, iron abundance, and disk inclination angle. The on-diagonal histograms demonstrate the MCMC posterior parameter distributions. The median and 68.3% confidence intervals are shown by vertical lines. The off-diagonal two-dimensional projections show MCMC contour plots for each pair of parameters. The dark, medium, and light blue areas represent 68.3%, 90%, and 95% confidence levels, respectively, with square symbols indicating the parameter medians.

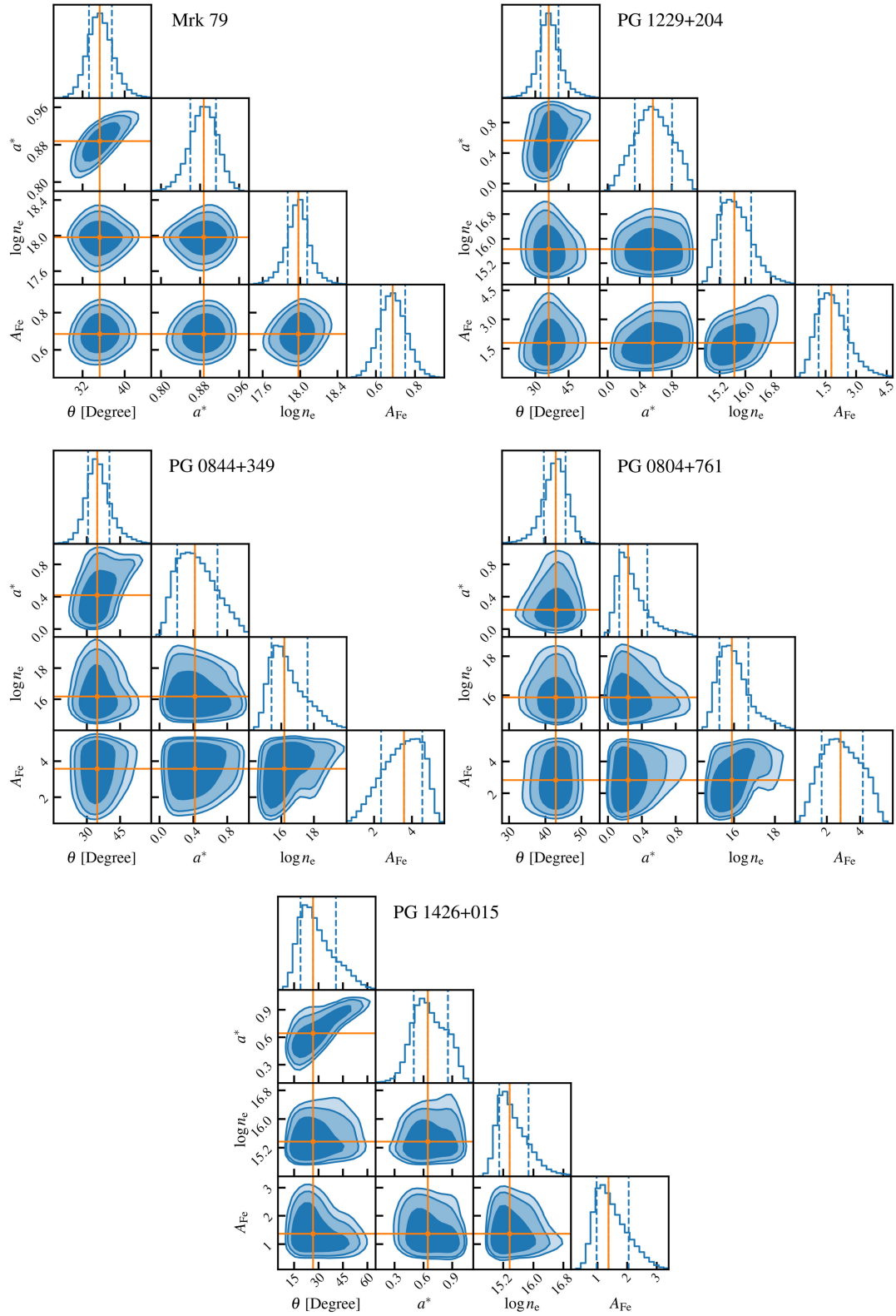


FIG. A6. Continued from previous page.

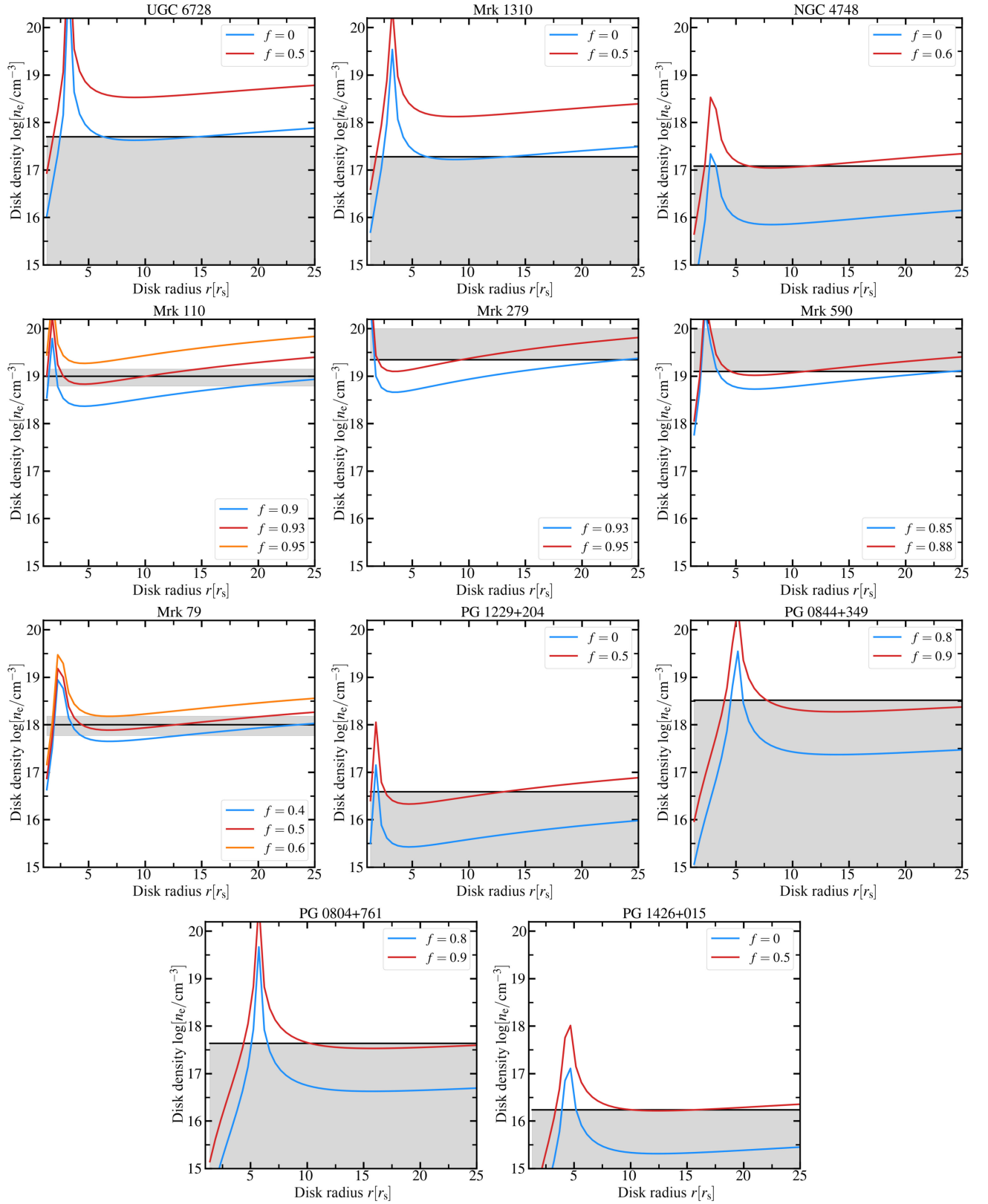


FIG. A7. The electron density ($\log[n_e/\text{cm}^{-3}]$) of the accretion disk is shown as a function of disk radius (r) and calculated for different values of the disk-to-corona power transfer fraction (f) for each source. The black solid line and gray shaded area represent the best-fit $\log n_e$ and associated 90% confidence intervals, respectively.

TABLE A1. Observing Log of the AGN sample. Observations marked with “*p*” are affected by pileup. To mitigate pileup effects, the inner radii of the annuli used for the affected observations are as follows: 8 arcsec for Obs. ID 0723100401 (Large Window mode) of NGC 4748; 12 arcsec for Obs. ID 0083960101 (Full Window mode) of Mrk 279; 8 arcsec for Obs. ID 0103660201 (Full Window mode) of PG 0844 + 349; 5 arcsec for Obs. ID 0852210101 (Small Window mode) of PG 1426 + 015 and 8 arcsec for Obs. ID 0102040501 (Full Window mode) of PG 1426 + 015.

Source (1)	Observatory (2)	Camera (3)	Obs. ID (4)	Start Time [MJD] (5)	Elapsed Time [ks] (6)	Net Exposure [ks] (7)	Net Count Rate [ct/s] (8)	Net Counts (9)
UGC 6728	<i>XMM-Newton</i>	EPIC-MOS1	0312191601	53789.8	12.15	8.22	0.87	7.16E + 03
		EPIC-MOS2			11.58	8.74	0.87	7.60E + 03
	<i>NuSTAR</i>	FPMA	60376007002	58039.8	68.39	51.80	0.40	2.07E + 04
		FPMB				50.73	0.37	1.86E + 04
Mrk 1310	<i>XMM-Newton</i>	FPMA	60160450002	57579.5	27.59	21.20	0.19	3.95E + 03
		FPMB				21.36	0.17	3.72E + 03
	<i>NuSTAR</i>	FPMA	60160465002	57556.5	38.57	21.13	0.23	4.86E + 03
		FPMB				21.08	0.21	4.49E + 03
NGC 4748	<i>XMM-Newton</i>	EPIC-pn	0723100401 ^p	56671.3	65.38	16.32	1.76	2.88E + 04
		EPIC-pn				33.74	0.15	4.97E + 03
	<i>NuSTAR</i>	FPMA	60663002002	59238.4	154.82	79.91	0.18	1.41E + 04
		FPMB				79.07	0.16	1.27E + 04
Mrk 110	<i>XMM-Newton</i>	EPIC-pn	0201130501	53324.2	46.97	32.30	21.18	6.84E + 05
		EPIC-pn	0852590101	58804.4	42.64	29.03	19.48	5.65E + 05
		EPIC-pn	0852590201	58946.0	46.64	31.69	15.23	4.83E + 05
		EPIC-pn	0840220801	58792.4	41.14	28.37	14.53	4.12E + 05
		EPIC-pn	0840220901	58794.4	38.74	26.35	15.48	4.08E + 05
		EPIC-pn	0840220701	58790.4	41.74	28.41	10.07	2.86E + 05
		EPIC-pn	0840220701	58790.4	41.74	28.41	10.07	2.86E + 05
	<i>NuSTAR</i>	FPMA	60201025002	57776.8	386.30	177.33	1.09	1.93E + 05
		FPMB				176.53	1.03	1.82E + 05
		FPMA	60502022002	58803.2	163.20	86.77	0.75	6.50E + 04
		FPMB				85.97	0.69	5.93E + 04
		FPMA	60502022004	58944.6	168.44	88.67	0.64	5.66E + 04
		FPMB				87.75	0.59	5.15E + 04
		FPMA	60502022004	58944.6	168.44	88.67	0.64	5.66E + 04
		FPMB				87.75	0.59	5.15E + 04
Mrk 279	<i>XMM-Newton</i>	EPIC-pn	0302480401	53689.7	59.37	40.58	19.33	7.85E + 05
		EPIC-pn	0302480501	53691.7	59.37	37.76	16.56	6.25E + 05
		EPIC-pn	0302480601	53694.0	37.77	21.76	17.50	3.81E + 05
		EPIC-pn	0872391301	59203.3	28.64	19.60	11.47	2.25E + 05
		EPIC-pn	0083960101 ^p	52401.5	30.07	13.00	3.22	4.19E + 04
		EPIC-pn	0083960101 ^p	52401.5	30.07	13.00	3.22	4.19E + 04
	<i>NuSTAR</i>	FPMA	60601011004	59066.5	317.27	199.41	0.16	3.21E + 04
		FPMB				197.85	0.14	2.77E + 04
		FPMA	60160562002	58785.4	37.75	27.28	0.66	1.81E + 04
		FPMB				27.07	0.63	1.71E + 04
		FPMA	60601011006	59073.0	79.61	52.38	0.21	1.09E + 04
		FPMB				51.97	0.19	9.94E + 03
		FPMA	60601011002	59064.9	97.04	58.24	0.17	9.74E + 03
		FPMB				58.05	0.15	8.58E + 03
Mrk 590	<i>XMM-Newton</i>	EPIC-pn	0201020201	53190.9	61.45	41.85	3.37	1.41E + 05
		EPIC-pn				29.23	3.17	9.27E + 04
		EPIC-pn	0865470301	59217.5	25.14	17.10	3.84	6.56E + 04
		EPIC-pn				17.25	3.23	5.58E + 04

(Table continued)

TABLE A1. (*Continued*)

Source (1)	Observatory (2)	Camera (3)	Obs. ID (4)	Start Time [MJD] (5)	Elapsed Time [ks] (6)	Net Exposure [ks] (7)	Net Count Rate [ct/s] (8)	Net Counts (9)
		EPIC-pn	0870840301	59981.2	36.14	24.73	2.01	$4.97\text{E} + 04$
		EPIC-pn	0912400101	59788.4	25.14	17.26	2.75	$4.75\text{E} + 04$
		EPIC-pn	0870840101	59437.1	8.94	6.04	5.65	$3.41\text{E} + 04$
		EPIC-pn	0870840401	59439.1	5.34	3.64	6.69	$2.43\text{E} + 04$
		EPIC-pn	0870840201	59603.1	23.94	16.34	1.12	$1.83\text{E} + 04$
		EPIC-pn	0109130301	52275.5	10.00	6.80	2.43	$1.65\text{E} + 04$
	<i>NuSTAR</i>	FPMA	80502630002	58726.4	136.75	65.02	0.35	$2.29\text{E} + 04$
		FPMB				64.05	0.31	$2.00\text{E} + 04$
		FPMA	80502630004	58869.5	97.75	50.03	0.36	$1.79\text{E} + 04$
		FPMB				49.64	0.33	$1.66\text{E} + 04$
		FPMA	80502630006	59224.5	84.21	41.26	0.17	$7.13\text{E} + 03$
		FPMB				41.00	0.16	$6.41\text{E} + 03$
		FPMA	80602604002	59571.0	114.55	53.05	0.13	$7.05\text{E} + 03$
		FPMB				52.42	0.12	$6.41\text{E} + 03$
		FPMA	60761012002	59444.2	39.76	18.65	0.34	$6.40\text{E} + 03$
		FPMB				16.96	0.32	$5.39\text{E} + 03$
		FPMA	80402610002	58418.4	38.50	21.07	0.25	$5.26\text{E} + 03$
		FPMB				20.96	0.23	$4.78\text{E} + 03$
		FPMA	80602604004	59981.1	84.75	40.58	0.12	$4.69\text{E} + 03$
		FPMB				39.88	0.11	$4.31\text{E} + 03$
		FPMA	90201043002	57724.5	95.57	50.62	0.08	$4.20\text{E} + 03$
		FPMB				50.45	0.07	$3.46\text{E} + 03$
		FPMA	60160095002	57423.7	39.03	21.21	0.08	$1.60\text{E} + 03$
		FPMB				21.16	0.07	$1.39\text{E} + 03$
	<i>XMM-Newton</i>	EPIC-pn	0870880101	59117.6	29.64	20.11	12.72	$2.56\text{E} + 05$
		EPIC-pn	0400070201	54008.5	20.67	13.84	15.11	$2.09\text{E} + 05$
		EPIC-pn	0400070401	54178.1	20.31	13.57	10.24	$1.39\text{E} + 05$
		EPIC-pn	0400070301	54040.6	19.97	13.92	9.95	$1.39\text{E} + 05$
		EPIC-pn	0103860801	51826.2	2.41	1.61	11.46	$1.84\text{E} + 04$
		EPIC-pn	0103862101	52025.8	5.12	3.52	5.21	$1.83\text{E} + 04$
		FPMA	60601010002	59115.5	125.42	64.44	0.63	$4.08\text{E} + 04$
		FPMB				64.45	0.58	$3.72\text{E} + 04$
		FPMA	60601010004	59117.2	73.09	38.19	0.65	$2.47\text{E} + 04$
		FPMB				38.10	0.59	$2.25\text{E} + 04$
	<i>XMM-Newton</i>	EPIC-pn	0301450201	53560.5	25.07	17.17	2.74	$4.71\text{E} + 04$
		FPMA	60061229002	57597.4	37.52	18.61	0.09	$1.75\text{E} + 03$
		FPMB				18.39	0.09	$1.58\text{E} + 03$
	<i>XMM-Newton</i>	EPIC-pn	0103660201 ^p	51853.0	21.24	5.98	2.38	$1.43\text{E} + 04$
		EPIC-pn	0554710101	54954.2	12.67	11.33	0.42	$4.72\text{E} + 03$
		FPMA	60463024002	59224.0	41.43	17.70	0.05	$8.14\text{E} + 02$
		FPMB				17.59	0.04	$7.46\text{E} + 02$
	<i>XMM-Newton</i>	EPIC-pn	0605110101	55265.5	44.50	16.18	10.36	$1.68\text{E} + 05$
		EPIC-pn	0605110201	55267.5	36.25	15.56	7.88	$1.23\text{E} + 05$
		FPMA	60160322002	57480.1	26.30	16.94	0.18	$3.13\text{E} + 03$
		FPMB				16.87	0.18	$2.96\text{E} + 03$

(*Table continued*)

TABLE A1. (Continued)

Source (1)	Observatory (2)	Camera (3)	Obs. ID (4)	Start Time [MJD] (5)	Elapsed Time [ks] (6)	Net Exposure [ks] (7)	Net Count Rate [ct/s] (8)	Net Counts (9)
PG 1426 + 015	<i>XMM-Newton</i>	EPIC-pn	0852210101 ^P	58872.5	106.14	70.48	3.05	2.15E + 05
		EPIC-pn	0102040501 ^P	51753.5	6.06	0.68	4.16	2.85E + 03
	<i>NuSTAR</i>	FPMA	60501049002	58871.9	202.55	94.12	0.17	1.56E + 04
		FPMB				93.82	0.16	1.48E + 04
		FPMA	60061254002	58340.4	62.21	32.48	0.20	6.62E + 03
		FPMB				32.32	0.19	6.05E + 03

TABLE A2. Details of the MLR and Bayesian statistical tests for quantitative assessment of the broad Fe K emission feature in the 5–7 keV band. Model 1 and Model 2 in columns (2) and (3) show the source models used to fit the 3–10 keV spectra before and after adding the *relxillCp* component for the broad Fe K emission. Columns (4) and (5) report the model statistics obtained from fitting the 3–10 keV spectra with Model 1 and Model 2, respectively. Column (6) reports the confidence level of the detected broad Fe K emission feature, as computed by the MLR test. Columns (7) and (8) present the Deviance Information Criteria, DIC_1 and DIC_2 , without and with the *relxillCp* component, respectively. Column (9) shows the difference between DIC_1 and DIC_2 , affirming the significance of the broad Fe K emission feature.

Source (1)	Model 1 (without <i>relxillCp</i>) (2)	Model 2 (with <i>relxillCp</i>) (3)	χ^2_1 d.o.f. ₁ (4)	χ^2_2 d.o.f. ₂ (5)	C.I.-MLR (%) (6)	DIC_1 (7)	DIC_2 (8)	ΔDIC_{12} $DIC_1 - DIC_2$ (9)
UGC 6728	<i>nthComp</i>	<i>nthComp+relxillCp</i>	$\frac{183.5}{174}$	$\frac{153.6}{165}$	99.95	200.2	187.6	12.6 [Very Strong]
Mrk 1310	<i>nthComp</i>	<i>nthComp+relxillCp</i>	$\frac{144.0}{104}$	$\frac{115.5}{95}$	99.92	151.5	145.4	6.1 [Strong]
NGC 4748	<i>nthComp+zGauss_N</i>	(<i>nthComp+zGauss_N</i> + <i>relxillCp</i>)	$\frac{123.0}{108}$	$\frac{103.4}{100}$	98.8	139.1	129.4	9.7 [Strong]
Mrk 110	<i>nthComp+zGauss_N</i>	(<i>nthComp+zGauss_N</i> + <i>relxillCp</i>)	$\frac{653.0}{558}$	$\frac{591.6}{545}$	>99.99	707.6	669.2	38.4 [Very Strong]
Mrk 279	(<i>nthComp+zGauss_N</i> + <i>zGauss2_N</i>)	(<i>nthComp+zGauss_N</i> + <i>zGauss2_N+relxillCp</i>)	$\frac{618.0}{513}$	$\frac{552.2}{498}$	>99.99	694.8	650.8	44.0 [Very Strong]
Mrk 590	(<i>nthComp+zGauss_N</i> + <i>zGauss2_N</i>)	(<i>nthComp+zGauss_N</i> + <i>zGauss2_N+relxillCp</i>)	$\frac{969.0}{945}$	$\frac{930.0}{921}$	97.27	1157.2	1151.8	5.4 [Positive]
Mrk 79	(<i>nthComp+zGauss_N</i> + <i>zGauss2_N</i>)	(<i>nthComp+zGauss_N</i> + <i>zGauss2_N+relxillCp</i>)	$\frac{446.2}{434}$	$\frac{411.8}{421}$	99.9	511.5	497.9	13.6 [Very Strong]
PG 1229 + 204	<i>nthComp</i>	<i>nthComp+relxillCp</i>	$\frac{111.1}{91}$	$\frac{95.6}{83}$	94.99	119.7	110.0	9.7 [Strong]
PG 0844 + 349	<i>nthComp</i>	<i>nthComp+relxillCp</i>	$\frac{106.3}{83}$	$\frac{69.7}{74}$	>99.99	133.2	103.0	30.2 [Very Strong]
PG 0804 + 761	<i>nthComp+zGauss_N</i>	(<i>nthComp+zGauss_N</i> + <i>relxillCp</i>)	$\frac{140.3}{148}$	$\frac{121.0}{139}$	97.72	164.2	159.1	5.1 [Positive]
PG 1426 + 015	<i>nthComp+zGauss_N</i>	(<i>nthComp+zGauss_N</i> + <i>relxillCp</i>)	$\frac{214.0}{168}$	$\frac{184.1}{159}$	99.95	237.1	225.7	11.4 [Very Strong]

TABLE A3. Details of the MLR and Bayesian statistical tests for quantitative assessment of the Compton hump. Model 1 and Model 2 in columns (2) and (3), respectively, represent the primary power-law continuum (nthComp) and Compton reflection plus the primary continuum (xillverCp+nthComp). Columns (4) and (5) show the model statistics obtained from fitting the 7–78 keV spectra with Model 1 and Model 2, respectively. Column (6) reports the confidence level of the Compton hump over the primary power-law continuum and is evaluated by the MLR test. Columns (7) and (8) show the Deviance Information Criteria, DIC₁ and DIC₂, for Model 1 and Model 2, respectively. Column (9) presents the difference between DIC₁ and DIC₂, verifying the relevance of Compton reflection hump over the primary continuum.

Source (1)	Model 1 (2)	Model 2 (3)	$\chi^2_{\text{d.o.f}_1}$ (4)	$\chi^2_{\text{d.o.f}_2}$ (5)	C.I.-MLR (%) (6)	DIC ₁ (7)	DIC ₂ (8)	ΔDIC_{12} DIC ₁ – DIC ₂ (9)
UGC 6728	nthComp	nthComp+xillverCp	$\frac{277.6}{236}$	$\frac{270.6}{232}$	86.41	300.0	304.3	–4.3 [Negative]
Mrk 1310	nthComp	nthComp+xillverCp	$\frac{104.5}{93}$	$\frac{102.0}{89}$	35.54	111.4	112.5	–1.1 [Negative]
NGC 4748	nthComp	nthComp+xillverCp	$\frac{144.7}{132}$	$\frac{133.1}{129}$	99.11	150.7	145.4	5.3 [Positive]
Mrk 110	nthComp	nthComp+xillverCp	$\frac{898.0}{801}$	$\frac{861.8}{793}$	>99.99	931.8	903.8	28.0 [Very Strong]
Mrk 279	nthComp	nthComp+xillverCp	$\frac{717.3}{630}$	$\frac{632.9}{620}$	>99.99	745.9	675.3	70.6 [Very Strong]
Mrk 590	nthComp	nthComp+xillverCp	$\frac{1037.4}{1026}$	$\frac{1010.6}{1007}$	89.06	1095.5	1094.9	0.6 [Neutral]
Mrk 79	nthComp	nthComp+xillverCp	$\frac{427.0}{432}$	$\frac{413.4}{424}$	90.72	451.7	449.4	2.3 [Positive]
PG 1229 + 204	nthComp	nthComp+xillverCp	$\frac{42.7}{54}$	$\frac{42.5}{51}$	2.24	50.3	56.3	–6.0 [Negative]
PG 0844 + 349	nthComp	nthComp+xillverCp	$\frac{24.4}{20}$	$\frac{22.5}{16}$	24.59	33.6	37.7	–4.1 [Negative]
PG 0804 + 761	nthComp	nthComp+xillverCp	$\frac{87.6}{98}$	$\frac{81.4}{94}$	81.53	97.6	99.4	–1.8 [Negative]
PG 1426 + 015	nthComp	nthComp+xillverCp	$\frac{245.9}{248}$	$\frac{237.3}{245}$	96.49	272.6	264.5	8.1 [Strong]

TABLE A4. Details of the MLR and Bayesian statistical tests to quantitatively assess the relativistic nature of the Compton reflection hump. Model 1 and Model 2 in columns (2) and (3) show the primary power-law continuum (nthComp) and relativistic disk reflection plus the primary continuum (rellxillCp+nthComp), respectively. Columns (4) and (5) report the model statistics obtained by fitting the 7–78 keV spectra with Model 1 and Model 2, respectively. Column (6) shows the confidence level of the relativistic Compton reflection hump, evaluated by the MLR test. Columns (7) and (8) present the Deviance Information Criteria, DIC₁ and DIC₂, for Model 1 and Model 2, respectively. Column (9) shows the difference between DIC₁ and DIC₂, revealing the relativistic nature of the Compton reflection hump.

Source (1)	Model 2 (3)	$\chi^2_{\text{d.o.f}_1}$ (4)	$\chi^2_{\text{d.o.f}_2}$ (5)	C.I.-MLR (%) (6)	DIC ₁ (7)	DIC ₂ (8)	ΔDIC_{12} DIC ₁ – DIC ₂ (9)
UGC 6728	nthComp+rellxillCp	$\frac{277.6}{236}$	$\frac{260.2}{227}$	95.72	300.0	285.8	14.2 [Very Strong]
Mrk 1310	nthComp+rellxillCp	$\frac{104.5}{93}$	$\frac{100.2}{84}$	10.94	111.4	113.2	–1.8 [Negative]
NGC 4748	nthComp+rellxillCp	$\frac{144.7}{132}$	$\frac{127.5}{124}$	97.19	150.7	146.5	4.2 [Positive]
Mrk 110	nthComp+rellxillCp	$\frac{898.0}{801}$	$\frac{848.8}{788}$	>99.99	931.8	911.7	20.1 [Very Strong]
Mrk 279	nthComp+rellxillCp	$\frac{717.3}{630}$	$\frac{627.8}{615}$	>99.99	745.9	677.5	68.4 [Very Strong]
Mrk 590	nthComp+rellxillCp	$\frac{1037.4}{1026}$	$\frac{1005.4}{1002}$	87.3	1095.5	1103.5	–8.0 [Negative]
Mrk 79	nthComp+rellxillCp	$\frac{427.0}{432}$	$\frac{406.6}{419}$	91.43	451.7	446.4	5.3 [Positive]
PG 1229 + 204	nthComp+rellxillCp	$\frac{42.7}{54}$	$\frac{40.8}{46}$	1.61	50.3	52.1	–1.8 [Negative]
PG 0844 + 349	nthComp+rellxillCp	$\frac{24.4}{20}$	$\frac{21.0}{11}$	5.37	33.6	41.3	–7.7 [Negative]
PG 0804 + 761	nthComp+rellxillCp	$\frac{87.6}{98}$	$\frac{75.7}{89}$	78.1	97.6	101.6	–4.0 [Negative]
PG 1426 + 015	nthComp+rellxillCp	$\frac{245.9}{248}$	$\frac{227.8}{239}$	96.6	272.6	258.9	13.7 [Very Strong]

TABLE A5. Details of the MLR and Bayesian statistical tests performed to examine the relevance of an extra compTT component in addition to the variable-density relxillCp model for the observed soft x-ray excess. Model 1 and Model 2 in columns (2) and (3), respectively, illustrate the source models used to fit the 0.3–78 keV spectra before and after the addition of an extra compTT component for the soft x-ray excess. Columns (4) and (5) report the model statistics obtained from fitting the 0.3–78 keV spectra with Model 1 and Model 2, respectively. Column (6) presents the confidence level of the warm coronal emission evaluated by the MLR test. Columns (7) and (8), respectively, show the Deviance Information Criteria, DIC₁ and DIC₂, without and with compTT. Column (9) reports the difference between DIC₁ and DIC₂, verifying the contribution of warm coronal emission to the observed soft x-ray excess.

Source (1)	Model 1 (without compTT) (2)	Model 2 (with compTT) (3)	$\frac{\chi^2_1}{\text{d.o.f}_1}$ (4)	$\frac{\chi^2_2}{\text{d.o.f}_2}$ (5)	C.I.-MLR (%) (6)	DIC ₁ (7)	DIC ₂ (8)	ΔDIC_{12} DIC ₁ – DIC ₂ (9)
UGC 6728	nthComp+relxillCp	(nthComp+relxillCp +compTT)	$\frac{514.7}{460}$	$\frac{507.8}{455}$	77.18	543.9	545.2	–1.3 [Negative]
Mrk 1310	nthComp+relxillCp	(nthComp+relxillCp +compTT)	$\frac{277.1}{228}$	$\frac{263.4}{223}$	98.24	308.9	297.9	11.0 [Very Strong]
NGC 4748	(nthComp+xillverCp +relxillCp)	(nthComp+xillverCp +relxillCp+compTT)	$\frac{281.0}{242}$	$\frac{258.3}{238}$	99.99	306.7	290.6	16.1 [Very Strong]
Mrk 110	(nthComp+xillverCp +relxillCp)	(nthComp+xillverCp +relxillCp+compTT)	$\frac{1904.4}{1434}$	$\frac{1647.0}{1425}$	> 99.99	1996.5	1752.8	243.7 [Very Strong]
Mrk 279	(nthComp+xillverCp +relxillCp)	(nthComp+xillverCp +relxillCp+compTT)	$\frac{1376.9}{1181}$	$\frac{1307.6}{1170}$	> 99.99	1463.2	1413.0	50.2 [Very Strong]
Mrk 590	(nthComp+zGauss_N +zGauss2_N +relxillCp)	(nthComp+zGauss_N +zGauss2_N +relxillCp+compTT)	$\frac{2030.2}{2028}$	$\frac{2002.0}{2008}$	89.53	2229.5	2236.9	–7.4 [Negative]
Mrk 79	(nthComp+xillverCp +relxillCp)	(nthComp+xillverCp +relxillCp+compTT)	$\frac{1138.6}{972}$	$\frac{1075.0}{963}$	> 99.99	1211.3	1196.3	15.0 [Very Strong]
PG 1229 + 204	nthComp+relxillCp	(nthComp+relxillCp +compTT)	$\frac{162.0}{154}$	$\frac{158.2}{150}$	56.63	188.2	201.9	–13.7 [Negative]
PG 0844 + 349	nthComp+relxillCp	(nthComp+relxillCp +compTT)	$\frac{192.0}{155}$	$\frac{158.7}{150}$	> 99.99	236.7	191.4	45.3 [Very Strong]
PG 0804 + 761	(nthComp+zGauss_N +relxillCp)	(nthComp+zGauss_N +relxillCp+compTT)	$\frac{301.6}{277}$	$\frac{252.2}{272}$	> 99.99	348.5	310.1	38.4 [Very Strong]
PG 1426 + 015	(nthComp+xillverCp +relxillCp)	(nthComp+xillverCp +relxillCp+compTT)	$\frac{524.9}{420}$	$\frac{477.8}{415}$	> 99.99	575.8	534.6	41.2 [Very Strong]

TABLE A6. The best-fit model parameters obtained from the joint fitting of the broadband (0.3–78 keV) *XMM-Newton*+*NuSTAR* spectra for each source. The errors represent 90% confidence intervals calculated through MCMC analyses. The “+” and “+” symbols mark simultaneous or quasimultaneous *XMM-Newton*/*NuSTAR* observations. The “*” symbol denotes the parameters tied between observations. F_{nth} , F_{rel} , and F_{xil} , respectively, indicate flux of the primary continuum (nthComp), relativistic disk reflection (relxillCp), and distant reflection (xillverCp) model components in units of 10^{-12} erg cm^{-2} s^{-1} , and are measured in the 0.3–50 keV range. The observation start time, T_{start} , is in Modified Julian Day (MJD). The disk inclination angle (θ), density (n_e), ionization (ξ), and hot coronal temperature (kT_e) are in units of degree, cm^{-3} , erg cm^{-1} , and keV, respectively. The inner emissivity index (q_{in}) and black hole spin (a^*) are dimensionless. The iron abundance, A_{Fe} , is calculated relative to the solar abundance. E_c and E_{c2} are the centroid energies of the narrow ($\sigma = 10$ eV) Fe K_{α} and Fe K_{β} emission lines in keV modeled using zGauss_N and zGauss2_N, respectively. F_{NGA} and F_{NGA2} indicate flux values of the narrow Fe K_{α} and Fe K_{β} emission lines, respectively, in units of 10^{-14} erg cm^{-2} s^{-1} and are measured in the 5–7 keV range. F_{wc} is in the unit of 10^{-12} erg cm^{-2} s^{-1} and denotes the warm corona flux measured in the range of 0.3–2 keV. kT_{wc} and τ_{wc} represent the warm coronal temperature and optical depth, respectively. $\chi^2_{\text{d.o.f}}$ shows the best-fit model statistic.

UGC 6728: Model $\equiv$ relxillCp [Relativistic Reflection] + nthComp [Hot Corona]													
T_{start}	q_{in}	θ	a^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	F_{c}	$\chi^2_{\text{d.o.f}}$	
EPIC 53789.8	≥ 3.3	50^{+4}_{-21}	$0.73^{+0.114}_{-0.61}$	≤ 17.7	≤ 4.3	$1.8^{+0.7}_{-1.2}$	$8.4^{+1.7}_{-1.7}$	$1.82^{+0.04}_{-0.09}$	≥ 42	$18.1^{+0.5}_{-0.5}$	$514.7^{+0.5}_{-0.5}$		
FPM 58039.8	—	50^*	0.73^*	—	—	1.8^*	$7.5^{+0.9}_{-0.9}$	$1.78^{+0.07}_{-0.05}$	—	$45.1^{+0.8}_{-0.9}$			
FPM 57579.5	—	50^*	0.73^*	—	—	1.8^*	$2.5^{+1.1}_{-1.1}$	$1.71^{+0.08}_{-0.12}$	—	$23.9^{+1.0}_{-1.0}$			
Mrk 1310: Model $\equiv$ relxillCp [Relativistic Reflection] + nthComp [Hot Corona] + CompTT [Warm Corona]													
T_{start}	q_{in}	θ	a^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	kT_{wc}	τ_{wc}	F_{wc}
EPIC 58487.3	≥ 3.5	49^{+7}_{-31}	$0.752^{+0.19}_{-0.605}$	≤ 17.3	$1.4^{+2.3}_{-0.7}$	$0.3^{+0.7}_{-0.2}$	$3.2^{+0.6}_{-0.6}$	$1.93^{+0.05}_{-0.11}$	≥ 26	$8.6^{+0.2}_{-0.2}$	$0.24^{+0.11}_{-0.05}$	$22.4^{+6.7}_{-7.4}$	≤ 0.3
EPIC 56635.4	—	49^*	0.752^*	—	1.4^*	0.3^*	$0.6^{+0.4}_{-0.4}$	$\leq 1.86^*$	—	$1.2^{+0.1}_{-0.1}$	0.24^*	22.4^*	≤ 0.1
FPM 57556.5	—	49^*	0.752^*	—	1.4^*	0.3^*	$4.7^{+1.0}_{-1.0}$	$1.91^{+0.26}_{-0.15}$	—	$27.3^{+1.2}_{-1.1}$	0.24^*	22.4^*	≤ 0.01
NGC 4748: Model $\equiv$ relxillCp [Relativistic Reflection] + nthComp [Hot Corona] + xillverCp [Distant Reflection] + CompTT [Warm Corona]													
T_{start}	q_{in}	θ	a^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	F_{xil}	kT_{wc}	F_{wc}
EPIC 56671.3	≥ 3.3	51^{+5}_{-33}	$0.796^{+0.123}_{-0.663}$	≤ 17.1	≤ 4.9	$3.1^{+0.8}_{-2.0}$	$3.5^{+1.5}_{-1.5}$	$1.77^{+0.11}_{-0.21}$	≥ 63	$23.2^{+1.0}_{-1.0}$	$1.0^{+0.6}_{-0.6}$	$1.8^{+0.2}_{-1.0}$	$12.9^{+2.5}_{-1.1}$
FPM 59238.4	—	51^*	0.796^*	—	—	3.1^*	$5.5^{+1.7}_{-2.4}$	$1.88^{+0.17}_{-0.13}$	—	$22.5^{+1.3}_{-1.1}$	≤ 0.9	1.8^*	≤ 0.6
Mrk 110: Model $\equiv$ relxillCp [Relativistic Reflection] + nthComp [Hot Corona] + xillverCp [Distant Reflection] + CompTT [Warm Corona]													
T_{start}	q_{in}	θ	a^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	F_{xil}	kT_{wc}	F_{wc}
EPIC 53324.2	$5.2^{+0.3}_{-0.4}$	≤ 24	$0.978^{+0.016}_{-0.024}$	$19.0^{+0.2}_{-0.2}$	$1.5^{+0.7}_{-0.4}$	$0.6^{+0.1}_{-0.2}$	$21.6^{+1.0}_{-1.0}$	$2.01^{+0.02}_{-0.03}$	≥ 118	$71.4^{+0.3}_{-0.3}$	$5.0^{+1.4}_{-1.4}$	$0.25^{+0.02}_{-0.02}$	$4.1^{+1.2}_{-0.6}$
EPIC 58804.4 [†]	5.2^*	—	0.978^*	19.0^*	1.5^*	0.6^*	$14.3^{+0.8}_{-0.8}$	$1.99^{+0.02}_{-0.03}$	—	$67.1^{+0.3}_{-0.3}$	$5.5^{+0.7}_{-0.7}$	0.25^*	$5.8^{+0.8}_{-0.7}$
EPIC 58946.0 [‡]	5.2^*	—	0.978^*	19.0^*	1.5^*	0.6^*	$9.5^{+0.7}_{-0.7}$	$1.90^{+0.02}_{-0.03}$	—	$62.9^{+0.3}_{-0.3}$	$4.6^{+0.7}_{-0.7}$	0.25^*	$4.0^{+0.7}_{-0.5}$

(Table continued)

TABLE A6. (Continued)

Mrk 110: Model≡ relxillCp [Relativistic Reflection] + nthComp [Hot Corona] + xillverCp [Distant Reflection] + CompTT [Warm Corona]															
T_{start}	q_{in}	θ	α^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	F_{xil}	kT_{wc}	τ_{wc}	F_{wc}	$\chi^2_{\text{d.o.f}}$
EPIC 58792.4	5.2*	—	0.978*	19.0*	1.5*	0.6*	$13.7^{+1.2}_{-1.2}$	$1.94^{+0.04}_{-0.05}$	—	$69.9^{+0.4}_{-0.4}$	$4.2^{+1.6}_{-1.6}$	0.25*	15.6*	$5.8^{+1.5}_{-1.0}$	
EPIC 58794.4	5.2*	—	0.978*	19.0*	1.5*	0.6*	$15.1^{+1.2}_{-1.2}$	$1.99^{+0.03}_{-0.05}$	—	$70.2^{+0.4}_{-0.4}$	$5.4^{+1.6}_{-1.6}$	0.25*	15.6*	$5.1^{+1.6}_{-0.9}$	
EPIC 58790.4	5.2*	—	0.978*	19.0*	1.5*	0.6*	$12.2^{+1.1}_{-1.1}$	$1.87^{+0.03}_{-0.08}$	—	$53.3^{+0.4}_{-0.4}$	$6.0^{+1.6}_{-1.6}$	0.25*	15.6*	$4.2^{+1.5}_{-0.8}$	
FPM 57776.8	5.2*	—	0.978*	19.0*	1.5*	0.6*	$13.4^{+1.0}_{-1.0}$	$1.94^{+0.01}_{-0.02}$	—	$117.9^{+1.1}_{-1.1}$	$7.1^{+0.8}_{-0.8}$	0.25*	15.6*	≤ 0.01	
FPM 58803.2 [†]	5.2*	—	0.978*	19.0*	1.5*	0.6*	14.3*	1.99*	—	67.1*	5.5*	0.25*	15.6*	5.8*	
FPM 58944.6 [†]	5.2*	—	0.978*	19.0*	1.5*	0.6*	9.5*	1.90*	—	62.9*	4.6*	0.25*	15.6*	4.0*	
Mrk 279: Model≡ relxillCp [Relativistic Reflection] + nthComp [Hot Corona] + xillverCp [Distant Reflection] + CompTT [Warm corona]															
T_{start}	q_{in}	θ	α^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	F_{xil}	kT_{wc}	τ_{wc}	F_{wc}	$\chi^2_{\text{d.o.f}}$
EPIC 53689.7	≥ 7.9	47^{+2}_{-11}	≥ 0.991	≥ 19.3	$4.0^{+0.8}_{-1.5}$	$2.3^{+0.1}_{-0.4}$	$7.9^{+0.3}_{-0.3}$	$1.81^{+0.02}_{-0.03}$	40^{+24}_{-14}	$80.0^{+0.4}_{-0.4}$	$5.1^{+0.6}_{-0.6}$	$0.45^{+0.11}_{-0.04}$	$12.3^{+1.3}_{-2.3}$	$7.6^{+2.1}_{-1.2}$	$\frac{1307.6}{1170}$
EPIC 53691.7	—	47*	—	—	4.0*	2.3*	$6.2^{+0.3}_{-0.3}$	$1.78^{+0.03}_{-0.03}$	40*	$75.7^{+0.4}_{-0.4}$	$4.9^{+0.7}_{-0.7}$	0.45*	12.3*	$6.2^{+1.6}_{-1.0}$	
EPIC 53694.0	—	47*	—	—	4.0*	2.3*	$6.3^{+0.4}_{-0.4}$	$1.79^{+0.03}_{-0.04}$	40*	$74.4^{+0.6}_{-0.6}$	$5.8^{+0.9}_{-0.9}$	0.45*	12.3*	$8.2^{+2.0}_{-1.2}$	
EPIC 59203.3	—	47*	—	—	4.0*	2.3*	$3.6^{+0.3}_{-0.3}$	$1.78^{+0.04}_{-0.04}$	40*	$48.2^{+0.5}_{-0.5}$	$3.1^{+0.8}_{-0.7}$	0.45*	12.3*	$5.2^{+1.5}_{-1.0}$	
EPIC 52401.5	—	47*	—	—	4.0*	2.3*	$4.4^{+0.8}_{-0.8}$	$1.76^{+0.08}_{-0.08}$	40*	$51.1^{+1.2}_{-1.2}$	$2.7^{+1.9}_{-1.8}$	0.45*	12.3*	$6.4^{+2.1}_{-2.1}$	
FPM 59066.5	—	47*	—	—	4.0*	2.3*	$6.2^{+1.4}_{-1.4}$	$1.78^{+0.04}_{-0.04}$	40*	$16.7^{+0.9}_{-0.9}$	$3.2^{+0.3}_{-0.2}$	0.45*	12.3*	≤ 0.01	
FPM 58785.4	—	47*	—	—	4.0*	2.3*	≤ 20.8	$1.90^{+0.03}_{-0.03}$	40*	$76.4^{+6.1}_{-7.5}$	$5.8^{+1.1}_{-1.1}$	0.45*	12.3*	≤ 1.1	
FPM 59073.0	—	47*	—	—	4.0*	2.3*	$3.4^{+3.7}_{-3.2}$	$1.74^{+0.04}_{-0.04}$	40*	$25.9^{+2.2}_{-2.7}$	$2.9^{+0.6}_{-0.6}$	0.45*	12.3*	≤ 0.3	
FPM 59064.9	—	47*	—	—	4.0*	2.3*	$12.2^{+3.8}_{-3.1}$	$1.82^{+0.07}_{-0.05}$	40*	$14.7^{+2.0}_{-2.5}$	$3.0^{+0.5}_{-0.5}$	0.45*	12.3*	≤ 0.01	
Mrk 590: Model≡ relxillCp [Relativistic Reflection] + nthComp [Hot Corona] + zGauss_N [Narrow Fe K α Emission Line]+ zGauss2_N [Narrow Fe K β Emission Line]															
T_{start}	q_{in}	θ	α^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	E_c	F_{NGa}	E_{c2}	F_{NGa2}	$\chi^2_{\text{d.o.f}}$
EPIC 53190.9	≥ 9.0	41^{+5}_{-4}	$0.89^{+0.041}_{-0.093}$	≥ 19.1	$2.6^{+0.5}_{-0.3}$	$3.0^{+0.1}_{-0.1}$	$1.9^{+0.1}_{-0.1}$	$1.68^{+0.01}_{-0.01}$	49^{+37}_{-19}	$22.4^{+0.2}_{-0.2}$	$6.39^{+0.01}_{-0.01}$	$9.2^{+1.8}_{-1.8}$	$6.99^{+0.04}_{-0.02}$	$5.5^{+1.9}_{-1.8}$	$\frac{2030.2}{2028}$
EPIC 59217.5	—	41*	0.89*	—	2.6*	3.0*	$2.6^{+0.2}_{-0.2}$	$1.66^{+0.02}_{-0.02}$	49*	$25.2^{+0.4}_{-0.4}$	6.39*	$10.9^{+3.2}_{-3.0}$	6.99*	≤ 5.3	
EPIC 59034.6	—	41*	0.89*	—	2.6*	3.0*	$1.9^{+0.2}_{-0.2}$	$1.69^{+0.02}_{-0.02}$	49*	$21.0^{+0.3}_{-0.3}$	6.39*	$10.8^{+3.0}_{-2.8}$	6.99*	$4.5^{+2.9}_{-2.7}$	
EPIC 59981.2 [†]	—	41*	0.89*	—	2.6*	3.0*	$1.5^{+0.1}_{-0.1}$	$1.64^{+0.01}_{-0.01}$	49*	$14.0^{+0.2}_{-0.2}$	6.39*	$7.5^{+1.4}_{-1.3}$	6.99*	≤ 3.7	
EPIC 59788.4	—	41*	0.89*	—	2.6*	3.0*	$1.6^{+0.1}_{-0.1}$	$1.66^{+0.02}_{-0.02}$	49*	$19.3^{+0.3}_{-0.3}$	6.39*	$8.2^{+2.7}_{-2.5}$	6.99*	≤ 4.7	
EPIC 59437.1	—	41*	0.89*	—	2.6*	3.0*	$4.6^{+0.4}_{-0.4}$	$1.74^{+0.03}_{-0.02}$	49*	$30.4^{+0.8}_{-0.7}$	6.39*	$9.1^{+6.0}_{-5.6}$	6.99*	≤ 10.9	
EPIC 59439.1	—	41*	0.89*	—	2.6*	3.0*	$5.6^{+0.5}_{-0.5}$	$1.73^{+0.03}_{-0.03}$	49*	$37.1^{+1.0}_{-1.0}$	6.39*	≤ 6.6	6.99*	≤ 9.6	
EPIC 59603.1	—	41*	0.89*	—	2.6*	3.0*	$0.9^{+0.1}_{-0.1}$	$1.63^{+0.03}_{-0.02}$	49*	$7.8^{+0.2}_{-0.2}$	6.39*	$9.1^{+2.3}_{-2.2}$	6.99*	$3.7^{+2.2}_{-2.0}$	
EPIC 52275.5	—	41*	0.89*	—	2.6*	3.0*	$1.8^{+0.2}_{-0.2}$	$1.67^{+0.03}_{-0.03}$	49*	$15.8^{+0.5}_{-0.5}$	6.39*	$13.3^{+4.5}_{-4.1}$	6.99*	≤ 7.3	
FPM 58726.4	—	41*	0.89*	—	2.6*	3.0*	≤ 3.1	$1.70^{+0.01}_{-0.01}$	49*	$47.8^{+0.4}_{-2.2}$	6.39*	$13.8^{+3.6}_{-3.5}$	6.99*	≤ 6.3	
FPM 58869.5	—	41*	0.89*	—	2.6*	3.0*	≤ 6.0	$1.71^{+0.01}_{-0.01}$	49*	$44.4^{+2.1}_{-2.4}$	6.39*	$18.8^{+3.9}_{-3.8}$	6.99*	$8.2^{+3.7}_{-3.6}$	

(Table continued)

TABLE A6. (Continued)

Mrk 590: Model $\equiv$ relxillCp [Relativistic Reflection] + nthComp [Hot Corona] + zGauss_N [Narrow Fe K α Emission Line] + zGauss2_N [Narrow Fe K β Emission Line]														
T_{start}	q_{in}	θ	a^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	E_c	F_{NGa}	E_{c2}	F_{NGa2} $\frac{\chi^2}{\text{d.o.f}}$
FPM	59224.5	—	41*	—	2.6*	3.0*	$4.0^{+2.9}_{-0.2}$	$1.69^{+0.02}_{-0.02}$	49*	$21.1^{+2.0}_{-2.0}$	6.39*	$12.9^{+3.4}_{-3.3}$	6.99*	≤ 5.4
FPM	59571.0	—	41*	—	2.6*	3.0*	$4.2^{+2.5}_{-2.5}$	$1.66^{+0.02}_{-0.02}$	49*	$16.4^{+1.7}_{-1.7}$	6.39*	$9.5^{+2.7}_{-2.6}$	6.99*	$3.9^{+2.5}_{-2.4}$
FPM	59444.2	—	41*	—	2.6*	3.0*	≤ 7.4	$1.71^{+0.02}_{-0.02}$	49*	$46.5^{+1.9}_{-4.1}$	6.39*	$17.0^{+6.9}_{-6.6}$	6.99*	≤ 11.1
FPM	58418.4	—	41*	—	2.6*	3.0*	≤ 5.2	$1.7^{+0.02}_{-0.02}$	49*	$34.2^{+1.0}_{-3.3}$	6.39*	$16.2^{+5.5}_{-5.3}$	6.99*	$7.3^{+5.2}_{-4.9}$
FPM	59981.1 [†]	—	41*	—	2.6*	3.0*	1.5*	1.64*	49*	14.0*	6.39*	7.5*	6.99*	—
FPM	57724.5	—	41*	—	2.6*	3.0*	≤ 3.3	$1.65^{+0.03}_{-0.03}$	49*	$10.9^{+1.2}_{-1.5}$	6.39*	$8.0^{+2.3}_{-2.2}$	6.99*	≤ 4.5
FPM	57423.7	—	41*	—	2.6*	3.0*	≤ 5.0	$1.63^{+0.04}_{-0.04}$	49*	$10.8^{+1.6}_{-2.4}$	6.39*	$5.4^{+3.5}_{-3.3}$	6.99*	≤ 6.4
Mrk 79: Model $\equiv$ relxillCp [Relativistic Reflection] + nthComp [Hot Corona] + xillverCp [Distant Reflection] + CompTT [Warm Corona]														
T_{start}	q_{in}	θ	a^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	F_{xil}	kT_{wc}	τ_{wc}	F_{wc} $\frac{\chi^2}{\text{d.o.f}}$
EPIC	59117.6 [†]	35^{+4}_{-3}	$0.88^{+0.048}_{-0.039}$	$18.0^{+0.2}_{-0.2}$	$0.7^{+0.1}_{-0.1}$	$2.9^{+0.1}_{-0.1}$	$17.8^{+0.6}_{-0.6}$	$1.88^{+0.01}_{-0.01}$	≥ 93	$56.6^{+1.2}_{-1.3}$	$8.8^{+1.1}_{-1.0}$	$0.69^{+0.20}_{-0.05}$	≥ 15.1	≤ 0.1 $\frac{1075.2}{963}$
EPIC	54008.5	6.5*	0.88*	18.0*	0.7*	2.9*	$41.1^{+0.9}_{-0.9}$	$1.88^{+0.02}_{-0.02}$	—	$40.8^{+2.0}_{-2.1}$	$13.4^{+2.5}_{-2.5}$	0.69*	—	$1.0^{+0.9}_{-0.4}$
EPIC	54178.1	6.5*	0.88*	18.0*	0.7*	2.9*	$31.4^{+0.8}_{-0.8}$	$1.80^{+0.02}_{-0.02}$	—	$32.8^{+1.9}_{-1.9}$	$17.3^{+2.7}_{-2.6}$	0.69*	—	$0.5^{+0.5}_{-0.4}$
EPIC	54040.6	6.5*	0.88*	18.0*	0.7*	2.9*	$19.4^{+0.7}_{-0.7}$	$1.75^{+0.03}_{-0.03}$	—	$47.6^{+1.9}_{-3.0}$	$17.9^{+3.1}_{-3.0}$	0.69*	—	$1.1^{+1.0}_{-0.3}$
EPIC	51826.2	6.5*	0.88*	18.0*	0.7*	2.9*	$17.4^{+2.5}_{-2.3}$	$1.91^{+0.05}_{-0.05}$	—	$46.6^{+4.2}_{-5.4}$	$18.3^{+6.6}_{-6.4}$	0.69*	—	≤ 1.0
EPIC	52025.8	6.5*	0.88*	18.0*	0.7*	2.9*	$16.9^{+1.0}_{-0.9}$	$1.62^{+0.04}_{-0.05}$	—	$26.7^{+3.1}_{-3.1}$	$19.0^{+6.5}_{-6.3}$	0.69*	—	$0.9^{+0.9}_{-0.4}$
FPM	59115.5	6.5*	0.88*	18.0*	0.7*	2.9*	$17.3^{+13.3}_{-13.3}$	$1.98^{+0.04}_{-0.03}$	—	$66.7^{+8.7}_{-8.7}$	$14.2^{+1.2}_{-1.3}$	0.69*	—	≤ 0.1
FPM	59117.2 [†]	6.5*	0.88*	18.0*	0.7*	2.9*	17.8*	1.88*	—	56.6*	8.8*	0.69*	—	—
PG 1229 + 204: Model $\equiv$ relxillCp [Relativistic Reflection] + nthComp [Hot Corona]														
T_{start}	q_{in}	θ	a^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	kT_{wc}	τ_{wc}	F_{wc}	$\frac{\chi^2}{\text{d.o.f}}$
EPIC	53560.5	≥ 3.2	37^{+8}_{-8}	$0.974^{+0.024}_{-0.075}$	≤ 16.6	$1.6^{+1.6}_{-0.8}$	≤ 0.3	$4.2^{+0.3}_{-0.3}$	$2.31^{+0.03}_{-0.05}$	≥ 64	$8.6^{+0.1}_{-0.1}$	$\frac{162.0}{154}$	—	—
FPM	57597.4	—	37*	0.974*	—	1.6*	—	$4.7^{+0.8}_{-0.8}$	$2.26^{+0.31}_{-0.30}$	—	$12.3^{+1.1}_{-1.1}$	—	—	—
PG 0844 + 349: Model $\equiv$ relxillCp [Relativistic Reflection] + nthComp [Hot Corona] + CompTT [Warm Corona]														
T_{start}	q_{in}	θ	a^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	kT_{wc}	τ_{wc}	F_{wc}	$\frac{\chi^2}{\text{d.o.f}}$
EPIC	51853.0	≥ 3.2	34^{+11}_{-7}	$0.292^{+0.555}_{-0.157}$	≤ 18.5	$4.5^{+0.4}_{-2.8}$	$2.9^{+1.4}_{-1.3}$	$2.17^{+0.20}_{-0.20}$	≥ 54	$14.9^{+0.4}_{-0.4}$	$0.26^{+0.07}_{-0.09}$	$12.0^{+5.8}_{-1.8}$	$4.1^{+1.6}_{-1.7}$	$\frac{158.7}{150}$
EPIC	54954.2	—	34*	0.292*	—	4.5*	$10.0^{+2.4}_{-2.3}$	≤ 1.87	—	$4.4^{+0.3}_{-0.3}$	0.26*	12.0*	$0.6^{+0.1}_{-0.2}$	—
FPM	59224.0	—	34*	0.292*	—	4.5*	$5.1^{+1.2}_{-1.2}$	$2.34^{+0.20}_{-0.20}$	—	$10.5^{+2.0}_{-2.0}$	0.26*	12.0*	≤ 18.9	—

(Table continued)

TABLE A6. (Continued)

PG 0804 + 761: Model≡ relxillCp [Relativistic Reflection] + nthComp [Hot Corona] + zGauss_N [Narrow Fe K α Emission Line] + CompTT [Warm Corona]																
T_{start}	q_{in}	θ	a^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	E_c	F_{NGa}	kT_{wc}	τ_{wc}	F_{wc}	$\chi^2_{\text{d.o.f}}$
EPIC	55265.5	≥ 3.1	44^{+4}_{-7}	≤ 17.6	$2.1^{+2.6}_{-1.0}$	≤ 0.7	$3.0^{+1.3}_{-1.3}$	$1.95^{+0.08}_{-0.10}$	≥ 52	$26.5^{+0.4}_{-0.4}$	$6.4^{+0.04}_{-0.04}$	$7.3^{+3.1}_{-3.9}$	≤ 0.99	≥ 5.5	$11.9^{+1.2}_{-1.5}$	$\frac{252.2}{272}$
EPIC	55267.5	—	44^*	—	2.1^*	—	$7.3^{+1.0}_{-0.9}$	$2.09^{+0.07}_{-0.19}$	—	$16.6^{+0.3}_{-0.3}$	6.4^*	$5.9^{+3.2}_{-3.9}$	—	—	$8.9^{+1.8}_{-1.4}$	
FPM	57480.1	—	44^*	—	2.1^*	—	$5.6^{+1.3}_{-1.1}$	$1.96^{+0.19}_{-0.20}$	—	$21.9^{+1.3}_{-1.6}$	6.4^*	≤ 9.4	—	—	≤ 0.7	
PG 1426 + 015: Model≡ relxillCp [Relativistic Reflection] + nthComp [Hot Corona] + xillverCp [Distant Reflection] + CompTT [Warm Corona]																
T_{start}	q_{in}	θ	a^*	$\log n_e$	A_{Fe}	$\log \xi$	F_{rel}	Γ	kT_e	F_{nth}	F_{xil}	kT_{wc}	τ_{wc}	F_{wc}	$\chi^2_{\text{d.o.f}}$	
EPIC	$58872.5^{\dagger}$	≥ 4.2	22^{+27}_{-6}	≤ 16.2	$1.6^{+0.9}_{-0.7}$	$3.0^{+0.2}_{-0.1}$	$3.9^{+0.9}_{-0.9}$	$1.81^{+0.04}_{-0.03}$	≥ 52	$16.6^{+0.6}_{-0.6}$	$2.4^{+0.4}_{-0.4}$	$0.26^{+0.03}_{-0.04}$	$11.6^{+2.2}_{-1.1}$	$2.7^{+0.3}_{-0.4}$	$\frac{477.8}{415}$	
EPIC	51753.5	—	22^*	—	1.6^*	3.0^*	≤ 15.9	$2.08^{+0.20}_{-0.25}$	—	$17.8^{+5.8}_{-5.7}$	$8.6^{+6.1}_{-5.6}$	0.26^*	11.6^*	$3.3^{+3.0}_{-3.1}$		
FPM	$58871.9^{\dagger}$	—	22^*	—	1.6^*	3.0^*	3.9^*	1.81^*	—	16.6^*	2.4^*	0.26^*	11.6^*	2.7^*		
FPM	58340.4	—	22^*	—	1.6^*	3.0^*	$7.1^{+4.8}_{-4.8}$	$1.96^{+0.15}_{-0.07}$	—	$21.7^{+3.0}_{-3.0}$	$3.4^{+1.4}_{-1.4}$	0.26^*	11.6^*	≤ 39.0		

TABLE A7. Details of the MLR and Bayesian statistical tests performed to assess the presence of relativistic disk reflection in the broadband spectral data. Model 1 and Model 2 in columns (2) and (3) show the source models used to fit the 0.3–78 keV spectra without and with the `relxillCp` component, respectively. In Model 1, a `compTT` component was always included for all sources, whereas in Model 2, with `relxillCp`, an extra `compTT` component was included only if it was statistically significant, as reported in Table A5. Columns (4) and (5) show the model statistics obtained from fitting the 0.3–78 keV spectra with Model 1 and Model 2, respectively. Column (6) reports the confidence level of relativistic disk reflection as computed by the MLR test. Columns (7) and (8) show the Deviance Information Criteria, DIC_1 and DIC_2 for Model 1 and Model 2, respectively. Column (9) reports the difference between DIC_1 and DIC_2 , reconfirming the importance of relativistic disk reflection in the broadband spectral data for all sources in the sample.

Source	Model 1 (without <code>relxillCp</code> and with <code>compTT</code>) (2)	Model 2 (with <code>relxillCp</code> and <code>compTT</code> if needed) (3)	$\frac{\chi^2}{d.o.f_1}$	$\frac{\chi^2}{d.o.f_2}$	C.I.-MLR (%)	DIC_1	DIC_2	ΔDIC_{12} $DIC_1 - DIC_2$
(1)			(4)	(5)	(6)	(7)	(8)	(9)
UGC 6728	<code>nthComp+compTT</code>	<code>nthComp+relxillCp</code>	$\frac{549.8}{464}$	$\frac{514.7}{460}$	> 99.99	575.8	543.9	31.9 [Very Strong]
Mrk 1310	<code>nthComp+compTT</code>	(<code>nthComp+compTT</code> + <code>relxillCp</code>)	$\frac{283.4}{232}$	$\frac{263.4}{223}$	98.21	305.0	297.9	7.1 [Strong]
NGC 4748	(<code>nthComp+xi1l1verCp</code> + <code>compTT</code>)	(<code>nthComp+xi1l1verCp</code> + <code>compTT+relxillCp</code>)	$\frac{275.2}{245}$	$\frac{258.3}{238}$	98.19	300.4	290.6	9.8 [Strong]
Mrk 110	(<code>nthComp+xi1l1verCp</code> + <code>compTT</code>)	(<code>nthComp+xi1l1verCp</code> + <code>compTT+relxillCp</code>)	$\frac{1954.0}{1437}$	$\frac{1647.0}{1425}$	> 99.99	2027.4	1752.8	274.6 [Very Strong]
Mrk 279	(<code>nthComp+xi1l1verCp</code> + <code>compTT</code>)	(<code>nthComp+xi1l1verCp</code> + <code>compTT+relxillCp</code>)	$\frac{1426.5}{1184}$	$\frac{1307.6}{1170}$	> 99.99	1504.2	1413.0	91.2 [Very Strong]
Mrk 590	(<code>nthComp+zGauss_N</code> + <code>zGauss2_N+compTT</code>)	(<code>nthComp+zGauss_N</code> + <code>zGauss2_N+relxillCp</code>)	$\frac{2045.0}{2032}$	$\frac{2030.2}{2028}$	99.49	2319.9	2229.5	90.4 [Very Strong]
Mrk 79	(<code>nthComp+xi1l1verCp</code> + <code>compTT</code>)	(<code>nthComp+xi1l1verCp</code> + <code>compTT+relxillCp</code>)	$\frac{1685.1}{975}$	$\frac{1075.0}{963}$	> 99.99	1746.3	1196.3	550.0 [Very Strong]
PG 1229 + 204	<code>nthComp+compTT</code>	<code>nthComp+relxillCp</code>	$\frac{188.7}{158}$	$\frac{162.0}{154}$	> 99.99	203.8	188.2	15.6 [Very Strong]
PG 0844 + 349	<code>nthComp+compTT</code>	(<code>nthComp+compTT</code> + <code>relxillCp</code>)	$\frac{190.4}{159}$	$\frac{158.7}{150}$	99.98	216.5	191.4	25.1 [Very Strong]
PG 0804 + 761	(<code>nthComp+zGauss_N</code> + <code>compTT</code>)	(<code>nthComp+zGauss_N</code> + <code>compTT+relxillCp</code>)	$\frac{287.8}{281}$	$\frac{252.2}{272}$	> 99.99	320.5	310.1	10.4 [Very Strong]
PG 1426 + 015	(<code>nthComp+xi1l1verCp</code> + <code>compTT</code>)	(<code>nthComp+xi1l1verCp</code> + <code>compTT+relxillCp</code>)	$\frac{527.4}{423}$	$\frac{477.8}{415}$	> 99.99	567.7	534.6	33.1 [Very Strong]

- [1] D. Lynden-Bell, *Nature (London)* **223**, 690 (1969).
- [2] V. G. Gurzadian and L. M. Ozernoi, *Nature (London)* **280**, 214 (1979).
- [3] M. J. Rees, *Annu. Rev. Astron. Astrophys.* **22**, 471 (1984).
- [4] A. E. Reines, J. E. Greene, and M. Geha, *Astrophys. J.* **775**, 116 (2013).
- [5] J. Frank, A. King, and D. Raine, *Accretion Power in Astrophysics*, 3rd ed. (Cambridge University Press, Cambridge, 2002).
- [6] L. Ferrarese and D. Merritt, *Astrophys. J. Lett.* **539**, L9 (2000).
- [7] K. Gültekin, D. O. Richstone, K. Gebhardt, T. R. Lauer, S. Tremaine, M. C. Aller, R. Bender, A. Dressler, S. M. Faber, A. V. Filippenko, R. Green, L. C. Ho, J. Kormendy, J. Magorrian, J. Pinkney, and C. Siopis, *Astrophys. J.* **698**, 198 (2009).
- [8] J. Kormendy and L. C. Ho, *Annu. Rev. Astron. Astrophys.* **51**, 511 (2013).
- [9] T. Di Matteo, V. Springel, and L. Hernquist, *Nature (London)* **433**, 604 (2005).
- [10] P. F. Hopkins and M. Elvis, *Mon. Not. R. Astron. Soc.* **401**, 7 (2010).
- [11] A. C. Fabian, *Annu. Rev. Astron. Astrophys.* **50**, 455 (2012).
- [12] T. M. Heckman and P. N. Best, *Annu. Rev. Astron. Astrophys.* **52**, 589 (2014).
- [13] J. Silk and M. J. Rees, *Astron. Astrophys.* **331**, L1 (1998).
- [14] F. Tombesi, R. M. Sambruna, J. N. Reeves, C. S. Reynolds, and V. Braitto, *Mon. Not. R. Astron. Soc.* **418**, L89 (2011).
- [15] M. L. Parker *et al.*, *Nature (London)* **543**, 83 (2017).
- [16] C. Pinto, W. Alston, M. L. Parker, A. C. Fabian, L. C. Gallo, D. J. K. Buisson, D. J. Walton, E. Kara, J. Jiang, A. Lohfink, and C. S. Reynolds, *Mon. Not. R. Astron. Soc.* **476**, 1021 (2018).
- [17] M. Elvis, B. J. Wilkes, J. C. McDowell, R. F. Green, J. Bechtold, S. P. Willner, M. S. Oey, E. Polonski, and R. Cutri, *Astrophys. J. Suppl. Ser.* **95**, 1 (1994).
- [18] E. A. Pier and J. H. Krolik, *Astrophys. J.* **418**, 673 (1993).
- [19] M. Cappi, F. Panessa, L. Bassani, M. Dadina, G. Di Cocco, A. Comastri, R. della Ceca, A. V. Filippenko, F. Gianotti, L. C. Ho, G. Malaguti, J. S. Mulchaey, G. G. C. Palumbo, E. Piconcelli, W. L. W. Sargent, J. Stephen, M. Trifoglio, and K. A. Weaver, *Astron. Astrophys.* **446**, 459 (2006).
- [20] F. Combes, S. García-Burillo, A. Audibert, L. Hunt, A. Eckart, S. Aalto, V. Casasola, F. Boone, M. Krips, S. Viti, K. Sakamoto, S. Muller, K. Dasyra, P. van der Werf, and S. Martin, *Astron. Astrophys.* **623**, A79 (2019).
- [21] R. Antonucci, *Annu. Rev. Astron. Astrophys.* **31**, 473 (1993).
- [22] H. Netzer, *Annu. Rev. Astron. Astrophys.* **53**, 365 (2015).
- [23] R. A. Sunyaev and L. G. Titarchuk, *Astron. Astrophys.* **86**, 121 (1980).
- [24] F. Haardt and L. Maraschi, *Astrophys. J.* **413**, 507 (1993).
- [25] Y.-F. Jiang, O. Blaes, J. M. Stone, and S. W. Davis, *Astrophys. J.* **885**, 144 (2019).
- [26] D. N. Page and K. S. Thorne, *Astrophys. J.* **191**, 499 (1974).
- [27] I. M. George and A. C. Fabian, *Mon. Not. R. Astron. Soc.* **249**, 352 (1991).
- [28] R. R. Ross and A. C. Fabian, *Mon. Not. R. Astron. Soc.* **358**, 211 (2005).
- [29] J. García and T. R. Kallman, *Astrophys. J.* **718**, 695 (2010).
- [30] A. C. Fabian, K. Iwasawa, C. S. Reynolds, and A. J. Young, *Publ. Astron. Soc. Pac.* **112**, 1145 (2000).
- [31] J. García, T. Dauser, A. Lohfink, T. R. Kallman, J. F. Steiner, J. E. McClintock, L. Brenneman, J. Wilms, W. Eikmann, C. S. Reynolds, and F. Tombesi, *Astrophys. J.* **782**, 76 (2014).
- [32] G. Matt *et al.*, *Mon. Not. R. Astron. Soc.* **439**, 3016 (2014).
- [33] A. Róžańska and J. Madej, *Mon. Not. R. Astron. Soc.* **386**, 1872 (2008).
- [34] L. W. Brenneman and C. S. Reynolds, *Astrophys. J.* **652**, 1028 (2006).
- [35] C. S. Reynolds, *Annu. Rev. Astron. Astrophys.* **59**, 117 (2021).
- [36] C. Bambi, L. W. Brenneman, T. Dauser, J. A. García, V. Grinberg, A. Ingram, J. Jiang, H. Liu, A. M. Lohfink, A. Marinucci, G. Mastroserio, R. Middei, S. Nampalliwar, A. Niedźwiecki, J. F. Steiner, A. Tripathi, and A. A. Zdziarski, *Space Sci. Rev.* **217**, 65 (2021).
- [37] L. Mallick, A. C. Fabian, J. A. García, J. A. Tomsick, M. L. Parker, T. Dauser, D. R. Wilkins, B. De Marco, J. F. Steiner, R. M. T. Connors, G. Mastroserio, A. G. Markowitz, C. Pinto, W. N. Alston, A. M. Lohfink, and P. Gandhi, *Mon. Not. R. Astron. Soc.* **513**, 4361 (2022).
- [38] M. Volonteri, P. Madau, E. Quataert, and M. J. Rees, *Astrophys. J.* **620**, 69 (2005).
- [39] E. Berti and M. Volonteri, *Astrophys. J.* **684**, 822 (2008).
- [40] M. Volonteri, M. Sikora, J. P. Lasota, and A. Merloni, *Astrophys. J.* **775**, 94 (2013).
- [41] F. Pacucci and A. Loeb, *Astrophys. J.* **895**, 95 (2020).
- [42] K. P. Singh, G. P. Garmire, and J. Nousek, *Astrophys. J.* **297**, 633 (1985).
- [43] K. A. Arnaud, G. Branduardi-Raymont, J. L. Culhane, A. C. Fabian, C. Hazard, T. A. McGlynn, R. A. Shafer, A. F. Tennant, and M. J. Ward, *Mon. Not. R. Astron. Soc.* **217**, 105 (1985).
- [44] M. Gierliński and C. Done, *Mon. Not. R. Astron. Soc.* **349**, L7 (2004).
- [45] L. Mallick, G. C. Dewangan, I. M. McHardy, and M. Pahari, *Mon. Not. R. Astron. Soc.* **472**, 174 (2017).
- [46] P. O. Petrucci, F. Ursini, A. De Rosa, S. Bianchi, M. Cappi, G. Matt, M. Dadina, and J. Malzac, *Astron. Astrophys.* **611**, A59 (2018).
- [47] S. Chalise, A. M. Lohfink, J. Chauhan, T. D. Russell, D. J. K. Buisson, and L. Mallick, *Mon. Not. R. Astron. Soc.* **517**, 4788 (2022).
- [48] J. Crummy, A. C. Fabian, L. Gallo, and R. R. Ross, *Mon. Not. R. Astron. Soc.* **365**, 1067 (2006).
- [49] L. Mallick and G. C. Dewangan, *Astrophys. J.* **863**, 178 (2018).
- [50] D. Porquet, J. N. Reeves, G. Matt, A. Marinucci, E. Nardini, V. Braitto, A. Lobban, D. R. Ballantyne, S. E. Boggs, F. E. Christensen, T. Dauser, D. Farrah, J. Garcia, C. J. Hailey, F. Harrison, D. Stern, A. Tortosa, F. Ursini, and W. W. Zhang, *Astron. Astrophys.* **609**, A42 (2018).
- [51] D. Porquet, S. Hagen, N. Grosso, A. Lobban, J. N. Reeves, V. Braitto, and C. Done, *Astron. Astrophys.* **681**, A40 (2024).

- [52] J. A. García, A. C. Fabian, T. R. Kallman, T. Dauser, M. L. Parker, J. E. McClintock, J. F. Steiner, and J. Wilms, *Mon. Not. R. Astron. Soc.* **462**, 751 (2016).
- [53] N. I. Shakura and R. A. Sunyaev, *Astron. Astrophys.* **24**, 337 (1973).
- [54] R. Svensson and A. A. Zdziarski, *Astrophys. J.* **436**, 599 (1994).
- [55] L. Mallick, W. N. Alston, M. L. Parker, A. C. Fabian, C. Pinto, G. C. Dewangan, A. Markowitz, P. Gandhi, A. K. Kembhavi, and R. Misra, *Mon. Not. R. Astron. Soc.* **479**, 615 (2018).
- [56] J. A. García, E. Kara, D. Walton, T. Beuchert, T. Dauser, E. Gattuzz, M. Balokovic, J. F. Steiner, F. Tombesi, R. M. T. Connors, T. R. Kallman, F. A. Harrison, A. Fabian, J. Wilms, D. Stern, L. Lanz, C. Ricci, and D. R. Ballantyne, *Astrophys. J.* **871**, 88 (2019).
- [57] F. Jansen, D. Lumb, B. Altieri, J. Clavel, M. Ehle, C. Erd, C. Gabriel, M. Guainazzi, P. Gondoin, R. Much, R. Munoz, M. Santos, N. Schartel, D. Texier, and G. Vacanti, *Astron. Astrophys.* **365**, L1 (2001).
- [58] F. A. Harrison *et al.*, *Astrophys. J.* **770**, 103 (2013).
- [59] J. Jiang, A. C. Fabian, T. Dauser, L. Gallo, J. A. García, E. Kara, M. L. Parker, J. A. Tomsick, D. J. Walton, and C. S. Reynolds, *Mon. Not. R. Astron. Soc.* **489**, 3436 (2019).
- [60] M. C. Bentz and S. Katz, *Publ. Astron. Soc. Pac.* **127**, 67 (2015).
- [61] R. V. Vasudevan and A. C. Fabian, *Mon. Not. R. Astron. Soc.* **381**, 1235 (2007).
- [62] S. I. Raimundo, A. C. Fabian, R. V. Vasudevan, P. Gandhi, and J. Wu, *Mon. Not. R. Astron. Soc.* **419**, 2529 (2012).
- [63] L. Mallick, D. R. Wilkins, W. N. Alston, A. Markowitz, B. De Marco, M. L. Parker, A. M. Lohfink, and C. S. Stalin, *Mon. Not. R. Astron. Soc.* **503**, 3775 (2021).
- [64] J. S. Kaastra and J. A. M. Bleeker, *Astron. Astrophys.* **587**, A151 (2016).
- [65] R. Willingale, R. L. C. Starling, A. P. Beardmore, N. R. Tanvir, and P. T. O'Brien, *Mon. Not. R. Astron. Soc.* **431**, 394 (2013).
- [66] NASA/IPAC Extragalactic Database (NED). NASA/IPAC extragalactic database (NED) (2019).
- [67] M. Batische, M. C. Bentz, S. I. Raimundo, M. Vestergaard, and C. A. Onken, *Astrophys. J. Lett.* **838**, L10 (2017).
- [68] K. A. Arnaud, in *Astronomical Data Analysis Software and Systems V*, Astronomical Society of the Pacific Conference Series Vol. 101, edited by G. H. Jacoby and J. Barnes (ASP, San Francisco, CA, 1996), p. 17.
- [69] J. Wilms, A. Allen, and R. McCray, *Astrophys. J.* **542**, 914 (2000).
- [70] D. A. Verner, G. J. Ferland, K. T. Korista, and D. G. Yakovlev, *Astrophys. J.* **465**, 487 (1996).
- [71] W. Cash, *Astrophys. J.* **228**, 939 (1979).
- [72] D. J. Spiegelhalter, N. G. Best, B. P. Carlin, and A. VanDerLinde, *J. R. Stat. Soc. Ser. B* **64**, 583 (2002).
- [73] H. Akaike, *IEEE Trans. Autom. Control* **19**, 716 (1974).
- [74] R. E. Kass and A. E. Raftery, *J. Am. Stat. Assoc.* **90**, 773 (1995).
- [75] H. Jeffreys, *Theory of Probability*, 3rd ed. (Clarendon Press, Oxford, 1961).
- [76] A. A. Zdziarski, W. N. Johnson, and P. Magdziarz, *Mon. Not. R. Astron. Soc.* **283**, 193 (1996).
- [77] A. C. Fabian, M. J. Rees, L. Stella, and N. E. White, *Mon. Not. R. Astron. Soc.* **238**, 729 (1989).
- [78] A. C. Fabian, D. R. Ballantyne, A. Merloni, S. Vaughan, K. Iwasawa, and T. Boller, *Mon. Not. R. Astron. Soc.* **331**, L35 (2002).
- [79] T. Dauser, J. García, M. L. Parker, A. C. Fabian, and J. Wilms, *Mon. Not. R. Astron. Soc.* **444**, L100 (2014).
- [80] J. García, T. Dauser, C. S. Reynolds, T. R. Kallman, J. E. McClintock, J. Wilms, and W. Eikmann, *Astrophys. J.* **768**, 146 (2013).
- [81] X. Zhao, S. Marchesi, M. Ajello, D. Cole, Z. Hu, R. Silver, and N. Torres-Albà, *Astron. Astrophys.* **650**, A57 (2021).
- [82] J. Goodman and J. Weare, *Commun. Appl. Math. Comput. Sci.* **5**, 65 (2010).
- [83] S. Vaughan, R. Edelson, R. S. Warwick, and P. Uttley, *Mon. Not. R. Astron. Soc.* **345**, 1271 (2003).
- [84] L. Mallick, G. C. Dewangan, P. Gandhi, R. Misra, and A. K. Kembhavi, *Mon. Not. R. Astron. Soc.* **460**, 1705 (2016).
- [85] C. Done, S. W. Davis, C. Jin, O. Blaes, and M. Ward, *Mon. Not. R. Astron. Soc.* **420**, 1848 (2012).
- [86] T. Dauser, J. Svoboda, N. Schartel, J. Wilms, M. Dovčiak, M. Ehle, V. Karas, M. Santos-Lleó, and H. L. Marshall, *Mon. Not. R. Astron. Soc.* **422**, 1914 (2012).
- [87] R. R. Ross and A. C. Fabian, *Mon. Not. R. Astron. Soc.* **381**, 1697 (2007).
- [88] D. R. Ballantyne, *Mon. Not. R. Astron. Soc.* **491**, 3553 (2020).
- [89] P. O. Petrucci, D. Gronkiewicz, A. Rozanska, R. Belmont, S. Bianchi, B. Czerny, G. Matt, J. Malzac, R. Middei, A. De Rosa, F. Ursini, and M. Cappi, *Astron. Astrophys.* **634**, A85 (2020).
- [90] N. Kawanaka and S. Mineshige, *Publ. Astron. Soc. Jpn.* **76**, 306 (2024).
- [91] A. Róžańska, J. Malzac, R. Belmont, B. Czerny, and P. O. Petrucci, *Astron. Astrophys.* **580**, A77 (2015).
- [92] D. Gronkiewicz, A. Róžańska, P.-O. Petrucci, and R. Belmont, *Astron. Astrophys.* **675**, A198 (2023).
- [93] G. Salvesen, J. B. Simon, P. J. Armitage, and M. C. Begelman, *Mon. Not. R. Astron. Soc.* **457**, 857 (2016).
- [94] C. Ricci *et al.*, *Astrophys. J. Suppl. Ser.* **233**, 17 (2017).
- [95] A. Akylas and I. Georgantopoulos, *Astron. Astrophys.* **655**, A60 (2021).
- [96] J.-L. Kang and J.-X. Wang, *Astrophys. J.* **929**, 141 (2022).
- [97] P. O. Petrucci, F. Haardt, L. Maraschi, P. Grandi, J. Malzac, G. Matt, F. Nicastro, L. Piro, G. C. Perola, and A. De Rosa, *Astrophys. J.* **556**, 716 (2001).
- [98] A. Tortosa, S. Bianchi, A. Marinucci, G. Matt, and P. O. Petrucci, *Astron. Astrophys.* **614**, A37 (2018).
- [99] A. C. Fabian, A. Lohfink, E. Kara, M. L. Parker, R. Vasudevan, and C. S. Reynolds, *Mon. Not. R. Astron. Soc.* **451**, 4375 (2015).
- [100] A. C. Fabian, *Astrophys. J. Suppl. Ser.* **92**, 555 (1994).
- [101] J. M. Bardeen, W. H. Press, and S. A. Teukolsky, *Astrophys. J.* **178**, 347 (1972).
- [102] R. V. Vasudevan, A. C. Fabian, C. S. Reynolds, J. Aird, T. Dauser, and L. C. Gallo, *Mon. Not. R. Astron. Soc.* **458**, 2012 (2016).
- [103] M. C. Weisskopf *et al.*, *J. Astron. Telesc. Instrum. Syst.* **8**, 026002 (2022).

- [104] M. Cruise, M. Guainazzi, J. Aird, F. J. Carrera, E. Costantini, L. Corrales, T. Dauser, D. Eckert, F. Gastaldello, H. Matsumoto, R. Osten, P.-O. Petrucci, D. Porquet, G. W. Pratt, N. Rea, T. H. Reiprich, A. Simionescu, D. Spiga, and E. Troja, *Nat. Astron.* **9**, 36 (2025).
- [105] J. Heyl, I. Caiazzo, S. Gallagher, K. Hoffman, and S. Safi-Harb, in *Space Telescopes and Instrumentation 2020: Ultraviolet to Gamma Ray*, Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series Vol. 11444, edited by J.-W. A. den Herder, S. Nikzad, and K. Nakazawa (2020), p. 114442A.
- [106] T. Di Matteo, Y. Ni, N. Chen, R. Croft, S. Bird, F. Pacucci, A. Ricarte, and M. Tremmel, *Mon. Not. R. Astron. Soc.* **525**, 1479 (2023).
- [107] N. Cappelluti, A. Foord, S. Marchesi, F. Pacucci, A. Ricarte, M. Habouzit, F. Vito, M. Powell, M. Koss, and R. Mushotzky, *Universe* **10**, 276 (2024).
- [108] F. Pacucci and A. Loeb, *Mon. Not. R. Astron. Soc.* **509**, 1885 (2022).
- [109] R. Barkana and A. Loeb, *Phys. Rep.* **349**, 125 (2001).
- [110] Y. Harikane, Y. Zhang, K. Nakajima, M. Ouchi, Y. Isobe, Y. Ono, S. Hatano, Y. Xu, and H. Umeda, *Astrophys. J.* **959**, 39 (2023).
- [111] F. Pacucci and R. Narayan, *Astrophys. J.* **976**, 96 (2024).
- [112] NASA High Energy Astrophysics Science Archive Research Center (HEASARC). [Nasa/gsfrc heasarc](https://heasarc.nasa.gov).